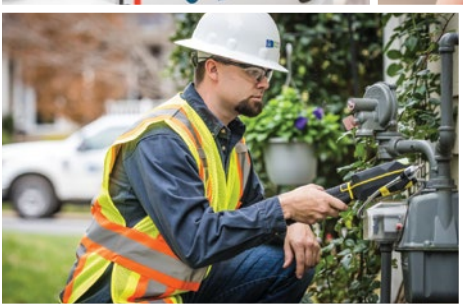
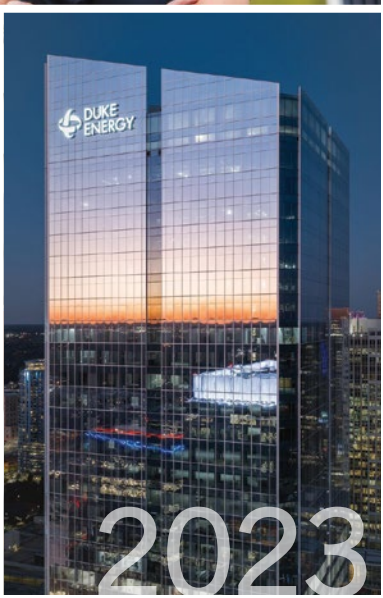
Environmental, Health and Safety Handbook





2023

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Duke Energy EHS Vision

Relentless drive to protect the safety of our employees, our communities and our environment.

Safety Principles and Employee Responsibilities

It takes all of us to be engaged in safety for every task, every day. Engaged employees will follow the Duke Energy safety principles:

Personal Accountability

- Be responsible for your own safety.
- Know and follow safety rules and standards at all times – even when no one is watching.

Active Caring

- Observe, stop and coach co-workers who are not in compliance or who exhibit at-risk behaviors.
- Take action to ensure the safety of others, and be willing to accept advice.

Hazard Recognition

- Be aware of your surroundings at all times.
- Take time to actively seek out and mitigate hazards.
- Participate actively in safety improvement efforts.
- Report immediately to supervision incidents and near misses of all types, including minor injuries and first-aid cases.
- Be fit for duty and capable to perform the task at hand.
- Participate actively in job briefings. Be fully aware of hazards associated with the job, work procedures involved, special precautions, energy source controls, chemical exposures, and personal protective equipment requirements.

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Responsibility for Environmental Protection

We are committed to being a good neighbor, which means we respect and honor our role as a steward of our environment. Sustainability is central to everything we do, and responsible management of our natural resources is critical to a cleaner environment, the quality of life in the communities we serve and Duke Energy's long-term business success.

What this means for you when it comes to environmental protection is that you will:

- Know, understand and comply with environmental rules and regulations applicable to the job.
- Do not take shortcuts that could result in damage to our environment or violate environmental procedures.
- Support sustainability efforts at your job site.
- Promptly report all incidents or near misses to supervision and ensure that you do not misrepresent the occurrence or severity of an incident.
- Implement job site best management practices and lessons learned to improve environmental performance and mitigate risks.
- Actively participate in job briefings and environmental requirement reviews. Be fully aware of the environmental aspects and impacts associated with the job, potential exposures, work procedures and controls, and roles and responsibilities.

Duke Energy Environmental, Health & Safety Policy

Duke Energy is committed to protecting the communities we live in and serve by achieving the highest levels of safety and environmental performance. By proactively addressing risk, empowering employees to be good environmental stewards, and providing an event-free workplace, we enhance the quality of life of our workforce, improve the lives of customers, and contribute to Duke Energy's long-term business success.

This Environmental, Health & Safety Policy establishes principles to fulfill this commitment. Through these, Duke Energy will:

Compliance

- Comply with applicable laws, regulations, obligations, and company policies and procedures, while also proactively addressing risk and building margin into our planning and operations.

Agility

- Be nimble as new risks and opportunities emerge. Act with a sense of urgency while ensuring accuracy in all our responses. Apply lessons learned effectively across the enterprise.

Accountability

- Set clear expectations, provide support and training, and hold employees accountable for understanding and incorporating their environmental and safety responsibilities into daily work activities. Actively engage the management of our contractors to comply with laws, regulations, obligations, and applicable company requirements, maintain a safe workplace, and be responsible stewards of the environment.

Managing Impacts

- Strive to provide an event-free workplace and be prepared to respond effectively if incidents occur. Empower all employees and contractors to actively identify risks and hazards, prevent and correct unacceptable conditions, and demonstrate expected environmental and safety behaviors.

Continual Improvement

- Set challenging goals to attain industry-leading performance and continually assess performance to improve environmental and health & safety management systems, processes and results. Collaborate across the enterprise to proactively address issues and explore opportunities to prevent environmental and safety events while providing affordable and reliable energy to our customers.

Stewardship and Transparency

- Achieve significant reductions in overall carbon emissions. Develop and maintain effective public safety programs. Engage in partnerships that address common issues and promote sound public policy. Communicate challenges and performance to maintain the trust and confidence of our stakeholders.

EHS Management System

Duke Energy has implemented an Environmental, Health and Safety Management System to ensure the consistency and effectiveness of its EHS management activities, including:

- Continually improving EHS risk management
- Preventing and detecting noncompliances
- Reducing adverse impacts to the environment
- Enhancing the company's reputation with external stakeholders

The EHS Management System is designed to be consistent with relevant international standards such as International Organization for Standardization guidelines and industry standards. However, it has been customized to apply to Duke Energy's unique operating environment, risk profile and culture.

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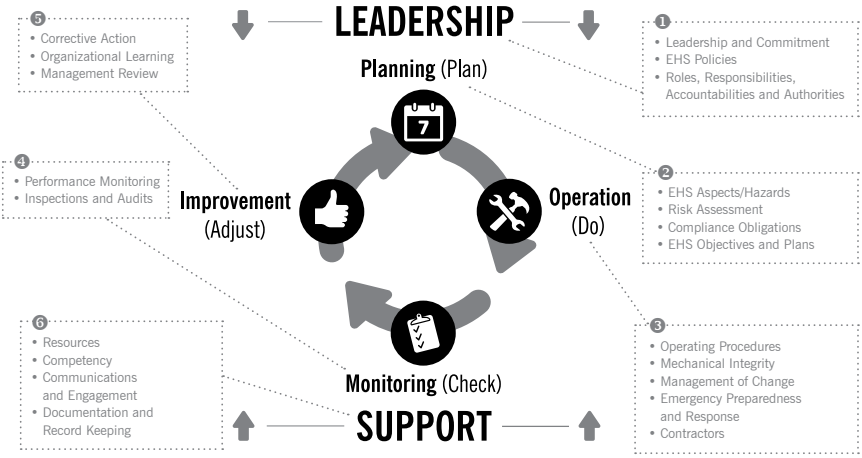

Scope

The EHS Handbook applies to all non-nuclear Duke Energy employees and contractors. Some sections or paragraphs may appear addressed specifically to employees or specifically to contractors, but employees and contractors are responsible for complying with the entire handbook. If it is unclear how to comply with a particular requirement in the handbook (e.g., if a requirement's instruction relies on internal Duke Energy contacts, forms or processes to which contractors do not have access), ask your Duke Energy contact for assistance on how to comply.

Environmental Awareness

Identify and discuss environmental risks during work planning and pre-job briefs to eliminate or avoid environmental risks.

1. Consider environmental risks and associated control measures in Job Hazard Analyses (JHAs).
2. Be prepared to respond effectively if an environmental incident occurs, and if it is safe to do so. Immediately control the area and contact your supervisor and the field environmental/EHS professional.
3. Adhere to site-specific environmental procedures or guidance. Contact your supervisor or the field environmental/EHS professional to discuss site-specific environmental requirements or training.
4. Comply with all environmental laws, regulations, permit obligations, and company policies and procedures.
5. Use materials and other resources efficiently to reduce waste generation, discharges and emissions.
6. Be aware of your proximity to water bodies and/or drains that could discharge materials directly to a water body.
7. Give extra focus and attention to environmental risks when working in or near sensitive environmental areas (e.g., on or near wetlands or water bodies).
8. Protect the sensitive area with engineered and/or administrative controls as necessary.
9. Avoid chemicals that create hazardous waste byproducts or may become hazardous waste if unused. Substitute with nonhazardous chemicals when possible. Ensure all chemicals are approved and have a Safety Data Sheet on file.
10. Notify management and the local EHS professional if an environmental regulator arrives on-site. If the EHS professional is not available, provide clear, concise and factual information as requested.

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EHS Sections

Chemical Management

Hazard Communication

1. Workers must know how to access Safety Data Sheets (SDS) for chemicals they are exposed to at work. Safety Data Sheets were formerly known as Material Safety Data Sheets (MSDS).
2. Notify supervision if label is missing or cannot be read.
3. Use only materials/chemicals that are appropriately labeled. Exception: Secondary containers do not require labels if under constant user control and used within the shift.
4. Follow label instructions for chemical use and storage unless otherwise instructed by H&S/EHS professional.
5. The handling of a chemical shall be in accordance with the precautions and personal protective equipment specified in the SDS for that chemical unless otherwise instructed by H&S/EHS professional.
6. Wear proper personal protective equipment when handling chemicals.
7. When working at another company's facility or alongside another company's workgroup, you have a right to request SDS for their chemicals to which you are potentially exposed. Likewise, they can request copies of SDS for chemicals being used by Duke Energy employees.
8. Chemical wastes generated during maintenance on systems or equipment containing hazardous chemicals shall be handled in accordance with environmental procedures.
9. Personnel shall be familiar with the location of first-aid supplies and ensure that emergency showers and eyewash basins in the immediate work area are operable before starting work.
10. Follow procedures for loading and unloading bulk hazardous chemicals from tanker trucks or rail cars.

General Chemical Safety

1. Before starting work or during the pre-job brief, identify all chemical products involved with the work task. Chemical products can be involved in the work process, materials used or coatings and insulation to be installed or removed.
2. Review the SDS for hazard information when there is a potential for chemical exposure.

3. When working with chemicals, be able to perform the following in an emergency situation:
    a. Identify an emergency situation.
    b. Know how and when to report the chemical emergency.
    c. Know local places of refuge/how to evacuate the area.
    d. Know appropriate decontamination procedures.
4. Observe the following work practices where hazardous chemicals are present:
    a. Do not eat or drink; do not use tobacco products.
    b. Wash hands and face at breaks. (When appropriate, shower at the end of work task or shift.)
    c. Never blow on or shake off contaminated clothing, and never use compressed air to clean it.
    d. Place contaminated clothing in plastic bags and label the bags.
    e. Clean the contaminated area. During the cleanup, use care to avoid further spread of the contamination.
    f. To prevent unnecessary personnel exposure, mark off the work area as necessary with ribbons, tapes, signs or barriers.
    g. When a splash hazard exists, verify availability and location of eyewash/shower facilities before performing tasks.
    h. Use personal protective equipment as specified by SDS, policy or job hazard analysis, etc.
    i. Protect equipment and the environment from spills by using spill containments.
    j. After completing a chemical transfer, remove, clean and store the transfer equipment. Ensure tanks, totes, drums or other containers are closed to prevent spills and exposures.
5. Conduct industrial hygiene monitoring as necessary to ensure a safe work environment.
6. Where hazardous chemicals are used, stored or disposed of in the workplace, use engineering controls (e.g., natural, forced or local exhaust ventilation) to eliminate or reduce airborne concentrations of chemicals where practical.
7. Respiratory protection equipment and personal protective equipment (including clothing) shall be specified by work procedures or qualified persons based on an evaluation of the hazard and the exposure levels.
8. Bulk loading or unloading of chemicals shall be performed by procedure and continuously observed.

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9. When moving chemicals, ensure the following precautions are observed:
    a. Ensure that chemical containers are closed tightly and protected from damage.
    b. Ensure that glass containers containing dry chemicals are protected from breakage by placing them in appropriate outer packaging.
    c. If moving highly hazardous liquid chemicals in glass containers, place the glass containers in protective carriers before movement.
    d. Secure chemicals being moved on vehicles or pallets against movement by means such as banding, strapping or taping.
    e. Comply with Department of Transportation requirements where applicable.
10. Contact the facility EHS field professional for appropriate procedures and guidelines related to chemical storage and segregation (flammables, oxidizers, toxic chemicals, etc.).

Acid and Caustics

1. When acids or caustics are mixed with water, the acid or caustic must be poured into the water, not the water into the acid or caustic.
2. Hydrazine is toxic and caustic and must never be handled without adequate ventilation. Skin or clothing contact and the breathing of vapors must be avoided.
3. Open flames and smoking are prohibited when working with or near acid in metal containers, such as tanks, condensers or boilers.
4. Storage of acids and caustics:
    a. Acids, in any quantity, must be kept in an approved container and prominently labeled. These containers must not be used for any other purpose.
    b. Acids must not be stored near heaters, steam pipes or other sources of heat.
    c. Acid containers must be securely sealed or covered.
    d. Workers handling acids, caustics or other corrosive and toxic chemicals must wear personal protective equipment such as gloves, aprons, eye and face protection, etc.
    e. Storage areas for acids and caustics must be posted with appropriate warning signs.
    f. Do not store acids and caustics together unless an adequately engineered barrier is in place.

5. Handling of acids and caustics:
    a. Personal protective equipment and clothing must be worn whenever acids or caustics in harmful quantities could spill, splash or drip on the person handling them. The quantity and type of acid or caustic handled will determine the kind and quantity of clothing and equipment.
    b. Should any acids or caustics come in contact with the eyes, they must be thoroughly washed out with a large amount of running water for 15 minutes. A physician must be consulted as soon as possible. Do not rub the eyes.
    c. Approved emergency eyewash/shower must be available where acids and caustics are handled.
    d. Workers handling acids, caustics or other corrosives (pH <2 or >12) and toxic chemicals must wear personal protective equipment specific to the chemical hazard in accordance with the SDS. Handling shall be defined as activities that can include tasks such as:
        i. Any activity or task inside a physical chemical barrier
        ii. Physical contact with chemicals or chemical-containing equipment
        iii. Manipulation of chemical-containing equipment
        iv. Handling chemical caps/lids, buckets, barrels, totes or manipulation of pumps, hoses, etc.
6. Before an acid or caustic container is lifted or moved, it must be examined carefully to see that it is not leaking or defective.

Chemical Control

1. All chemical products purchased by Duke Energy must be evaluated through Duke Energy's Chemical Control Program and approved prior to purchase.
2. Safety Data Sheets, approval status and ordering numbers can be found via the MSDS Search tool on the DAE or Duke Energy Portal.
3. Purchase chemical products via supply chain using a Maximo Item Number, Nuclear Catalog ID, a service contract or approved supply chain e-stock catalog.
4. Credit cards cannot be used to purchase chemicals, except in emergency situations such as outages or situations that impact plant operability. Contact the field environmental/EHS professional for assistance submitting an Emergency Purchase Form.

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5. Submit requests for new chemicals by establishing a Maximo Item Number. For business units that do not use Maximo, submit an SDS Add Request Form. In all cases, provide the SDS and allow two weeks for approval. Contact the field environmental or Chemical Control team for assistance with new chemicals.
6. The following ingredients are highly restricted from purchase:
   - MDA or 4, 4-Methylenedianiline
   - Methylene chloride or dichloromethane
   - Asbestos
   - Paints or coatings containing lead or hexavalent chromium
7. Seek approval from senior leadership to procure a product that contains a highly restricted ingredient. Contact the field environmental/EHS professional for assistance submitting a Highly Restricted Exemption Form.
8. Current Safety Data Sheets can be found by using the MSDS Search tool.
9. The field EHS professional coordinates chemical inventory tasks to ensure accurate chemical inventory data and reports are provided to regulatory agencies. Alert the field environmental/EHS professional to any activities that have the potential to increase or decrease on-site chemical quantities, such as:
   - Changes in site processes involving chemical products (e.g., fuel handling, air pollution controls, water treatment, plant operations and maintenance)
   - Addition or removal of bulk containers (e.g., tanks, totes, drums)
   - Contractor activities involving chemical products brought on-site (e.g., during outage work, maintenance activities, construction projects)
   - New site construction, building projects, or demolition
10. Ensure products are being used in the proper application. All containers must have the manufacturer's label or OSHA Hazard Communication compliant secondary container label attached at all times.
11. Do not pour or put any chemical into a drain, toilet, trash receptacle or onto the ground for disposal purposes.
12. Contact your supervisor or the field environmental/EHS professional if you have unusable chemicals that require disposal.
13. Immediately report all chemical spills to your supervisor and the field environmental/EHS professional.

Contractor Chemical Control

1. At least two weeks before arriving at the site, work with your site sponsor or project manager to provide the field environmental/EHS professional a list of the products to be brought on-site. Include the maximum quantity of each to be stored, Safety Data Sheets for each product and the number of days each product will be stored.
2. Ensure the field environmental/EHS professional has evaluated products prior to bringing the products on-site. Field professional will submit certain products for additional review and approval to Duke Energy's Chemical Team as applicable due to contractor chemical requirements.
3. Avoid chemicals that create hazardous waste byproducts or become hazardous waste if unused. Substitute with nonhazardous chemicals when possible.
4. Familiarize yourself with the chemical hazards, required personal protective equipment (PPE), precautions, usage restrictions, exposure effects, storage requirements, and proper disposal methods prior to using any product. Do not handle any chemicals without the proper PPE.
5. Ensure products are being used in the proper application. All containers must have the manufacturer's label or OSHA Hazard Communication compliant secondary container label attached at all times.
6. Remove all usable chemical commodities from the site that are not consumed during the job. Waste materials must remain on-site for proper disposal.
7. Do not pour or put any chemical into drain, toilet or trash receptacles or onto the ground for disposal purposes.
8. Contact the field environmental/EHS professional if you have unusable chemicals that require disposal.
9. Ensure that preventive measures, such as containment, are in place to minimize the risk of a chemical release.
10. Know where your closest spill kit is located in the event of a spill.
11. Immediately report all spills to facility personnel.
12. Follow the SWIM process for any chemical spills (Secure the area, Warn others, Inform appropriate personnel and Maintain the area until help arrives) if you are not formally trained in spill response.

Emergency Response

1. Be familiar with emergency procedures, including site-specific emergency plans.
2. Provide environmental or safety observations or concerns to your supervisor or the H&S/EHS professional.
3. In the event of a chemical emergency, as necessary, inform the control room/ site management, and then do the following if it is safe to do so:
   - Secure the area.
   - Warn others to leave the area.
   - Inform the field Environmental/EHS professional immediately.
   - Monitor the situation until qualified emergency response personnel arrive.
4. Where applicable, be able to recognize the emergency evacuation signals.
5. Be familiar with and use designated evacuation routes and exits.
6. When an alarm or other notification is activated, report to your assigned assembly area, or proceed to your designated area of safe refuge and remain there until instructed otherwise.
7. Know and understand the site process for reporting emergencies (e.g., site emergency phone numbers vs. 911).
8. Report all fires immediately, regardless of the size of the fire.
9. Follow the instructions of all emergency response personnel.
10. If you are a member of an emergency response team (HAZMAT, Fire Brigade, Confined Space Rescue, etc.):
    a. Ensure that your required training is current.
    b. Follow the procedures in appropriate response plans. Inspect emergency response equipment prior to use.
    c. Inform the response team leader of any medical or other personal issues that would prevent you from safely participating in a response.

Environmental, Health and Safety Event Reporting and Investigation

1. At Duke Energy, the safety of our employees, contractors and the environment is of utmost importance. When accidents or events occur, it is essential to understand what happened and why so that we can learn and improve our operations to prevent event recurrence.
2. Employees shall report all environmental, health and safety incidents to supervision immediately. This includes work-related injuries and illnesses, near-miss incidents, vehicle accidents, fire, explosions and other situations. Supervisors shall report incidents to their management and H&S/EHS professional.
3. The EHS Event Reporting and Investigation Program (ADMP-EHS-EHS-00004) includes an event reporting and investigation process. This process ensures that EHS events are correctly categorized, properly reported and adequately investigated to determine the cause(s) and to ensure that recommendations and corrective actions are implemented to prevent recurrence. This program is applicable to all EHS events across the enterprise that affect employees and contractors working on Duke Energy property, equipment or products.
4. Take action when an event occurs:
    a. Ensure that personnel in the immediate area are safe.
    b. Take steps to stabilize the event location and surrounding area if it is safe to do so.
    c. Take steps to minimize environmental impacts if it is safe to do so.
    d. Obtain medical treatment for yourself or others as needed.
    e. Immediately report the event. Call the control room (if applicable), your supervisor and the field environmental/EHS professional.
5. Report all environmental events to your supervisor or the field environmental/EHS professional as soon as possible so that any necessary notifications to outside agencies can be made within strict regulatory deadlines (some are as short as 15 minutes):
    a. Preserve any evidence for investigation while ensuring personal safety and the environment.
    b. Provide accurate information, as required, to support incident investigation.
6. Contractors shall report EHS events and make notifications in accordance with the applicable business unit EHS Supplemental Requirements.

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OSHA Compliance Personnel

1. If an OSHA compliance inspector arrives on-site, employees should notify management and the site or local H&S/EHS professional that the inspector has arrived.
2. Ask the inspector not to initiate the inspection until the H&S/EHS professional arrives.
3. For mobile crews, ask the inspector not to initiate the inspection until the selected management representative arrives at the job site.
4. Funnel all communications to and from inspectors through the H&S/EHS professional or management representative.

Job Briefing

General

1. Conduct a job briefing prior to any activity involving hands-on work, material movement, infield inspections or work in a potentially hazardous environment.
2. The purpose of a job briefing is to validate that workers understand the scope of the work, have discussed specific roles and responsibilities, have fully identified potential hazards and risk and are properly prepared to perform their assigned work tasks safely and event-free.
3. The employee in charge or designated crew member will conduct the job briefing with the workers involved before they start each job.
4. An interactive job briefing is always required that involves engaged participants.
5. The time and location selected for conducting a job briefing should minimize distractions that reduce the effectiveness of the briefing.
6. Worker proficiency with equipment to be used or worked on should be discussed as well as the defined roles of each worker.
7. Each job briefing will clearly define the job scope, including detail of items that are not within the scope and any specific stopping points.
8. Individuals should come prepared for the job briefing. The preparation required is specific to the job performed but may include tasks such as reviewing the Job Hazard Assessment (JHA), qualifications, work documents, job site walk-down, etc.

Timing

1. Conduct job briefings prior to work starting for the job or task involved.
2. Additional job briefings may be conducted at any time during the performance of a job and are a valuable tool to refocus a team when moving from one task to another.
3. Upon arrival at the worksite, a Take-A-Minute/2-Minute Drill shall be conducted to ensure job-site hazards are assessed and mitigated. See business unit guidance.
4. In the case of an emergency where potential adverse impact to the safety of personnel or the public requires immediate response, job-site hazards should be assessed and mitigated, to the extent possible.
    a. The performance of a job briefing may be delayed until the first-possible opportunity following the immediate event response. A job briefing must be conducted prior to any non-emergency work being performed.
5. If the work or operations to be performed during the workday or shift are repetitive and similar, at least one job briefing shall be conducted before the start of the first job of each day or shift.
6. If the work is stopped for an extended break, interruption or delay, the work team shall review the job briefing prior to work start.

Stop Work Criteria

1. Stop work, contact supervision or employee in charge and conduct additional job briefings if:
    a. Significant changes that might affect the safety of the workers occur during the course of the work
    b. Changes to the job scope that impact the adequacy of the initial job briefing occur (i.e., changes in location, additional permits, equipment used, needed energy source control boundary or work methods, troubleshooting, etc.)
    c. Unexpected or unsafe conditions present at the work location that would require a change to the job scope described above to mitigate
    d. Unexpected impacts to the environment, health and safety of the public or customers occur
    e. Changes to the team composition or roles of the team members occur
    f. The employee in charge may evaluate whether a new job briefing must be performed with the entire team or only those that have changed roles

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2. The employee in charge may determine whether non-significant changes to job scope or conditions that do not impact safety may allow work to continue with a Take-A-Minute/2-Minute Drill.
3. If a work stoppage is called, the employee in charge must give approval prior to restarting work activities following the 2-Minute Drill or additional job briefing.

Job Briefing Content

1. All documented and undocumented pre-job briefs shall include, at a minimum:
    a. Job Scope and Preparation
        i. Job scope
        ii. Roles and responsibilities
        iii. Qualifications and proficiency of workers
        iv. Review stop work criteria
        v. Work instructions (SDS review)
        vi. Identify critical steps/tasks, if applicable
        vii. Human performance tool use
    b. Safety Awareness and Mitigation
        i. Personal safety and Keys to Life
        ii. Actions to take in the event of an emergency
    c. Hazard Analysis and Mitigation
        i. Existing (current) conditions
        ii. Personal protective equipment (PPE)
        iii. Job hazards
        iv. Energy source controls
        v. Risks to the environment
        vi. Error-likely situations with worst thing that can happen
    d. Operating Experience and Lessons Learned
        i. Any applicable operating experience/lessons learned (OE/LL) associated with the job
        ii. Special precautions
2. Only those items listed above that are applicable to the job need to be discussed.

3. Potential/Stored Energy not controlled by Lock-Out/Tag-Out must be assessed, mitigated and documented or discussed within the job briefing.
4. Additional content may be added to the job briefing, as directed by business unit-specific standards.

Extent of Pre-Job Brief

1. A brief discussion is satisfactory if the work involved is routine and if the worker, by virtue of training and experience, can reasonably be expected to recognize and avoid the hazards involved in the job.
2. A more extensive discussion shall be conducted if the work is complicated or particularly hazardous, or the worker cannot be expected to recognize and avoid the hazards involved in the job.

Electrical Lines and Equipment

1. The existing characteristics and conditions of electric lines and equipment that are related to the safety of the work to be performed shall be determined before work is started.
2. Discuss the following existing conditions during the job briefing where the job scope includes work on energized lines and equipment that are energized or that may become energized prior to the completion of work:
    a. Nominal voltages of lines and equipment
    b. Maximum switching transient voltages
    c. Presence of hazardous induced voltages
    d. Presence of protective grounds and equipment grounding conductors
    e. Location of electrical circuits and equipment and fire-protective signaling circuits
    f. Condition of protective grounds and equipment grounding conductors
    g. Physical condition of poles utilized for the suspension of electrical lines or equipment
    h. Environmental conditions relating to safety (i.e., presence of oil or contaminant, etc.)
3. This information can be determined through work planning processes such as work execution/work order packages, line drawings, planning meetings, written procedures, etc.

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Documentation

1. Job briefings are required to be documented unless a specific exception has been established by the business unit. In general, documentation is not required for simple, low-risk tasks.
2. Undocumented pre-job briefs require use of an approved form or standard or visual aid, such as a visor card, lanyard card or handbook. Undocumented pre-job briefs may not be performed from memory.

Working Alone

1. Individuals working alone must conduct a job briefing.
2. The job briefing must include thinking through each task in the work activity and how to complete the job safely.
3. Documentation is not required unless specified by the business unit.

Special Situations

1. For traffic control situations (i.e., work zones), conduct job briefings away from traffic with a view of possible vehicle intrusion – even if working within a road closure. Work zone contractors must communicate with other job briefing participants and understand the following: Work zone setup must comply with federal, state, county, city and local regulations; the flagger warning signal device that will be used to warn of danger; and the work zone contractors may stop work.
2. For vegetation management activities, job briefing shall include: a discussion of all personal protective equipment required including chain saw resistant leg protection; communication systems to be used (verbal, hand signals, radios); and the drop or fell zone.

Human Performance Tools

Human performance principles and tools can prevent undesirable events. The four fundamental human performance tools are listed here. Business units may provide additional human performance principles and tools. Contractors may use their company's Human Performance program tools with owner approval.

Pre-Job Brief

- Deliberate meeting of job participants that focuses on understanding job scope, requirements, hazards, risks and defenses. Ensures needed preparation has been performed and workers are ready to perform the task safely and error-free.

2-Minute Drill/Take-A-Minute

- Final situational check before beginning a task to ensure conditions reflect those that are expected or were discussed during the pre-job brief and that the workers are re-focused on the critical aspects of the job.

Procedure Use and Adherence

- Ensures that work is performed safely and error-free by understanding and following procedures as written. Ensures that work is stopped when procedure guidance cannot be followed to correct the issue prior to continuing.

STAR – Stop/Think/Act/Review (Self-Checking)

- Ensures conscious and deliberate review of intended actions and expected results before performing a task. Ensures right action is performed, on the right component, at the right time. Verifies expected results are understood and achieved.

Regulatory Communications

Interpretive questions related to laws and permits should be thoroughly vetted by EHS, external affairs and legal prior to approaching regulators for guidance. Contractors should not communicate directly with regulators on Duke Energy's behalf without direction and approval from a Duke Energy EHS professional.

Health & Safety Sections

Abrasive Blasting/Spray Coating

Abrasive Blasting

1. Inspect abrasive blasting equipment before use. Check the static pressure drop of the exhaust ventilation system for the abrasive blasting room or booth. Refer to manufacturer's instructions.
2. Frequently check exhaust ventilation filters. Replace per manufacturer's instructions.
3. Ensure dust does not accumulate on the floor and aisles outside the blasting enclosure. Keep walkways clear of steel shot or similar abrasives, which may create a slipping hazard.
4. Use signs and/or ribbons to designate temporary blasting areas.
5. Before removing blasting hood, brush, dust or vacuum abrasives or other contaminants from the hood and cape.
6. At the end of the work shift, shower and change clothes when necessary to remove contaminants.
7. Before performing work to prevent buildup of static electricity, ensure blast hose is grounded to the pot.
8. Ensure blast equipment is maintained by qualified personnel familiar with the hazards.
9. Use the following personal protective equipment (PPE): work gloves, hearing protection, safety glasses, NIOSH-approved air-supplied blast hoods with capes or aprons.
10. Refer to the "Lead in Construction" and "Silica" programs in the H&S Manual where exposure to lead is an issue. Check with your environmental professional for disposal guidance.

Spray Coatings

1. Be familiar with the Safety Data Sheet (SDS) for spray coating materials.
2. Ensure ventilation is adequate or wear respiratory protection.
3. Before each use, check the static pressure drop of the exhaust ventilation system for the spray room or booth. Follow manufacturer's instructions.
4. Designate temporary spray painting areas by using signs or warning tape.
5. Eliminate ignition sources in spray painting areas.

6. Inspect spray painting equipment before use.
7. At the end of the work shift, shower and change clothing when necessary to remove contaminants.
8. Ensure airless spray guns that atomize paints and fluids at high pressures (1,000 pounds or more per square inch) are equipped with automatic or visible manual safety devices to prevent release of the paint or fluid until the safety device is manually released.
9. Use the following personal protective equipment: eye protection, respiratory protection as necessary and solvent-resistant gloves.

Aerial Lifts (Bucket Trucks and Digger Derricks)

1. Workers shall be trained prior to operating aerial lifts.
2. Workers shall conduct pre-flight inspection.
3. Use tag lines. Do not guide the load by pulling on the rigging.
4. Loads shall not be moved over personnel, and workers shall not walk under a suspended load.
5. Booms and attachments shall not be loaded beyond manufacturer's recommendations. This includes vertical and horizontal loading.
6. This equipment shall be stable in accordance with manufacturer's recommendations before it is operated. Outriggers, if equipped, shall be used.
7. The area on which outriggers will be placed shall be examined to determine if it is capable of supporting the load. Where necessary, outrigger pads or cribbing shall be added to ensure stability of the equipment.
8. When work is completed at any given location, the boom shall be properly racked and secured, outriggers fully retracted and wheel chocks stowed prior to moving the vehicle.
9. If the boom must be uncradled to position the vehicle on the job site to perform work, the vehicle shall be stable during positioning.
10. Level indicators, if equipped, shall be operable.
11. For additional details, see the applicable business unit work practices.
12. For operations in high winds, see Motorized Equipment, Vehicle, and Trailer Operations – Operations in High Winds.

Asbestos Management

1. Personnel working in facilities must be aware of the types of materials that are known or suspected to contain asbestos.
2. Personnel performing work activities in facilities where they have the potential to contact, but not disturb, asbestos-containing materials must have current Class IV asbestos training. Examples include generating plants, substations and underground vaults.
3. Treat any suspect materials as asbestos-containing unless documentation and bulk samples are on file disputing the presence of asbestos.
4. Stop work if at any time you are unsure that the material you are working on contains asbestos. Do not resume work until the material has been sampled and verified as non-asbestos.
5. Ensure you have current training in the appropriate asbestos discipline (supervisor, worker, inspector and project designer). Some states may use different terms for similar functions.
6. When required by the applicable state agency, maintain current accreditations and/or licenses per discipline.
7. During the planning process, project managers/work planners shall consider the possibility of disturbing suspect asbestos-containing building materials by ensuring an inspection of the materials is performed before the work begins.
8. If you are working on known or suspect asbestos, use identified work practices, engineering controls and PPE including respirators based on the work activity being performed.
9. Regardless of exposure levels always:
    a. Use high-efficiency particulate air (HEPA) vacuums to collect dust/debris.
    b. Use wet methods to control exposures unless the methods create an electrical or slipping hazard in roofing operations.
    c. Promptly clean up asbestos-containing material (ACM).
10. Regardless of exposure levels, the following asbestos work activities are prohibited:
    a. High-speed abrasive disk saws without HEPA-filtered exhaust
    b. Compressed air to remove asbestos
    c. Dry sweeping of ACM
    d. Worker rotation to reduce asbestos exposures

11. Regulated areas, including appropriate signage, must be established for Class I, II or III work activities.
12. Supervision of asbestos-related jobs must be conducted by a competent person based on the level of asbestos work.
13. Depending upon the level of asbestos work activity and the material being disturbed, showers may be required at the completion of the task or completion of the shift. Wastewater from asbestos showers must be collected and filtered.
14. Employees 1) engaged in Class I, II or III work for a combined total of 30 or more days per year; 2) exposed at or above the permissible exposure or excursion limit or 3) have a history of asbestos exposure considered significant by a Duke Energy physician shall be included in an asbestos medical surveillance program.
15. If required to use a respirator, follow the requirements for your respiratory protection program.
16. Seal waste in leak-tight containers.
17. All ACM waste must be labeled appropriately, stored in secure ACM waste storage areas and disposed of in Duke Energy-approved landfills.
18. All projects involving the demolition or renovation of a structure must be inspected for asbestos-containing building materials and permitted by the appropriate state agency.
19. Contractors performing asbestos abatement or monitoring services shall follow the Duke Energy Asbestos Abatement and Air Monitoring Standard.

Avoiding Illness from Insects, Ticks and Snakes

1. When working outside, be aware of the hazards of insects, wasps, yellow jackets and ticks.
2. Use insect repellent according to the manufacturer's instructions.
3. When working in an outside environment where ticks are likely to be present, avoid tick-related illness by:
    a. Dressing appropriately. Tuck pant legs in and wear long-sleeved shirts.
    b. Using tick repellent. Apply DEET to skin and permethrin to clothing. Do not apply DEET to flame-resistant clothing.
    c. Removing embedded ticks. Wash area with warm soapy water.
    d. Reporting tick bites to your supervisor. This aids in diagnosis of symptoms later.

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   e. Disposing of live tick by putting it in alcohol, placing it in a sealed bag/container, wrapping it tightly in tape or you can flush it down the toilet.

4. Be on the lookout for yellow jackets. Do not reach into areas where yellow jackets may be nesting.
5. Seek medical treatment if you exhibit symptoms of illness or allergic reaction to bites or stings.
6. When working in areas where snakes could be present, take the following precautions:
   a. Wear above-the-ankle boots or protective snake-proof chaps/gaiters.
   b. Be alert when walking through underbrush or areas obscured by foliage. Walk slowly and give snakes time to get out of your way.
   c. Be careful when placing feet and hands. Do not put hands under objects or materials that obscure your ability to see snakes that may be underneath.
   d. Be cautious when moving rocks or moving around or over rocky terrain.
   e. Probe areas before stepping over logs or piles of brush or debris.
   f. If bitten by a snake, get immediate medical attention.

Battery Energy Storage Systems

1. Duke Energy operates Battery Energy Storage Systems (BESS) across our territory. BESS systems are monitored remotely from a central office within Duke Energy.
2. Batteries remain energized, even if all of the contactors, breakers and switches have been opened.
3. BESS are unique in that they can act as both a generation source and a load. BESS typically consist of lithium-ion (Li-ion) batteries arranged inside of a container. Under normal operating conditions Li-ion batteries are safe. However, when an abnormal condition exists, the batteries can vent flammable and potentially explosive gases that can build up inside of the containers.
4. An Emergency Response Station is located at the entrance gate of the locked site fence to allow First Responders to work with Duke Energy when an abnormal condition exists.
5. An Emergency Action Plan (EAP) that is specific to the site is located at the Emergency Response Station and is to be used to manage abnormal

conditions as well as provide emergency contacts. BESS sites are surrounded by a locked fence to protect the public and allow only necessary access to the systems for maintenance.

NOTE: Under no circumstances is it necessary or allowed for unauthorized personnel to enter the site without a Duke Energy escort. If an abnormal condition is observed, refer to the site EAP found at the Emergency Response Station.

Bicycle/Tricycle/Scooter Safe Operation

1. Carry only the number of persons for which the bicycle/tricycle is designed and equipped.
2. Operators and passengers shall ride only on seats.
3. Ride only bicycles/tricycles that are equipped with an audible warning device (bell or horn) and brakes.
4. Operate bicycles/tricycles before sunrise or after sunset only if equipped with a lamp on the front and lamp or reflector on the rear.
5. Avoid distractions. Do not carry objects or utilize radios or mobile devices while operating a bicycle/tricycle.
6. Be observant of pedestrians and obey posted traffic signage.
7. Wear bicycle helmet or hard hat with chin strap and reflective apparel.
8. Riding a rental scooter/bicycle while on company time is prohibited.

Bloodborne Pathogens

1. Prevent contact with the eyes, mouth, mucous membranes and non-intact skin with blood or other potentially infectious body fluids. Follow established procedures and universal precautions to prevent contact.
2. If possible, assist injured personnel without exposing yourself to fluids that can be infected with bloodborne pathogens.
3. At a scene with bloody conditions, remain calm, place a barrier between you and the injured person, control bleeding and notify emergency personnel.
4. Do not handle bloody clothing or contact contaminated surfaces unless you are trained and use proper personal protective equipment.
5. Cleanup shall only be performed by trained and properly equipped personnel.
6. Use the prescribed engineering controls and personal protective equipment.
7. Report to supervisor any contact with potentially infectious body fluid or material.

Compressed Gas Cylinders

1. Handling and moving gas cylinders:
    a. Where removable caps are provided for valve protection, such caps shall be kept on cylinders at all times, except when cylinders are in use.
    b. Do not lift cylinders by the cap.
    c. Never drop cylinders or permit them to strike against other objects.
    d. Avoid dragging or sliding cylinders.
    e. Never use cylinders as roller supports or for any purpose other than to contain the content as received.
    f. Never handle the cylinder with a lifting magnet. Slings, ropes or chains shall not be used unless provisions have been made on the cylinder for appropriate lifting devices, such as lugs.
    g. Use suitable hand trucks, fork truck roll platform or similar device with cylinder firmly secured for transporting and unloading.
    h. Open cylinder valve slowly. Point the valve opening away from you and other persons. Avoid the use of a wrench on valves equipped with hand wheels. Never hammer the valve wheel when attempting to open or close a valve.
    i. Personal protective barricading facilities shall be provided when transferring gases from one cylinder to another.
    j. Cylinder valves shall be closed before moving cylinders, when work is finished and when cylinders are empty. Before regulator is removed from the cylinder, release all pressure from the regulator.
    k. Empty cylinders shall be marked “Empty” or “MT,” segregated from full cylinders and returned to the supplier with all valves closed and valve protection caps in place.
2. Storage of gas cylinders:
    a. Cylinders shall be stored in accordance with SDS or other guidance provided for a specific gas.
    b. Cylinders shall not be stored near highly flammable substances.
    c. Cylinders shall be kept away from radiators or other sources of heat.
    d. Do not store cylinders where they could become part of an energized circuit.
    e. Cylinders shall be stored in assigned places, away from elevators, stairs or gangways. Assigned storage places shall be located where

cylinders will not be knocked over or damaged by passing or falling objects.

f. Cylinders shall not be stored in unventilated enclosures.

g. Compressed gas cylinders not designed to be used or stored horizontally shall be tightly secured above the midpoint in an upright position at all times except, if necessary, for short periods of time while cylinders are being moved with appropriate equipment.

h. Cylinders stored on mobile vehicles shall be secured in an upright or horizontal position and guarded to prevent any damage or movement. Regulators and gauges shall be removed from compressed gas cylinders while in transit unless designed for that purpose.

i. Cylinders shall be protected from rusting.

j. Cylinders shall be hydrostatically tested according to standards of Compressed Gas Association Inc.

k. Oxygen cylinders in storage shall be separated from fuel gas cylinders or combustible materials (especially oil or grease) by a minimum distance of 20 feet, or by a noncombustible barrier at least 5 feet high with a fire-resistance rating of at least one-half hour.

l. Oxygen and acetylene can be stored on dolly cart with regulators attached if the cart has a fire-rated barrier separating the cylinders. Without a fire-rated barrier, store cylinders on carts with regulators attached only if it is anticipated that they will be used within 24 hours.

3. When using compressed air for cleaning, reduce air pressure to 30 psi or less or ensure that safety nozzle is installed on the blow down device.
4. Compressed air hoses with an inside diameter exceeding one-half inch must have a safety device at the source of supply or branch line to reduce pressure in case of hose failure or disengagement of a connection.
5. Petroleum-based products shall not be used on any equipment where oxygen passes. Keep valves, torches and regulators of oxygen and fuel gas cylinders free of oils, grease and other combustibles.

Confined Space Entry

1. All workers involved in confined space entries must be trained to perform their confined space entry duties.
2. Follow the applicable business unit or site program for the location involved.

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3. Do not enter any confined space without (at minimum) authorization and a pre-entry briefing from an entry supervisor. Do not enter a confined space unless there is a business need to do so.
4. Entry into a confined space shall not be performed unless a comprehensive hazard evaluation has been completed and the applicable site operating experience has been reviewed for the confined space, including lockout/tagout isolation.
5. Prior to entry into a permit-required confined space:
    a. Entry supervisor shall be designated to oversee entry and shall fulfill Duke Energy's duty to reduce any serious hazard to the extent feasible or eliminate it.
    b. Entry supervisor shall complete and sign an entry permit to authorize entry.
    c. Rescue shall be pre-planned and rescue equipment shall be put in place.
6. Contractors and subcontractors may use their company program if approved in advance by a Duke Energy representative.

Cranes, Lifting Equipment and Rigging

General

1. Workers shall be trained and qualified prior to operating cranes. In addition, certain mobile crane operators are required to be certified.
2. Operators shall inspect cranes and hoists prior to operation and operate equipment according to operator's manual.
3. All lifts require planning. A pre-lift planning session shall be held for all personnel involved with the lift.
    a. Load weight, size and distance shall be determined to ensure that the proper equipment is available to make the lift safely.
    b. Riggers and equipment operators shall determine the weight of the load to be handled and the capacity of handling devices before moving the load.
    c. Lifting plans must be documented as described in the H&S Manual Cranes, Lifting Equipment and Rigging Program.
4. Outriggers shall be extended according to the manufacturer's load charts.
5. Cribbing shall be used as needed to level the crane and ensure stability based on soil class.

6. Crane setup location shall be established to ensure safe lifting capabilities and avoidance of obstacles.
7. Loads shall be guided by tag lines and not by pulling on the rigging.
8. Tag lines shall be of sufficient length to allow for proper control without placing worker under the load.
9. When extending or retracting the manual fly section (dead section), the operator shall ensure that it is properly pinned in place.
10. Drivers and/or operators of mobile cranes and trucks shall ensure that aerial ladders, lifts, booms, pole derricks, automatic lift gates and outriggers are in the proper position for work or travel.
11. When work is completed at any given location, the boom shall be lowered, retracted and secured and outriggers fully retracted before moving.

Housekeeping

1. Crane cabs and operator walk paths shall be kept clean and all litter removed at the end of each operation period.
2. Areas around the load pickup and setdown points shall be kept free of loose objects, material, etc. that may interfere with the safe operation of the lift.

Using Cranes to Lift Individuals

1. Lifting individuals using personnel baskets attached to a crane is allowed but shall be in strict compliance with OSHA requirements found in 1926.1431.

Keeping Clear of the Load

1. Hoist routes that minimize exposure of workers to hoisted loads shall be used.
2. While the operator is not moving a suspended load, no worker shall be within the fall zone, except for workers:
    a. Engaged in hooking, unhooking or guiding a load
    b. Engaged in the initial attachment of the load to a component or structure
3. When workers are engaged in hooking, unhooking or guiding the load or in the initial connection of a load to a component or structure and are within the fall zone, the materials being hoisted must be rigged to prevent unintentional displacement; hooks with self-closing latches are required; and the materials must be rigged by a qualified rigger.

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Rigging and Flagging

1. Workers shall be trained and qualified prior to conducting rigging and/or flagging.
2. The rigger is responsible for inspection and assembly of the rigging hardware and lifting equipment, as well as securing the load to the rigging hardware and lifting equipment. All rigging shall be inspected prior to use.
3. The rigger shall inform the flagger that the load is ready to lift.
4. The flagger is responsible for ensuring that the load is rigged satisfactorily, either by personal inspection or by verification with the rigger.
5. The flagger is responsible for directing the lift operation and communication with the lift equipment operator.
6. The flagger should wear a reflective vest and/or reflective gloves.
7. Flaggers for mobile cranes shall be qualified signal persons.
8. Roller block, hoist, lifting and mechanical pulling hooks shall be fitted with a safety latch or secured so that rigging components remain under control. Hooks designed to have a safety latch shall have the latch installed and, if inoperable or missing, the hook shall not be used.
9. Crane hooks shall not be painted.
10. Each shackle shall have permanently affixed and legible identification markings by the manufacturer to show: a) the name or trademark of the manufacturer, b) rated load, c) size, and d) each shackle pin shall also have manufacturer markings and grade, material type or load rating.
11. Slings shall be used in accordance with the manufacturer's permanently attached identification markings.
12. If a rope is used as a sling for rigging and attached to a mechanical device, it shall have a splice in lieu of a knot. Anything being lifted by hand (on a hand line) can be lifted using a rope with a knot as long as the rope (including the knot reduction) is rated for the weight being lifted and it can be done safely.
13. When wrapping slings around sharp corners or edges, softeners or padding shall be used to prevent damage.
14. The manufacturer's recommendations for safe loading of ropes, cables, chains and slings shall not be exceeded.

Flagging and Emergency Stop

1. Throughout the entire lift, the equipment operator shall accept communications only from the designated flagger, except in emergency situations.
2. The operator shall accept an emergency stop signal from anyone.

3. When there is a concern as to safety, the mobile crane operator has the authority to stop and refuse to handle loads until a qualified person has determined that safety has been ensured.

Nighttime Operations

1. Nighttime operations shall be conducted with adequate illumination of the pickup area, setdown area, the flagger, the boom tip and the load path (if the load path is not previously determined to be clear).
2. Nighttime crane operations where contact with energized electrical lines is possible shall not occur unless such lines can be adequately illuminated, de-energized or protected.
3. The crane operator shall determine the required illumination needed for safe operation.

Mobile Crane Setup

1. Cranes must not be assembled or used unless ground conditions are firm, drained and graded to a sufficient extent so that the equipment manufacturer's specifications for adequate support and degree of level are met.
2. The controlling entity (i.e., prime contractor, general contractor, construction manager or owner) must ensure that ground preparations necessary are provided and inform the crane operator of the location of hazards beneath the equipment setup area such as voids, tanks or utilities if those hazards are identified in drawings.
3. Adequate foundations for outriggers shall be determined by observation of the location area, available drawings and/or discussions with engineering.
4. Before extending the outriggers, check to see that all personnel are clear of area where the outrigger foot will be placed. To avoid injury to co-workers from contact with the outrigger, take actions such as discussions to cover safe work distance from the outriggers, verbal warnings and spotters to observe the outrigger movement.
5. Operators shall ensure that the crane setup area and outriggers are not located directly over embedded tanks, trenches, cableways, pipelines, basements, etc.
6. All foundations and surroundings shall be observed for changes and/or acceptability prior to lifting.
7. Bearing mats and blocking materials that satisfy the crane manufacturer's recommendations shall be provided with each mobile crane.
8. Assembly/disassembly of a mobile crane must be directed by a qualified person, such as an experienced crane operator.

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9. An experienced crane operator must perform a functional test and complete post-assembly inspection after assembly of a mobile crane before it is put back into service.

Power Line Safety

For operations near overhead power lines, a mobile crane operator must comply with these rules. This section does not apply to work on transmission and distribution lines and equipment.

1. Define the work zone.
2. Make a power line hazard assessment.
3. If any part of crane can get closer than trigger distance (20 feet for lines under 350 kilovolts (kV); 50 feet for lines over 350 kV), take additional steps. Select Option 1, 2 or 3:
    a. Option 1: Have the lines de-energized and visibly grounded at the worksite.
    b. Option 2: Maintain minimum clearance distance of 20/50 feet AND implement the “encroachment prevention measures.”
    c. Option 3: Maintain distances in Table 1 below. Determine the line’s voltage by asking the utility AND implement the “encroachment prevention measures.”
4. If Option 2 or 3 is selected, implement these encroachment prevention measures:
    a. Conduct a planning meeting with the operator and other workers who will be in the area.
    b. If used, tag lines must be nonconductive.
    c. Erect elevated warning lines, barricades or line of signs.
    d. In addition to the three measures above, also select one of these measures:
        i. Use a proximity alarm.
        ii. Use dedicated spotter (must be a qualified signal person).
        iii. Use a device that automatically warns the operator to stop (range control warning device).
        iv. Use device that limits range of movement.
        v. Use an insulating link/device between the end of the load line and the load.

Table 1: Required Clearances for Operations Near Overhead Power Lines

Normal Voltage, kV (phase-to-phase)	Minimum Required Clearance (feet)
0 to 50	10
Over 50 to 200	15
Over 200 to 350	20
Over 350 to 500	25
Over 500 to 750	35

Note: Minimum required clearance is 10 feet plus 4 inches for every 10 kV over 50 kV.

Cranes Traveling Under Power Lines

Cranes traveling under a power line with no load must comply with these rules. This does not apply to T&D crane operations.

1. The boom/mast and boom/mast support system must be lowered sufficiently to meet the requirements below.
2. The clearances specified in Table 2 must be maintained.
3. The effects of speed and terrain on crane movement (including movement of the boom/mast) are considered so that those effects do not cause the minimum clearance distances specified in Table 2 to be breached.
4. If any part of the equipment will get closer than 20 feet of the power line while traveling, the employer shall ensure that a dedicated spotter who is in continuous contact with the operator is used. The dedicated spotter shall:
    a. Be positioned to effectively gauge the clearance distance.
    b. Where necessary, use equipment that enables the dedicated spotter to communicate directly with the operator.
    c. Give timely information to the operator so that the required clearance distance can be maintained.
    d. Be a qualified signal person, completing signal person training and written and practical tests.
5. When traveling at night, or in conditions of poor visibility, the operator shall ensure that:
    a. The power lines are illuminated or another means of identifying the location of the lines shall be used.
    b. A safe path of travel is identified and used.

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Table 2: Minimum Clearance Distances While Traveling With No Load and Boom/Mast Lowered

Voltage (nominal, kV, alternating current)	While Traveling Minimum Clearance Distance (feet)
Up to 0.75	4 (while traveling/boom lowered)
Over 0.75 to 50	6 (while traveling/boom lowered
Over 50 to 345	10 (while traveling/boom lowered)
Over 345 to 750	16 (while traveling/boom lowered)
Over 750 to 1,000	20 (while traveling/boom lowered)

Inclement Weather Mobile Crane Operation

1. When a local storm warning has been issued, a competent person must determine if it is necessary to implement manufacturer's recommendations for securing the equipment.
2. For adverse weather conditions, observe operating wind velocity limits of the equipment based on manufacturer's recommendations that consider wind loads on the surface area of the crane and the lifted load.
3. When inclement weather is imminent or anticipated, develop a lift plan and operating procedures that include termination of lifting activities at specified conditions, anchoring procedures, the lowering and tying down of boom using cribbing and the requirements for attaching load line to suitable anchors.
4. Discontinue operations when lightning is present and when visibility is insufficient.

Planned Engineered Lifts

1. For overhead bridge and gantry cranes only, lifts in excess of the rated load must be conducted in compliance with ASME B30.2, Section 2-3.4.
2. These lifts are limited to cranes with load rating of 5 tons and above. Load shall not exceed 125% of rated capacity. Frequency shall not exceed two in a continuous 12-month period.
3. Crane manufacturer shall be consulted if the planned engineered lift exceeds 125% of rated load or if frequency exceeds two during a continuous 12-month period.
4. Documented lift plan approved by an engineer is required.

5. For mobile cranes:
    a. The equipment must not be operated in excess of its rated capacity.
    b. Weight of load must be determined OR
    c. Operator must begin hoisting the load to determine, using a load weighing device, load moment indicator, rated capacity indicator or rated capacity indicator if it exceeds 75% of the maximum rated capacity at the longest radius that will be used. If it does, the operator must not proceed with the lift until he verifies weight of the load.

Capacity of Slings and Shackles

1. Do not use any sling or shackle in excess of its rated capacity.
2. Do not use any sling unless it is marked with a tag that identifies size and rated capacity for types of hitches (e.g., vertical, basket, choker, etc.).
3. Do not use a wire rope sling unless it is marked with a tag that identifies the sling size, rated capacity for types of hitches, the angle upon which capacity is based and the number of legs if more than one. Remove unmarked wire rope slings from service until the capacity can be determined by the manufacturer, a vendor or a Duke Energy rigger and a new tag is applied.

Dog Bite Prevention

1. Always be aware of your surroundings and look for any signs of a dog being present. Signs can include but are not limited to:
    - Slightly open garage door
    - Fenced-in yard
    - Dog in yard/gate
    - Doggie doors/dog houses
    - Dog toys/dog chains or leash
    - Trail/path in yard that dog frequently walks
    - Dog waste in yard/food bowls
2. If there are no visual signs of a dog, it is still important to make yourself heard as it may help to avoid any unexpected encounters with a dog. Below are a few examples:
    - Shake or rattle the fence where you will enter
    - Tap on the gate with a tool that would get a dog's attention
    - Whistle/call for a dog
    - Rattle your keys
    - Ask the customer if they have a dog

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3. To help protect from a potential dog bite, use the approved bite-prevention devices per the business unit (i.e., dog spray, umbrella).
4. If a customer has a dog in their yard or home where an employee will need to gain access to perform any work, the safest option is to ask the customer to put the dog away until all work is completed and there is no threat of being bitten by the dog.

Dropped Object Prevention in Regulated & Renewable Energy Generating Stations

1. Contractors performing work at Duke Energy facilities shall implement a dropped object prevention program. Contractor's dropped object prevention program shall be submitted to Duke Energy for review and acceptance. Contractor shall address equipment/materials/tools security and dropped-prevention system and control requirements in the site-specific health and safety plan.
2. When workers are exposed to falling objects, each worker shall wear a hard hat and shall implement one of the following measures.
    a. Install barricading: Red Danger barricade with appropriate signage shall be established in areas where objects could fall from elevated work areas. The barricade shall prohibit workers from entering the affected area and shall be large enough to confine dropped objects.
    b. Where the potential for dropped objects exists on elevated work platforms, perimeter netting or plywood is required to be installed from the toe board to the top of the guardrail system. Plywood should be considered where objects greater than 50 lbs. have the potential to fall through perimeter netting. Perimeter netting is required to be secured to the top rail in a manner to not create an additional hazard.
3. Additionally, where objects are stored on platforms and scaffolds, the following measures shall be implemented to prevent objects from falling from the platform.
    a. Erect toe boards, screens or guardrail systems to prevent objects from falling from higher levels.
    b. Erect a canopy structure and keep potential falling objects far enough from the edge of the higher level so that those objects would not go over the edge if they were accidentally displaced.
    c. Barricade the area to which objects could fall, prohibit workers from entering the barricaded area, and keep objects that may fall far enough away from the edge of a higher level so that those objects would not go over the edge if they were accidentally displaced.

4. Falling object protection shall comply with the following provisions:
    a. Toe boards, when used as falling object protection, shall be erected along the edge of the overhead walking/working surface for a distance sufficient to protect workers below.
    b. Toe boards shall be capable of withstanding, without failure, a force of at least 50 pounds applied in any downward or outward direction at any point along the toe board.
    c. Toe boards shall be a minimum of 3 ½ inches vertical height from their top edge to the level of the walking/working surface. They shall have not more than ¼-inch clearance above the walking/working surface. They shall be solid or have openings not over 1 inch in greatest dimension.
    d. Where tools, equipment or materials are piled higher than the top edge of a toe board, paneling or screening shall be erected from the walking/working surface or toe board to the top of a guardrail system's top rail or midrail, for a distance sufficient to protect workers below.
    e. Guardrail systems, when used as falling object protection, shall have all openings small enough to prevent passage of potential falling objects.
    f. Debris netting shall be installed on guardrail systems where work is being performed.
    g. Debris netting shall have openings no greater than ½ inch to prevent objects from passing through.
    h. Debris netting utilized for below work protection shall have openings small enough to prevent passage of potential falling objects.
5. Prevent dropping of tools and equipment:
    a. Secure hand tools and equipment when working at heights.
    b. If using tethers, ensure they are intended for the purpose and inspected before and after each use.
    c. Tools weighing more than 5 pounds should not be attached to a person.
    d. Ensure tethers are nonconductive when working around electrical equipment.
    e. Do not use tethers around rotating equipment.

Dropped Object Prevention Plan for Other Business Units

1. Contractors performing work near or adjacent to Duke Energy employees or other contractors shall implement a dropped object prevention program to mitigate “Line of Fire” hazards that includes, but is not limited to, the barricading of work zones; securing tools and materials; and providing netting or canopies below the work areas.
2. Contractors shall protect against “struck by” hazards including falling, flying and moving objects. Protection for these hazards shall be discussed and mitigated prior to starting work. Three-way communication shall be maintained when entering and exiting work areas.

Electrical Safety for Transmission, Distribution and Metering

Designated Observer

1. An individual assigned to perform specific observation duties who has training and/or knowledge of the task(s) being observed to recognize related hazards, maintains ongoing effective communication and is always prepared to call an “All Stop” for unsafe condition(s) that could result in employee injury.
2. The Designated Observer (DO) must be alert, fully engaged and call an “All Stop” for any observed unsafe behavior or condition. The DO must not disengage from the assigned task to perform other duties.
3. For additional DO expectations, refer to business unit specific procedures or contractor supplemental requirements.

Cardinal Electrical Safety Rules (CESR)

1. Follow the Cardinal Electrical Safety Rules in Transmission and Customer Delivery.
    a. Cover Up/Insulate and Isolate
    b. Grounding/Zero Energy Check
    c. Electrical Safety PPE

Medical Services and First Aid

1. For transmission and distribution work, when two or more workers are working on exposed energized equipment, at least two persons trained in first aid/CPR shall be present.
2. First-aid kits in weatherproof containers shall be readily accessible at all T&D work locations. For line crews, first-aid kits shall be inspected at least annually.

3. When the eyes or body parts may be exposed to corrosive materials or chemicals, emergency eyewash facilities shall be available.

Personal and Electrical Protective Equipment

1. All personal protective equipment shall be properly worn, per manufacturer's instructions.
2. Wear hard hats where there is possible danger of head injury from impact, bumping, falling or flying objects, electrical shocks or burns. In addition, hard hats are required when there is a need for public identification of a company employee.
3. Wear safety glasses where there is possible danger of eye injury from impact, flying objects, electrical flash or burns. Safety glasses must comply with ANSI Z87.1.
4. Before each day's use, inspect electrical protective equipment for visible defects.
5. Air test rubber gloves before each day's use and immediately after any incident that is suspected of causing damage.
6. Electrical protective equipment used by Duke Energy employees shall be tested as shown. Contractors follow OSHA testing frequency. Do not use equipment if the test date is exceeded.

   Test Frequency:

   Rubber gloves ..... 4 months

   Rubber sleeves ..... 4 months

   Blankets ..... 12 months

   Line hose..... 12 months
7. If a defect is found during inspection of personal protective equipment, electrical protective equipment or live-line tools, the equipment must be marked defective and not used. Return rubber goods for retest if they are suspected to be defective.
8. Wear leather glove protectors over rubber gloves at all times. Do not use leather glove protectors for any other purpose.
9. Employees wear rubber gloves or gloves and sleeves in work situations as described in detail in the T&D Work Methods Manuals. Contractors follow employer PPE or work practice requirements.
10. Work shoes should have substantial uppers made of leather or the equivalent, cover the entire foot with no openings, provide ankle support, and have slip-resistant soles. Office environment footwear should minimize the potential for slipping or tripping. Wear EH (electrical hazard) shoes as required by the

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Work Methods Manual. Contractors follow employer PPE or work practice requirements.

Fall Protection

1. Before climbing a wood pole, workers must determine if the pole is safe to climb as described in the Inspecting and Testing Poles and Structures section of this handbook.
2. Before climbing towers, carefully examine to ensure they are safe and inspect base for corrosion or structural defects.
3. Workers shall use fall protection (harness and lanyard connected to lifeline or anchorage point) at all times when working at elevated locations and when ascending, descending or changing locations on steel towers, steel poles, concrete poles and substation structures.
4. Use full body harness with shock-absorbing lanyard when working out of a bucket truck, on an aerial platform, or on top of transformers or other equipment.
5. Workers climbing distribution and transmission class wood poles shall use fall protection equipment while ascending, descending and changing locations on a pole. This shall consist of a BuckSqueeze or similar equipment. Work positioning equipment alone is not allowed.

Live-line Tools

1. Visually inspect live-line tools and wipe them clean before use each day. If a defect or contamination could adversely affect the insulating quality or the mechanical integrity of the tool, remove it from service and have it tested before returning to service.
2. Every two years, live-line tools shall be examined, cleaned, waxed and repaired if necessary and electrically tested as indicated in item 3.
3. Test all live-line tools manufactured to ASTM F711 used by Duke Energy employees every two years. This includes:
    a. Telescopic sticks and measuring sticks
    b. Shotgun sticks
    c. Switch sticks, hot cutters, T-handles and link sticks
    d. Extendo sticks

Materials Handling and Storage

1. In areas not restricted to qualified workers, do not store materials closer than 10 feet from energized lines or exposed energized equipment (50 kV or less). For lines over 50 kV, the distance is 10 feet plus 4 inches for every 10 kV over 50 kV. CAPACITORS: Based on industry experience, manufacturer's

requirements must be followed when transporting and storing capacitor banks. One must familiarize themselves of the capacitor bank's setup before installing leads.

Working On or Near Exposed Energized Parts – General

1. Consider electric lines and equipment energized unless they have been properly isolated, tested for voltage and grounded.
2. Only qualified workers shall work on or with exposed energized lines or parts and in areas containing unguarded, uninsulated energized lines or parts of equipment at 50 volts or more.
3. When working around exposed energized parts, workers shall use proper protective equipment and work practices and comply with minimum approach distances.
4. Avoid positions where a shock or slip could expose the body to equipment at a potential different from the body.
5. When installing or removing fuses with one or both terminals energized at more than 300 volts or with exposed parts over 50 volts, use the appropriate tools or gloves. When installing expulsion-type fuses with one or both terminals energized over 300 volts, wear safety glasses or goggles, stay clear of the fuse barrel exhaust path and use tools rated for the voltage.
6. Ensure devices used to open circuits under load are designed to interrupt the current involved.
7. At least two workers shall be present for this work:
    a. Installation, removal or repair of lines over 600 volts
    b. Installation, removal or repair of de-energized lines if a worker is exposed to contact with other parts over 600 volts
    c. Installation, removal or repair of transformers, capacitors and regulators if the worker is exposed to parts over 600 volts
    d. Work involving the use of mechanical equipment, other than insulated aerial lifts, near parts over 600 volts
8. Workers may work alone for:
    a. Work involving equipment energized at 600 volts or less
    b. Routine switching of circuits, if conditions at the site allow the work to be done safely
    c. Work done with live-line tools if the worker is not within reach or otherwise exposed to contact with energized parts
    d. Emergency repairs (e.g., power restoration) to the extent necessary to safeguard the general public

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Table 3: Minimum Approach Distances for Voltages of 72.5 kV and Less

Nominal voltage phase-to-phase	Distance	
	Phase-to-ground exposure	Phase-to-phase exposure
50 to 300 volts	Avoid contact	Avoid contact
301 to 750 volts	1 ft. 1 in.	1 ft. 1 in.
751 volts to 5.0 kV	2 ft. 1 in.	2 ft. 1 in.
5.1 to 15.0 kV	2 ft. 2 in.	2 ft. 3 in.
15.1 to 36.0 kV	2 ft. 7 in.	2 ft. 11 in.
36.1 to 46.0 kV	2 ft. 10 in.	3 ft. 3 in.
46.1 to 72.5 kV	3 ft. 4 in.	4 ft.

Table 4: Minimum Approach Distances for Voltages of More than 72.5 kV

Voltage range phase-to-phase (kV)	Distance	
	Phase-to-ground exposure	Phase-to-phase exposure
72.6 to 121.0 kV	3 ft. 9 in.	4 ft. 8 in.
121.1 to 145.0 kV	3 ft. 10 in.	4 ft. 10 in.
145.1 to 169.0 kV	4 ft. 4 in.	5 ft. 5 in.
169.1 to 242.0 kV	5 ft. 8 in.	8 ft. 5 in.
242.1 to 420.0 kV	9 ft. 2 in.	14 ft. 9 in.
420.1 to 550.0 kV	11 ft. 11 in.	20 ft. 4 in.
550.1 to 800.0 kV	15 ft. 10 in.	27 ft. 10 in.

Notes:

1. Minimum Approach Distances based on worksites at elevations of 3,000 feet or less. Altitude correction is required over 3,000 feet.
2. Phase-to-ground distances assume using live-line tools in the air gap.
3. Phase-to-phase distances assume no live-line tool spans the air gap and no large conductive object is in the air gap. If a live-line tool spans the air gap or a large conductive object is in the air gap between phases, the required phase-to-phase distance can be determined by increasing the applicable voltage range Phase-to-Phase Exposure Distance by 10% or by requesting an engineering evaluation.

e. Substation work not involving direct contact with live parts or climbing on structures
f. Opening disconnects with live-line tools

Working On or Near Exposed Energized Parts – Making Connections

1. If connecting de-energized equipment or lines to an energized circuit using a conducting wire or device, attach the wire to the de-energized part first.
2. If disconnecting, remove the source end first, and keep loose conductors away from exposed energized parts.
3. When lines or equipment are connected to or disconnected from energized circuits, keep loose conductors away from exposed energized parts.

Working On or Near Exposed Energized Parts – Minimum Approach Distances

1. No worker shall approach or take any conductive object closer to exposed energized parts than the minimum approach distances unless:
    a. The worker is insulated from the energized part with rubber gloves or rubber gloves and sleeves AND the worker has positive control of the energized part.
    b. The energized part is insulated from the worker and from any other conductive object at a different potential OR
    c. The worker is insulated from any other exposed conductive object in accordance with requirements for live line bare hand work.
2. For voltages of 72.5 kV and less, utilize the minimum approach distances in Table 3 at elevations less than 3,000 feet. For voltages over 72.5 kV, utilize the minimum approach distances in Table 4 at elevations less than 3,000 feet.
3. Altitude correction is required over 3,000 feet elevation.
4. When a worker uses rubber gloves, he or she shall also use rubber sleeves UNLESS:
    a. Exposed energized parts on which the worker is not working are insulated from the employee AND
    b. When installing insulation for purposes of isolation, the worker installs the insulation from a position that does not expose his or her upper arm to contact with other energized parts.
5. See T&D Work Methods Manual for requirements for the use of rubber gloves. Contractors follow employer PPE or work practice requirements.
6. Low-voltage gloves (Class 0) with protectors are allowed on secondary voltages with management approval.

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Working On or Near Exposed Energized Parts – Protective Cover-up

1. When working on or near energized or de-energized conductors, neutrals or equipment, cover conductors or equipment you may contact. See the Work Methods Manual for examples of proper work methods for using protective insulating equipment. Contractors follow employer work practice requirements.
2. Inspect protective cover-up equipment visually before using to ensure they are in good condition and suitable for the purpose and within test dates.
3. Remove defective equipment from service and mark as defective.

Working On or Near Exposed Energized Parts – Apparel/Arc Flash Protection

1. Wear flame-resistant (FR) clothing for electrical hazards as described in the Work Methods Manual. Contractors follow the Supplemental Requirements.
2. Wear natural fiber garments under FR clothing. Do not wear undergarments made of acetate, nylon, polyester, rayon or polypropylene.
3. Outer layers of clothing shall be flame-resistant under any of these conditions:
    a. Worker is exposed to contact with circuits energized at more than 600 volts or
    b. Electric arc could ignite flammable material in the work area that could ignite the worker's clothing or
    c. Molten metal or electric arcs from faulted conductors in the work area could ignite the worker's clothing or
    d. Estimated incident energy exceeds 2.0 cal/cm^2.
4. Wear hand protection for electrical hazards. Hand protection shall consist of rubber gloves with leather protectors where there is a contact hazard or for work inside minimum approach distance. For work with an arc flash hazard but no contact hazard, arc-rated or heavy-duty leather work gloves with a weight of at least 407 gm/m^2 (12 oz/yd^2) shall be worn for exposures up to 14 cal/cm^2. Over 14 cal/cm^2, arc-rated gloves are required.
5. Where flame-retardant clothing is worn for exposures exceeding 2 cal/ cm^2, long sleeves are required.

Jewelry and Conductive Articles

1. No jewelry or conductive articles shall be worn by workers within the work zone while performing or observing physical work, other than in an office type environment or while driving. This includes loading and unloading material and any work at a job site or substation. This includes but is not limited to: finger rings, necklaces, earrings, body piercings, chains, wrist

bands, wrist watches, key rings/chains, metal or plastic bracelets, or studs. Refer to business unit-specific procedures for additional guidance.

2. Eyeglasses and MedicAlert necklaces and bracelets are not considered jewelry and are exempt from this requirement. However, MedicAlert necklaces and bracelets shall be worn under FR clothing or rubber protective gloves if working on energized conductors.
3. If there is a business need for a Bluetooth device to be worn on a job site, workers must conduct a hazard assessment to ensure that wearing the device will not put the worker(s) at risk for injury or cause distraction if a call is received. Refer to business unit guidance on wearing Bluetooth devices within minimum approach distance, when working from an aerial lift or aloft, or in a gaseous atmosphere. No cellphone calls are to be taken or made within or near minimum approach distance energized space.

Working On or Near Exposed Energized Parts – Face Protection

1. In addition to hard hat and safety glasses, arc-rated face and head protection is required for tasks exceeding the exposures shown here.
    a. Single-phase arc: Arc-rated face shields over 9 cal/cm^2. Arc-rated hood or face shield with balaclava over 13 cal/cm^2.
    b. Three-phase arc: Arc-rated face shield over 5 cal/cm^2. Arc-rated hood or face shield with balaclava over 9 cal/cm^2.
2. Face shields and approved safety glasses with side shields shall be used for these tasks when the systems are energized:
    a. Setting or removing three-phase self-contained meters (non-T-Rated)
    b. Working on all K-type meter bases
    c. Connecting or disconnecting underground parallel services

Insulated Hand Tools

1. If hand tools are used in, on or near exposed energized conductors or parts and the hand tool could make contact with other differences of potential (i.e., conductors, grounded cabinets, etc.), the hand tool shall be an approved insulated type.
2. Before using insulated tools, visually inspect them to ensure the insulating material is not damaged.
3. Use insulated tools only for their designated purposes. Do not use insulated hand tools on circuits over 1,000 volts.
4. Insulated tools shall be marked with double triangles by the manufacturer to indicate they were manufactured and tested in compliance with ASTM F1505, Standard Specification for Insulating and Insulated Hand Tools.

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5. Coverings on the handles of tools not marked with double triangles shall not be considered as insulation.
6. Wear rubber gloves with insulated hand tools if the work could cause the minimum approach distance to be violated.
7. Additional business unit guidance may apply.

De-energizing Lines and Equipment for Worker Protection

1. Before considering lines and equipment to be worked on as de-energized, ensure they are de-energized as described below, tested for voltage and grounded.
2. For systems under the control of a dispatcher or operator, obtain a clearance to de-energize the lines according to standard operating procedures.
3. De-energize lines and equipment by opening the proper switching device such as disconnects, interrupters, circuit breakers, reclosers, line switches or fuses.
4. For systems not under the control of a dispatcher or operator, isolate lines by doing one or more of the following:
    a. Removing fuses
    b. Disconnecting recloser leads
    c. Opening switches
    d. Removing jumpers
5. Render inoperable as design permits and tag the following:
    a. Automatically and remotely controlled switches that could cause the opened disconnection means to close (tag at the point of control)
    b. All switches, disconnects, jumpers, taps and other means through which electricity may be supplied to the lines and equipment
6. Tags used in accordance with jurisdictional switching and tagging manuals shall prohibit operation of the disconnecting means and indicate that workers are at work.
7. If design permits, render the recloser inoperable and tag it; remove source or load-side leads of the recloser or open source or load-side disconnects of the recloser for a visual open point. For parking the leads, use stand-off tool.
8. Render inoperable any disconnecting means that are open to protect workers and that are accessible to the public.
9. Test to verify that the lines and equipment are de-energized.
10. Install grounds as outlined in “Grounding” in this document.

11. To release a clearance, the worker in charge shall:
    a. Notify the crew.
    b. Determine that everyone is clear of the lines and equipment.
    c. Remove all protective grounds.
    d. Report this information to the system operator and release the clearance.
12. Do not re-energize lines until all grounds have been removed; all crews have released their clearances; all workers are clear of the lines and equipment; and all tags have been removed.
13. If two or more crews will be working on the same lines or equipment
    a. The crews shall coordinate their activities with a single employee in charge of the clearance for all of the crews and follow the requirements of this section as if all of the workers formed a single crew OR
    b. Each crew shall independently comply with this section and, if there is no system operator in charge of the lines or equipment, shall have separate tags and coordinate de-energizing and re-energizing the lines and equipment with the other crews.

Grounding for the Protection of Workers

1. Before considering lines and equipment to be worked on as de-energized, ensure they are de-energized as described above, tested for voltage and grounded as described here.
2. If installing a ground is impractical or would create greater hazards, treat the lines and equipment as de-energized if all the following conditions are met:
    a. The lines and equipment are de-energized as required.
    b. Contact with another energized source is not possible.
    c. No hazard of induced voltage exists.
3. Before installing grounds on a supposedly de-energized circuit, visually inspect the grounding equipment.
4. To verify that the circuit has been de-energized, use an approved voltage testing device to check for voltage on the conductors to be grounded.
5. Place temporary protective grounds at such locations and arrange in such a manner as to prevent each worker from being exposed to hazardous differences in electrical potential.

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6. Ensure protective grounding equipment:
    a. Is capable of conducting the maximum fault current that could flow at the point of grounding for the time necessary to clear the fault
    b. Has an ampacity greater than or equal to that of No. 2 AWG copper (Duke Energy requires 2/0 in most applications)
    c. Has an impedance-to-ground low enough to cause immediate operation of protective devices in case of accidental energizing of the lines or equipment
7. If there is no previously installed ground, test lines and equipment before installing any ground to ensure they are free of nominal voltage.
8. If attaching grounds to lines or equipment, attach the ground-end connection first; then attach the other end using live-line tools.
    a. Do not allow your body to come in contact with the grounding cable.
    b. Do not hold a grounding cable or clamp in your hand while installing grounds with a live-line tool.
9. When a ground is to be removed, the grounding device shall be removed from the line or equipment using a live-line tool before the ground-end connection is removed.
10. If work is performed on a cable at a location remote from the cable terminal and the hazardous transfer of potential is possible should a fault occur, do not ground the cable at the cable terminal.
11. Grounds may be removed temporarily during tests. During the test, use insulating equipment and other measures necessary to protect exposed workers in case the lines and equipment become energized.
12. When attaching grounds to and when removing grounds from de-energized lines or equipment, wear hard hat, safety glasses, rubber gloves and flame-resistant clothing. Note: Rubber gloves are not required in Transmission when using live-line tools to attach grounds.

Mechanical Equipment

1. Inspect the critical safety components of mechanical elevating and rotating equipment on each shift the equipment is used.
2. Do not operate vehicular equipment with an obstructed rear view on off-highway job sites that expose workers to hazards created by the moving vehicle unless the vehicle has a reverse signal alarm audible above the surrounding noise level or it is backed up only when a designated worker signals that it is safe to do so.
3. If the suspended load of a line truck endangers any workers, do not leave the controls of the truck.

4. If vehicular equipment includes outriggers:
    a. Operate with the outriggers extended and firmly set as necessary to keep the vehicle stable.
    b. Do not extend or retract outriggers outside of operator's clear view unless all workers are outside the range of possible equipment motion.
5. Use lifting equipment within its maximum-load rating and other design limitations for the work conditions.
6. Mechanical equipment shall be operated so that the minimum approach distances in Tables 3 and 4 are maintained from exposed energized lines and equipment.
    a. However, the insulated portion of an aerial lift operated by a qualified worker in the lift is exempt from this requirement if the applicable minimum approach distance is maintained between the uninsulated portions of the aerial lift and exposed objects having a different electrical potential.
7. Ensure a designated worker other than the operator observes the approach distance to exposed lines and gives timely warnings before the approach distance is reached (not required if operator can safely determine distance alone).
8. If mechanical equipment could become energized when used around overhead power lines, ground the equipment as described in Work Methods Manual. Contractors follow employer grounding requirements.

Overhead Lines

1. Before climbing, installing or removing any equipment, determine the amount of additional or unbalanced stress a pole or tower can handle; if necessary, support the pole or tower with braces.
2. When poles are set, moved or removed near overhead conductors, avoid direct contact of the pole with energized conductors.
3. When handling poles near overhead power lines, wear appropriate electrical protective equipment.
4. Install pole covers or cover conductors when setting a pole in or near conductors energized at primary voltage.
5. Do not contact the poles with uninsulated body parts.
6. Guard pole holes in areas where employees are working or provide an attendant.
7. To minimize the possibility that conductors and cables being installed or removed will contact energized lines or equipment, take precautions (e.g., using barriers or the tension-stringing method).

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8. If installing or removing conductors that cross over energized conductors in excess of 600 volts, render inoperative the automatic reclosing feature for the energized lines if permitted by the design of the circuit-interrupting devices protecting the lines.
9. Determine the voltage to be induced in the new lines before lines are installed parallel to existing energized lines.
10. If hazardous voltage induction is possible, do the following:
    a. Ground each bare conductor so that no point is more than 2 miles from a ground.
    b. Do not remove the grounds until the conductor installation is completed between dead ends.
    c. Remove grounds as the last phase of aerial cleanup.
    d. If working on bare conductors, install grounds at:
        i. Each work location where the workers are working.
        ii. All open dead-end or catch-off points or the next adjacent structure.
        iii. If splicing two bare conductors, bond and ground the conductors before splicing.
11. Keep reel handling equipment, including pulling and tensioning devices:
    a. In safe operating condition
    b. Leveled
    c. Aligned
12. Do not exceed load ratings of stringing lines, pulling lines, conductor grips, load-bearing hardware and accessories, and rigging and hoists.
13. Repair or replace defective pulling lines and accessories.
14. Do not use conductor grips on wire rope, unless the grip is specifically designed for the application.
15. Maintain reliable communications, through two-way radio or the equivalent, between the reel tender and the pulling-rig operator.
16. Operate the pulling rig only when it is safe to do so.
17. While the conductor or pulling line is being pulled (in motion) with a power-driven device, do not stand directly under overhead operations or on the cross arm, except as necessary to guide the stringing sock or board over or through the stringing sheaf.

Towers and Structures

1. Unless assisting workers working above, do not stand under a tower or structure while work is in progress.
2. Unless it is more hazardous, use tag lines or other similar devices to maintain control of tower sections being raised or positioned.
3. Do not detach the load line from a member or section until the load is safely secured.
4. Discontinue work when adverse weather conditions make the work unusually hazardous, except during emergency power restoration procedures.

Underground Electrical Installations

1. Use protective equipment and follow work practices when working in underground electrical installations.
2. When accessing manholes and subsurface vaults:
    a. Use ladders or other climbing devices to enter and exit manholes and subsurface vaults more than 4 feet deep.
    b. Do not step on cables or hangers to climb into or out of manholes or vaults.
3. When lowering equipment into manholes:
    a. Use equipment capable of supporting appropriate weight to lower materials and tools into manholes or vaults.
    b. Check equipment used to lower materials for defects before use.
    c. Clear the area directly under the opening before lowering hot solder or other hot compounds into the manhole or vault opening.
4. Ensure a worker trained in first aid/CPR is available on the surface to render any necessary emergency assistance while work is performed in a manhole containing energized electric equipment.
5. Allow the worker trained in first aid/CPR to briefly enter a manhole to assist if there are no atmospheric or traffic hazards.
6. If work can be performed safely in a manhole with energized cables or equipment, allow a worker working alone to enter a manhole briefly to inspect, housekeep, take readings, etc.
7. Maintain communications, through two-way radios or the equivalent, among all employees involved in the job.
8. Install duct rods in the least hazardous direction.
9. Ensure the required clearance distances are maintained by stationing a worker at the far end of the duct line being rodded.

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10. Inspect energized cables for defects.
11. When working on multiple cables:
    a. Identify the cable to be worked on by electrical means, unless its identity is obvious by reasons of distinctive appearance or location, or by other readily apparent means of identification.
    b. Protect cables other than the one being worked on from damage.
12. Do not work in a manhole where energized cables appear defective (e.g., oil or compound leaking from cable joint, broken cable sheaths or joint sleeves, hot localized surface temperatures of cables or joints, swollen joints where circumference exceeds 3 ½ times the sleeve diameter).
13. Refer to the Work Methods Manual for precautions to be taken for work in a manhole or vault that could cause a fault in a cable.
14. Cover all energized splices in manholes with Kevlar blankets.
15. Maintain metallic sheath continuity when work is performed on buried cable and cable in manholes or treat the cable sheath as energized.

Electric Vault and Manhole Entry

1. Workers entering electric vaults and manholes shall complete Confined Space Training and shall be trained in rescue techniques.
2. Anytime work is performed in a manhole or vault, an attendant trained in first aid/CPR shall be present outside. For brief entries to perform inspections, housekeeping, or to take readings, an attendant is not required. The attendant can enter briefly to assist in the work if air monitoring indicates no atmospheric hazard, a work zone for road and street work has been set up and the traffic patterns do not present a hazard.
3. Conduct atmospheric testing prior to entry. Ventilate or monitor continuously during entry.
4. Rescue equipment shall be available at the job site anytime work is done in a manhole or vault. Rescue equipment is not required for brief entries for inspections, housekeeping or taking readings.
5. Workers may not enter a manhole or vault while it contains a hazardous atmosphere unless they comply with permit-required entry requirements of the Confined Space Entry program.

Ground Fault Protection for Personnel

1. Use ground fault circuit interrupter (GFCI) on all 120-volt, single-phase 15, 20 and 30 amp receptacles that are not part of the permanent wiring of a building or structure.

Electrical Safety for Generating, Commercial and Natural Gas Business Unit Facilities

General

1. Only qualified persons shall be allowed to work near exposed energized electrical equipment.
2. De-energize equipment or circuits before working on or near them, unless:
    a. De-energizing increases hazards.
    b. De-energizing is infeasible.
    c. Energized equipment/circuits operate at less than 50 volts-to-ground and exposure to electrical burns or explosion caused by electrical arcs is not increased.
3. Unqualified persons shall not open enclosures or panels that contain exposed energized electrical parts or equipment.
4. When doing electrical work, alert others of potential hazards.

Minimum Approach Distances

1. Qualified persons shall not approach or take any conductive object without an insulating handle closer to exposed energized parts than the minimum approach distances shown in the tables below unless the:
    a. Worker is insulated from the energized part with rubber gloves or rubber gloves and sleeves and the worker has positive control of the energized part OR
    b. Energized part is insulated from the worker with line hose/blanket/etc.
2. For voltages of 72.5 kV and less, utilize the minimum approach distances in Table 5 at elevations less than 3,000 feet. For voltages over 72.5 kV, utilize the minimum approach distances in Table 6 at elevations less than 3,000 feet.
3. Altitude correction is required over 3,000 feet.
4. All unqualified persons shall maintain these distances from overhead power lines:
    a. For lines under 50 kV – 10 feet
    b. For lines over 50 kV – 10 feet plus 4 inches for every 10 kV over 50 kV

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Table 5: Minimum Approach Distances for Voltages of 72.5 kV and Less

Nominal voltage phase-to-phase	Distance	
	Phase-to-ground exposure	Phase-to-phase exposure
50 to 300 volts	Avoid contact	Avoid contact
301 to 750 volts	1 ft. 1 in.	1 ft. 1 in.
751 volts to 5.0 kV	2 ft. 1 in.	2 ft. 1 in.
5.1 to 15.0 kV	2 ft. 2 in.	2 ft. 3 in.
15.1 to 36.0 kV	2 ft. 7 in.	2 ft. 11 in.
36.1 to 46.0 kV	2 ft. 10 in.	3 ft. 3 in.
46.1 to 72.5 kV	3 ft. 4 in.	4 ft.

Table 6: Minimum Approach Distances for Voltages of More than 72.5 kV

Voltage range phase-to-phase	Distance	
	Phase-to-ground exposure	Phase-to-phase exposure
72.6 to 121.0 kV	3 ft. 9 in.	4 ft. 8 in.
121.1 to 145.0 kV	3 ft. 10 in.	4 ft. 10 in.
145.1 to 169.0 kV	4 ft. 4 in.	5 ft. 5 in.
169.1 to 242.0 kV	5 ft. 8 in.	8 ft. 5 in.
242.1 to 420.0 kV	9 ft. 2 in.	14 ft. 9 in.
420.1 to 550.0 kV	11 ft. 11 in.	20 ft. 4 in.
550.1 to 800.0 kV	15 ft. 10 in.	27 ft. 10 in.

Notes:

1. Minimum Approach Distances based on worksites at elevations of 3,000 feet or less. Altitude correction is required over 3,000 feet.
2. Phase-to-ground distances assume using live-line tools in the air gap.
3. Phase-to-phase distances assume no live-line tool spans the air gap and no large conductive object is in the air gap. If a live-line tool spans the air gap or a large conductive object is in the air gap between phases, the required phase-to-phase distance can be determined by increasing the applicable voltage range Phase-to-Phase Exposure Distance by 10% or by requesting an engineering evaluation.

Personal Protective Equipment

Table 7: Personal Protective Equipment for Electrical Work

This table describes personal protective equipment for electrical work.

	Work on Exposed Energized Parts This includes but is not limited to: maintenance, repair, cleaning and installing/removing fuses	Voltage Testing Voltage Testing (see notes below for glove use guidance)
50 to 600 volts	• Hard hat and safety glasses • Rubber gloves w/leather protectors	• Hard hat and safety glasses
Over 600 volts	PPE same as 50 to 600 volts. Additional requirement: Work in pairs (work alone is allowed for routine switching, work with live-line tools and emergency repairs necessary to safeguard the general public).	PPE same as 50 to 600 volts. Additional requirement: Work in pairs (work alone is allowed for routine switching, work with live-line tools and emergency repairs necessary to safeguard the general public).

Notes:

1. If there is a possibility for inadvertent electrical contact, then personnel shall wear appropriately rated rubber gloves with leather protectors.
2. If there is a possibility of an arc flash hazard, but no electrical contact hazard, then use the following guidance: a) Personnel shall use an arc-rated work glove with ≥ 8 cal/cm^2 arc rating for normal operation of equipment. b) If arc-rated work gloves are not available, then personnel shall wear appropriate voltage-rated rubber gloves with leather protectors. c) If a task requires the use of arc flash PPE, then personnel shall wear arc-rated gloves that meet or exceed the incident energy level, but not < 8 cal/cm^2. If arc-rated gloves are not available that meet or exceed the calculated incident energy level, then appropriately voltage-rated rubber gloves with leather protectors shall be worn.
3. See “Arc Flash Protection” section below for information on FR clothing and face shield requirements for electrical work.

Electrical Protective Equipment

1. The maximum-use voltage for electrical protective equipment (e.g., rubber gloves, rubber sleeves, rubber blankets, matting and line hose) is as follows:
    a. Class 0 – 1,000 volts
    b. Class 1 – 7,500 volts

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c. Class 2 – 17,000 volts
d. Class 3 – 26,500 volts
e. Class 4 – 36,000 volts

2. Electrical protective equipment shall be tested before first issue and retested as required by OSHA. Contractors follow OSHA testing frequency. Equipment tested will be tested on these frequencies. Business unit-specific exceptions may apply.

   Test Frequency:

   Rubber gloves ..... 4 months

   Rubber sleeves ..... 4 months

   Blankets ..... 12 months

   Line hose..... 12 months

3. Visually inspect and air test rubber gloves before use.
4. Visually check rubber blankets and sleeves for flaws before use.
5. Always wear leather protectors over rubber gloves. Do not use protectors for any other purpose. Exception: Class 0 rubber gloves can be used without protectors under limited-use conditions where small equipment and parts manipulation necessitates unusually high finger dexterity. Take extra care in the visual examination of each glove and in avoiding handling sharp objects. Visually inspect and air test gloves before and after use.
6. When protective equipment selection depends on voltage levels, use the phase-to-phase voltage on multiphase circuits. Phase-to-ground voltage can be used if there is no multiphase exposure or if energized conductors have been covered with rubber protective equipment, eliminating any possible multiphase exposure.

Jewelry

1. Do not wear jewelry (such as rings, earrings, bracelets, necklaces or pendants) where contact with energized electric lines or equipment is possible.
2. Take extra caution if wearing keys, wire rim glasses, identification badges, metal belt buckles or metal buttons on garments.

Insulated Tools

1. If hand tools are used near exposed energized conductors or parts AND the hand tools could make contact with the conductors or parts, the hand tools shall be the insulated type.

2. Before using insulated tools, visually inspect them to ensure the insulating material is not damaged.
3. Use insulated tools only for their designated purposes. Do not use insulated hand tools on circuits over 1,000 volts.
4. Insulated tools shall be marked with double triangles by the manufacturer to indicate they were manufactured and tested in compliance with ASTM F1505, Standard Specification for Insulating and Insulated Hand Tools.
5. Coverings on the handles of tools not marked with double triangles shall not be considered as insulation.
6. Wear rubber gloves with insulated hand tools if the work could cause the minimum approach distance to be violated.

Testing for Voltage

1. Consider lines and equipment as energized until they have been isolated, tested for voltage and grounded where applicable.
2. When work requires direct contact with electrical circuits, use voltage testing devices to verify that the equipment is de-energized.
3. Be aware that equipment can be fed by more than one source or can be actuated from more than one location.
4. If any unusual condition is encountered (e.g., voltage present when it should not be), immediately stop work, place the equipment in a safe condition and contact your supervisor.
5. Test for voltage initially, at shift change and after other work stoppage.
6. If the circuit to be tested is over 600 volts, check the test equipment for proper operation immediately before use. Follow the manufacturer's instructions for the device.
7. If there is a possibility of the re-accumulation of stored energy to a hazardous level, verify safe work conditions periodically throughout the shift until work is completed or until the possibility of such accumulation no longer exists.
8. Before use, visually inspect test instruments, leads, cables, power cords, probes and connectors for external damage. If there is any doubt as to the equipment's integrity, do not use it.
9. Verify that voltage testing devices are operating properly and that appropriate settings are used for the parameter and type of voltage to be checked.
10. Verify that the testing device is being used for the correct application and in the proper configuration. Follow the instructions in the operator's manual for the specific make and model being used.

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11. Do not exceed the maximum safe voltage for the test equipment. Follow manufacturer's guidelines and/or group procedures for specific limitations of equipment used.

Working Position

1. Workers shall work in a position from which a slip or shock will not bring the worker's body into contact with exposed energized parts.
2. For exposed energized parts over 600 volts, work from a position where you cannot reach into the minimum approach distance unless wearing rubber gloves or using live-line tools.

Arc Flash Protection

1. The following items apply to workers who are exposed to arc flash hazards.
2. Refer to detailed arc flash protection requirements in business unit-specific procedures.
3. Do not wear clothing that could melt onto the skin or that could ignite and continue to burn when exposed to heat energy from electric arcs. This includes acetate, nylon, polyester, rayon and polypropylene. These fibers are allowed when blended with FR fabrics (such as Nomex/rayon blends).
4. Outer layers of clothing shall be arc-rated flame-resistant under any of these conditions:
    a. Worker is exposed to contact with circuits energized at more than 600 volts.
    b. Electric arc could ignite flammable materials in the work area that could ignite the worker's clothes.
    c. Molten metal or electric arcs from faulted conductors could ignite the employee's clothing.
    d. Estimated incident energy exceeds 2.0 cal/cm^2 as determined by business unit arc flash analysis. FR clothing shall have an arc rating greater than the heat energy estimated as determined by business unit arc flash analysis.
5. Hand protection shall consist of rubber gloves with leather protectors where there is hazard of electrical contact. Where there is flash hazard but not contact hazard, workers shall wear arc-rated or heavy-duty leather work gloves with a weight of at least 407 gm/m^2 (12 oz/yd^2). The work gloves must be arc-rated for exposures over 14 cal/cm^2.
6. Wear heavy-duty work shoes or boots. Footwear does not have to be arc-rated.

7. In addition to hard hat and safety glasses, arc-rated face and head protection is required for these exposures:

Table 8: Exposures Requiring Arc-Rated Face and Head Protection

	No head or face protection required	Arc-rated face shield with minimum rating of 8 cal/cm^2	Arc-rated hood or face shield with balaclava
Single-phase exposure in air*	< 8 cal/cm^2	9 to 12 cal/cm^2	13 cal/cm^2 or higher
Three-phase exposure	< 4 cal/cm^2	5 to 8 cal/cm^2	9 cal/cm^2 or higher

* Single-phase and three-phase refer to the exposure – not the system voltage. Business units will define the tasks with these exposures.

8. Where flame-retardant clothing is worn for exposures exceeding 2 cal/cm^2, long sleeves are required.
9. When operating circuit breakers, switches or disconnect devices locally (open or closed) on an energized bus that has an arc flash hazard, when practical perform the following to minimize potential exposure to electrical fault blast energy:
    a. Stand clear to one side, hinge side preferred
    b. Stand with your side to the potential blast area (less surface area subject to the blast)
    c. Stand outside an imaginary line set to 45 degrees from opening (outside probable blast area)
    d. Increase working distance from the opening (decrease potential blast force)
    e. Wear ear plugs or ear muffs

Grounding

1. When grounding lines and equipment for the protection of workers, ensure safety ground leads are not less than No. 2 AWG flexible stranded copper rubber-covered cable or its equivalent and capable of conducting the maximum fault current potential and duration necessary to clear the fault. Where generation equipment has been de-energized according to lockout/tagout procedures, grounds do not have to be sized for maximum fault current.

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2. When installing and removing grounds, wear rubber gloves with leather protectors, safety glasses, hard hats and flame-resistant clothing.
3. Inspect grounding cables and connecting clamps before installation to ensure that all connections are solid and wires or connectors are not frayed or corroded.
4. Before any ground is installed, lines and equipment shall be tested and found absent of nominal voltage, unless a previously installed ground is present.
5. When a ground is to be attached to a line or to equipment, the ground-end connection shall be attached first, and then the other end shall be attached.
6. When a ground is to be removed, the grounding device shall be removed from the line or equipment before the ground-end connection is removed.
7. In switchyards and on T&D installations, grounds shall be attached to and removed from the line or equipment with a live-line tool.

✪ Tools, Equipment and Fixtures

1. When working near energized circuits, use only flashlights with exposed parts made of nonconductive material.
2. When working on or near electrical equipment, use ladders with nonconductive (fiberglass) side rails.
3. Before replacing fuses, verify that ratings and types are correct.
4. Appropriate tools shall be used for installing and removing fuses.
5. Before working on energized equipment, try to dry wet floors. If floors cannot be dried, use rubber gloves, insulating blankets and a wooden foundation to elevate the blanket above the moisture.
6. Secure or remove doors, hinged panels, etc. to prevent them from swinging into a worker and causing contact with exposed energized parts.
7. Do not obstruct access to electrical equipment.

Extension Cords

1. Before using extension cords, inspect them for loose parts, damaged pins and defective insulation. Replace damaged cords.
2. Only use extension cords with grounding conductors.
3. Do not raise and lower equipment with extension cords connected to the equipment.
4. Do not fasten extension cords with staples, nails, etc. or hang them in a fashion that could damage the outer jacket, insulation or conductors.
5. Elevate extension cords passing through work areas to protect them from damage and to eliminate tripping hazards.

6. Be sure your hands are dry when plugging or unplugging extension cords.
7. Properly secure locking connectors after connection, if applicable.
8. If an extension cord is wet from immersion, wear rubber gloves if it is still energized.
9. Protect extension cords from vehicular traffic (e.g., forklifts, manlifts, tractors).
10. When using power strips, follow manufacturer's recommendations. Do not overload outlets. Do not connect multiple power strips in series.

Electrical Equipment in Conductive Work Locations

1. Portable electric equipment used in highly conductive work locations (such as those inundated with water or other conductive liquids) or in job locations where workers are likely to contact water or conductive liquids shall be approved for those locations.
2. In job locations where workers are likely to contact or be drenched with water or conductive liquids, ground fault circuit interrupter protection for personnel shall also be used.

Batteries

1. Do not smoke or create sparks, arcs or flames in battery areas. Post signs accordingly.
2. Take extreme caution when carrying or using conductive materials around batteries. To prevent shocks, avoid physical contact with exposed conductors on batteries.
3. Do not remove vent plugs from cells, unless specific maintenance work is being performed. Immediately reinstall vent plugs after work is completed.
4. Before performing battery-related tasks, check the location of the nearest eyewash or eyewash/shower. If the plumbed eyewash/shower has not been tested within the last week, test it before working on the batteries.
5. Battery-related tasks may be performed by persons working alone. Ask for assistance if necessary.
6. Provide adequate ventilation in battery rooms.
7. When charging, repairing or servicing batteries, follow manufacturer's recommendations.
8. Wear appropriate personal protective equipment for work on batteries.
    a. Eye protection, full-face shield, chemical resistant gloves, full-body apron and protective footwear to:
        i. Add/remove electrolytes (acid)

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ii. Move lead acid cells
iii. Check specific gravity
iv. Wash/clean battery cells

b. Eye protection to:
i. Read cell battery voltage
ii. Adjust intercell connecting hardware

Switchyards at Generating Plants

1. Consider all conductors and equipment as energized until tested for voltage and grounded.
2. Report any unusual conditions observed on substation structures, equipment, ground wires, busses or wiring to supervision or the system coordinator.
3. When entering an occupied substation, report your presence to the employee in charge.
4. Lock substation entrances that are not observed by an attendant.
5. Post DANGER KEEP OUT signs (or the equivalent) to warn unauthorized persons.
6. Provide guarding around live parts over 150 volts-to-ground that do not have an insulating cover unless they are located to prevent inadvertent contact.
7. Ensure switching operations are performed only by qualified persons.
8. For all live-line tools:
    a. Wipe clean and inspect daily before use. Properly wax as needed.
    b. Every two years, dielectrically test at 75,000 volts per foot for one minute. Use appropriate tester.
    c. Do not place on the ground.
    d. When refinished, repaired or failing a visual inspection, dielectrically test before returning to service.
9. Carry or place conductive material so as to prevent contact with energized lines, equipment or busses.
10. Do not use metallic cloth tapes, metal tapes, metal rules or other conductive material near energized conductors or equipment.
11. Before applying grounds, de-energize static capacitors for five minutes.
12. Attach fall protection only to designated substantial anchorage points; do not attach to conductors or cables.
13. Do not use portable metal or other conductive ladders in substations.

14. Ensure transformer cooling fans that are less than 7 feet above the ground are provided with guards that have openings less than ½ inch.

Ground Fault Protection for Personnel in Construction

1. Persons must be protected from ground fault hazards as described below during activities involving construction, remodeling, maintenance, repair or demolition of buildings, structures, equipment or similar activities.
2. Temporary wiring: Use ground fault circuit interrupter (GFCI) on all 120-volt, single-phase 15, 20 and 30 amp receptacles that are not part of the permanent wiring of a building or structure.
3. Portable generators: Follow National Electric Code requirements. All 125-volt and 125/250-volt, single-phase, 15-, 20- and 30-ampere receptacle outlets that are a part of a 15-kW or smaller portable generator shall have listed GFCI protection for personnel. Cord sets or devices incorporating GFCI protection for personnel identified for portable use shall be permitted, provided the generator was manufactured prior to 1/1/2015. Exception for unbonded (floating neutral) generators: GFCI protection shall not be required where the 125-volt receptacle outlet(s) is interlocked such that it is not available for use when any 125/250-volt receptacle(s) is in use.
4. For cord sets, extension cords, and cord- and plug-connected tools, comply with one of the following options:
    a. Option 1: Use GFCI on all cord sets, extension cords, and cord- and plug-connected tools. When using portable GFCIs, install the GFCI between the receptacle and the cord set, not between the tool and the cord set.
    b. Option 2: Implement an assured equipment grounding program.
        i. Visually inspect cord sets, extension cords and tools before each day's use for external defects.
        ii. Test extension cords and grounded tools for continuity before first use, quarterly and following repairs. Testing is not required for double-insulated tools or for cords with illuminated ground continuity monitors.

Electromagnetic Fields (EMF) and Medical Devices

1. High electromagnetic fields (EMF) may interfere with the operation of medical devices. High EMF levels may be present in electrical equipment located in generating station switchyards, station main bus lines, magnetic separators, generator brush rooms and other locations such as substations. Office areas do not present a risk.

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2. Workers who have medical devices such as internal pacemakers, defibrillators, insulin pumps and pain modification devices are encouraged to consider voluntarily reporting such devices to supervision and H&S/EHS professionals.
    a. Management cannot require workers to report medical devices, but they should consider doing so.
    b. While cochlear implants and hearing aids may be affected, they should not present a life-threatening risk.
3. Personnel with medical devices should not enter generating plant switchyards due to potential for EMF.
4. If there appears to be an immediate hazard related to medical device operation and high levels of EMF:
    a. Stop work.
    b. Move away from the high EMF source.
    c. Contact your supervisor.
5. Additional guidance for supervisors is available through the H&S Portal and from H&S/EHS professionals.

Elevating Mobile Work Platforms

General

1. This section outlines requirements for the operation of boom-supported elevating work platforms, self-propelled elevating work platforms and manually propelled elevating work platforms (JLGs, Genies and scissor lifts).
2. The operator shall ensure the operating and maintenance manuals are available.
3. The operator shall comply with manufacturer's warnings and instructions.
4. Personal fall arrest equipment is required for workers working from boom-supported elevating work platforms.
5. Workers working in scissor lifts shall be protected by guardrails or personal fall arrest if guardrails are not installed.
6. Workers shall be trained and qualified prior to operating or riding in elevating work platforms.
7. Plan movements to avoid hazards. Use spotters when necessary.

Pre-start Inspection

1. Before use each day, pre-start inspection shall be conducted in accordance with operator's manual.
2. Any problems or malfunctions shall be repaired prior to use of the platform.

Workplace Inspection

1. Before the elevating work platform is used and during use, the operator shall check the area for possible hazards including:
    a. Drop-offs, holes and slopes
    b. Bumps and floor obstructions
    c. Debris
    d. Overhead obstructions and electrical conductors
    e. Hazardous atmospheres
    f. Inadequate surface and support to withstand load forces imposed by the aerial platform
    g. Wind and weather conditions
    h. Presence of unauthorized persons
    i. Other possible unsafe conditions
2. Prior to each operation, the operator shall ensure:
    a. Outriggers are used as required
    b. Guardrails are installed and access gates closed
    c. Load and its distribution on the platform are in accordance with manufacturer's rated capacity
    d. All personnel on the platform have PPE for the work and environment envisioned

Driving Requirements

1. Before and during driving while the platform is elevated, the operator shall:
    a. Maintain a clear view of the support surface and route of travel.
    b. Ensure personnel in the worksite area that may be affected are aware of the movement, communicating and maneuvering the platform as required to protect against personal injury.
    c. Maintain a safe distance from obstacles, debris, drop-offs, holes, depressions, ramps and other hazards to ensure safe travel.
    d. Maintain a safe distance from overhead obstacles.

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   e. Maintain the minimum approach distance from energized power lines. For operators who are not qualified T&D line technicians, the distances are:
      i. For lines under 50 kV – 10 feet
      ii. For lines over 50 kV – 10 feet plus 4 inches for every 10 kV over 50 kV
   f. Cease operation of the elevating platform should the operator encounter any suspected malfunction of the elevating platform or any hazard or potentially unsafe condition.
   g. The elevating platform shall not be positioned against another object to steady the platform or improve stability.
   h. If the elevating platform becomes caught, snagged or otherwise prevented from normal motion by adjacent structures or other obstacles such that control reversal does not free the platform, all personnel shall be removed from the platform before attempts are made to free the platform using lower controls.
2. Operation from ground with a remote controller includes driving requirements above and:
   a. Maintain safe distances between the operator, machine and fixed objects.
   b. Be aware of the direction the machine will travel when using the controller and do not place yourself in the machine's path.

Eyewash and Emergency Showers

1. Personnel shall ensure that emergency showers and eyewash basins in the immediate work area are operable before starting work.
2. Where the eyes or body of any person may be exposed to injurious corrosive materials, suitable facilities for quick drenching or flushing of the eyes and body shall be easily accessible, unobstructed and able to be located within 10 seconds. For a strong acid or caustic, the eyewash should be immediately adjacent to the hazard.
3. Know the effects of chemicals with which you are working. Read, ask questions about and understand SDS for each chemical with which you work.
4. Always wear personal protective equipment suitable for the material.
5. Prior to starting work, flush plumbed eyewash/showers if they have not been flushed in the last week.
6. Immediately wash off even small amounts of chemicals.

7. In case of chemical exposure, flush skin and eyes with cool water for at least 15 minutes. Do not rub.
8. Hold your eyes open with your hands while using an eyewash to be sure water reaches the eyes.
9. Remove contaminated clothing after the shower has been activated.
10. Get medical assistance immediately following flushing.
11. If possible, continue flushing while on the way to medical help.
12. Keep covers on nozzles to prevent contamination.

Ergonomics

Work Practices to Avoid Musculoskeletal Disorders

1. Workers should be cognizant of the primary risk factors that can lead to the development of musculoskeletal disorders (MSD):
    a. Awkward postures
    b. Excessive force
    c. Excessive repetition
    d. Contact stress (hard edge/pressure)
    e. Heavy vibration (increased risk at cold temperatures)
2. Workers should recognize the signs of musculoskeletal disorders:
    a. Decreased range of motion
    b. Deformity or swelling
    c. Decreased grip strength
    d. Loss of function (e.g., cannot close hand)
3. Workers should recognize the symptoms of musculoskeletal disorders:
    a. Persistent numbness
    b. Burning sensation
    c. Pain
    d. Tingling
    e. Cramping
    f. Stiffness
4. Suggestions for avoiding ergonomic injuries include:
    a. Alternate work patterns and tasks as much as possible throughout the day to allow rest for different muscle groups.

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   b. Perform stretching exercises periodically throughout the workday to relieve stressed muscles.
   c. Keep your body in “neutral” positions as much as possible.
   d. Always use good posture.
   e. Begin or continue a physical fitness program after consulting your health care provider.
5. Report MSD signs and symptoms through normal incident reporting process.

Workstations

1. When working at a computer workstation, workers should consider the following:
   a. Adjust working height of chair, desk and keyboard.
   b. Sit in an upright position using good posture.
   c. Adjust monitor and copy stand side by side and at a comfortable viewing angle.
   d. Set angle of monitor and copy to reduce glare.
   e. Keep wrists and hands in line while using keyboard and mouse. Avoid bending wrists forward or backward.
   f. When using a video display terminal, blink frequently to maintain eye surface moisture.
   g. To prevent eye fatigue, momentarily focus eyes on a distant object.
   h. When using a video display terminal for prolonged periods, frequently stretch and move head, neck, shoulders and arms to prevent buildup of muscle tension.
   i. Ensure chairs are easily and fully adjusted, and allow the body to shift position to the greatest extent possible. Use footrests when adjustments to the chair height do not relieve pressure under the thigh.
   j. When continuously and simultaneously using the telephone and computer, use telephone headrest, headset or speakerphone to prevent injury.
   k. Organize work areas to avoid stretching/twisting to reach items.

Manual Transport of Objects and Materials

1. When performing manual transport or storage operations, employees should consider:

a. When lifting, holding or pushing, workers should avoid strains and sprains caused by incorrect posture, lack of proper assistance and/or lifting aids.
b. When handling materials, workers should use the following techniques if mechanical equipment is not available:
    i. Lifting
        1. Keep the load close to your body.
        2. Bend your knees and hips.
        3. Lift with your legs.
        4. Avoid twisting as you lift.
        5. Get help when needed.
    ii. Bending
        1. Kneel on one knee.
        2. Bend knees and hips, not your back.
        3. When leaning forward, move your whole body, not just your arms.
    iii. Repetitive motions
        1. Keep the load small.
        2. Turn your whole body instead of twisting.
        3. Get close to the load; do not reach and lift.
        4. Lift with your arms and legs, not your back.
        5. Change positions frequently.
    iv. Reaching
        1. Reach only as high as is comfortable; do not stretch.
        2. If you need to reach beyond your comfort level, use a ladder.
        3. Test the weight of the load before lifting. Let your arms and legs do the work, not your back.
    v. Pushing and pulling
        1. Stay close to the load; do not lean forward.
        2. Push the load rather than pulling.
        3. Use both arms.
        4. Get help when needed.

Fall Protection and Walking/Working Surfaces

General Fall Protection Requirements

1. Prior to beginning each job, an assessment must be performed to determine if fall hazards are present and if protective measures are needed. Fall hazards and protective measures shall be included in job planning as appropriate.
2. Locations will have plans in place to rescue workers promptly if they fall while wearing fall arrest equipment. Rescue planning should minimize the amount of time a person is suspended in a fall arrest harness and may include provisions for self-rescue, calling local emergency services, or use of plant emergency responders.
3. When personal fall arrest equipment is required, it shall include a full-body harness with a shock-absorbing lanyard, retracting lanyard or fall arrestor such as a rope grab. It may include other equipment for anchorage or attachment.
4. For maintenance activities, in elevated areas 4 feet or more above a lower level and where guardrails or safety nets are not practical, workers shall use personal fall arrest equipment to prevent injury due to a fall.
5. For construction activities, see Table 10: Fall Protection Requirements for Specific Construction Activities.
6. Fall protection for wood poles, towers and other transmission and distribution structures is described in the Work Methods Manual.
7. Workers must rig fall protection equipment so that free fall is minimized (6 feet or less) and hitting obstructions or a lower level is avoided.
8. Anchorage points must be capable of withstanding 5,000 pounds of force per worker attached or shall be selected by an engineer as part of a fall protection system with a safety factor of two.
9. On vertical lifelines, each worker must have a separate lifeline with a breaking strength of at least 5,000 pounds.
10. Snaphooks, when used, shall be the locking type.
11. All fall protection equipment must be inspected before each use. Any defective pieces must be removed from service. Inspect for these defects:

Table 9: Fall Protection Equipment Inspection Guidelines

Inspect	For evidence of defect or damage including:
Hardware, D-rings, Buckles, Connectors, etc.	Cracks, sharp edges, deformation, corrosion, chemical attack, excessive wear, alterations, excessive heating
Ropes, Straps, Lines, Cables	Fraying, unsplicing, unlaying, kinking, knotting, broken or pulled stitches, excessive elongation, chemical attack, abrasion, excessive wear, excessive lubrication
Mechanical devices, Rope grabs, Connectors	Improper functioning, absence of parts

12. Body belts may be used for work positioning and travel restraint. Body belts may also be used with ladder safety devices where the point of attachment to the harness is 9 inches or less. Do not use body belts for fall arrest.
13. Horizontal lifelines with a safety factor of two shall be designed, installed and used under the supervision of a qualified person.
14. Workers who are on surfaces at any height above dangerous equipment shall be protected by guardrails or fall arrest equipment.
15. Skylights shall be guarded by standard skylight screens or guardrails or personnel shall wear personal fall arrest equipment.
16. For personnel platforms secured to the forks of forklift trucks, ensure:
    a. Workers are protected from moving parts of the truck and falling objects overhead.
    b. Platforms are at least 18 inches wide.
    c. The platform has guardrails or the worker wears fall arrest equipment.
    d. Platforms are secured to the forklift according to manufacturer's recommendations.
17. For the nonworking side of loading docks with a fall hazard of 4 feet or greater, install guardrails or ensure that workers wear fall arrest equipment. The working side of loading docks does not require a guardrail if it prevents work from being performed.
18. Workers working in the bucket of a bucket truck shall wear personal fall arrest equipment.
19. Workers working on top of tanker trucks and rail cars shall use personal fall arrest equipment.

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20. Personal fall arrest equipment is required for workers working from boom-supported elevating work platforms.
21. Workers working in scissor lifts shall be protected by guardrails or personal fall arrest equipment if guardrails are not installed.
22. Protect lifelines against damage.
23. Do not use lanyards or lifelines that have knots.
24. Immediately remove from service personal fall arrest systems and components subjected to impact loading. Do not reuse until a competent person inspects them and determines them undamaged and suitable for reuse. Work with your H&S/EHS professional for evaluation.
25. Do not attach personal fall arrest systems to guardrail systems or to hoists unless approved by a qualified person.
26. When a personal fall arrest system is used at hoist areas, rig it to allow the movement of the worker only as far as the edge of the walking/working surface.
27. When wearing a fall arrest harness, take precautions to ensure that the lanyard does not create a tripping hazard or get caught in rotating or moving equipment.

Walking and Working Surfaces

1. Before allowing workers to work on walking/working surfaces, ensure that such surfaces have the strength and structural integrity to support workers safely.
2. All floor surfaces must be kept clean, dry and free of protruding nails, splinters, loose boards or grating, holes or any other projections.
3. Drainage must be provided in areas of wet processes and mats should be provided where practical.
4. Every open-sided floor platform or runway 4 feet or more above an adjacent floor or ground level must be protected by a standard railing consisting of a guardrail, midrail and toe board.
5. Every stairway with four or more risers must have stair railings.
6. Floor holes, hatchways, pits, chutes or other floor openings must be covered or guarded by guardrails and toe boards.
7. A floor hole that is less than 1 foot in the least dimension and that provides for the passage of machinery, piping or other equipment that may expand, contract, vibrate and/or move shall be at least guarded by a toe board to prevent a worker's feet from entering the hole and tools or debris from falling on workers below.

8. Ensure floor hole guards are in place at all times unless it is necessary to remove the guards temporarily. If the floor hole guards are removed, use guardrails or personal fall arrest equipment to protect workers from falling through holes that are more than 4 feet above lower levels.
9. Do not exceed the weight capacity of a floor hole cover.
10. On stairs, hold handrails where available.

Customer Attic Safety/Walking Surface Safety

1. Ensure worker and equipment do not exceed weight rating of steps and ladder.
2. Inspect ladder, steps/rungs and hardware prior to use.
3. Verify acceptable lighting prior to entry (including head lamp).
4. Validate an accessible entry point and walkway to work area/appliance.
5. Mitigate thermal temperature hazards (heat/cold).
6. Verify safe, properly secured, solid walking surface, free from sharp edges and tripping hazards.
7. Scan flooring to predetermine surface edges and existence of unfloored gaps/holes.
8. Deploy stop work authority if potential hazards are encountered.

Ladders

1. Use ladders only for the purpose for which they were designed.
2. When ascending or descending, workers shall face the ladder, use at least one hand to grasp the ladder and not carry anything that could cause loss of balance or a fall. Use three points of contact.
3. Before each use, ladders must be inspected for defects or damage. Defective ladders must not be used, and they must be immediately removed and tagged out of service.
4. Ladders used to gain access to roofs, floors, platforms, landings, scaffolds, etc. must extend at least 3 feet above the access point or be secured at the top and provided with a grasping device to assist workers in mounting and dismounting the ladder.
5. Portable metal ladders must not be used in areas containing exposed energized electrical lines or equipment.
6. Ladders must not be placed against movable objects, and they must not be placed in doorways opening toward the ladder unless the door is open, locked or guarded.
7. Ladders must be securely placed, held or tied to prevent slipping and falling.

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8. The area around the top and base of ladders must be kept free of tripping hazards, such as loose materials, trash, cords, hoses, loads, etc.
9. If ladders have to be set up in aisles or other passageways, they must be protected from fork trucks, material-handling equipment and other traffic.
10. Working load on ladder must not exceed load limits of the ladder. Labels on portable ladders must be legible.
11. Portable ladders must be equipped with non-skid feet and, where appropriate, self-leveling feet. Ladders must not be placed on unstable bases such as boards, bricks, boxes, barrels, etc.
12. Portable ladders must be placed so that the distance between the bottom legs of the ladder and the supporting points is one-fourth of the ladder length between supports.
13. Two-section extension ladders must not exceed 48 feet in total length, and ladders with more than two sections must not exceed 60 feet in total length.
14. Portable ladders must not be used horizontally as scaffolds, runways, platforms or as guys, braces or skids, or for anything other than their designed uses.
15. When using a portable single or extension ladder, the ladder shall be secured at the top, if possible. If the work requires both hands while working on a secured ladder, use personal fall arrest equipment or work positioning equipment (body belt).
16. Stepladders are to be used only with the legs fully extended and the spreader bar locked in place. Stepladders must not be used as straight ladders.
17. The top or top step of stepladders must not be used, except for platform ladders that are specifically designed for that purpose.
18. Cages, wells, ladder safety devices or personal fall arrest equipment is required for fixed ladders more than 24 feet in height.
19. Fall protection shall be provided at fixed ladder access openings.
20. Do not overload ladders. Pay attention to the duty rating of portable ladders:
    a. **Type III**: Light duty, household use, capable of supporting 200 pounds
    b. **Type II:** Medium duty, commercial use, capable of supporting 225 pounds
    c. **Type 1:** Heavy duty, industrial use, capable of supporting 250 pounds
    d. **Type 1A:** Extra heavy duty, industrial use, capable of supporting 300 pounds
    e. **Type 1AA:** Extra heavy duty, industrial use, capable of supporting 375 pounds

Vehicle repair pits

1. Vehicle repair pits and assembly pits over 4 feet but less than 10 feet deep do not need to be protected by a fall protection system, provided that the following requirements are met:
    a. Access within 6 feet of the edge of the pit is limited to authorized workers.
    b. Authorized workers are trained to recognize and avoid the hazards involved with work around the pit area.
    c. Floor marking in colors contrasting to that of the surrounding area are applied, or rope, wire or chain with support stanchions are placed at a distance of at least 6 feet from the edges of the pits.
    d. Caution signs stating “Restricted Area,” “Authorized Personnel Only” or a similar legend are used to limit entry into the area to authorized workers.

Mobile ladder stands

1. Do not move occupied units. Do not overload. Do not store materials or equipment on the step or platform.
2. Do not attempt to gain additional height by adding any type of extension or object on the unit.
3. Use handrails while ascending and descending.
4. Face the ladder when ascending or descending when the slope of the ladder is greater than 50 degrees above the horizontal.
5. Do not place occupied units in front of a door unless the door is secured.
6. Do not overreach. Keep the unit in close proximity to the work.
7. Use only on a level surface.
8. Access to or egress from any step or platform from any other elevated surface is prohibited unless the unit has been positively secured against movement.

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Specific requirements for construction activities

Table 10: Fall Protection Requirements for Specific Construction Activities

For construction work on this surface:	The fall protection system required is:	Additional information
Surfaces with unprotected sides and edges 6 feet or more above lower levels. These are surfaces that are not listed below.	Guardrail system, safety net system or personal fall arrest system.	
Leading edges 6 feet or more above lower levels for workers engaged in the leading edge work. Erecting **precast concrete members** (including erection of wall panels, columns, beams, and floor and roof "tees") and when performing related operations 6 feet or more above lower levels. Fall protection plan is allowed ONLY for leading edge work, precast concrete work and residential construction. **Residential** construction activities 6 feet or more above lower levels.	Guardrail system, safety net system or personal fall arrest system.	Exception: If these are infeasible or create a greater hazard, develop and implement a fall protection plan.
Workers who are working on a walking/working surface 6 feet or more above a lower level where **leading edges** are under construction, but who are not engaged in the leading-edge work.	Guardrail system, safety net system or personal fall arrest system.	Note: When a guardrail system is chosen and when a controlled access zone has already been established for leading-edge work, use, if desired, the control line instead of a guardrail along the edge that parallels the leading edge.

For construction work on this surface:	The fall protection system required is:	Additional information
Hoisting areas more than 6 feet above lower levels.	Guardrail system or a personal fall arrest system.	Note: If guardrail systems are removed during hoisting operations, ensure workers leaning through the access opening or over the edge use personal fall arrest systems.
When working around **holes** (including skylights) more than 6 feet above lower levels. This includes holes created by temporarily removing grating.	Personal fall arrest system, cover or guardrail system erected around the holes.	Cover holes (including skylights) to protect workers from tripping in or stepping into or through holes and objects falling through holes.
Formwork and reinforcing steel 6 feet or more above lower levels.	Personal fall arrest systems, safety net systems or positioning device systems.	
Ramps, walkways and runways 6 feet or more above lower levels.	Guardrail systems.	
Edges of excavations 6 feet or more in depth and when the excavations are not readily seen because of plant growth or other visual barrier.	Guardrail systems, fences or barricades.	
Edges of wells, pits, shafts and similar excavations 6 feet or more in depth.	Guardrail systems, fences, barricades or covers.	
When working **less than 6 feet** above **dangerous equipment.**	Guardrail systems or equipment guards.	

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For construction work on this surface:	The fall protection system required is:	Additional information
When engaged in overhand **bricklaying** and related work 6 feet or more above lower levels.	Use guardrail systems, safety net systems, personal fall arrest systems or controlled access zones. When reaching more than 10 inches below the level of the surface being worked on, use a guardrail system, safety net system or personal fall arrest system.	Note: This does not apply to bricklaying done from scaffolds. (See OSHA, 1926, Subpart L.)
When doing **roofing work on low-slope roofs** with unprotected sides and edges 6 feet or more above lower levels.	Guardrail systems, safety net systems, personal fall arrest systems or a combination of: • Warning line system and guardrail system when workers are outside the warning line • Warning line system and safety net system when workers are outside the warning line • Warning line system and personal fall arrest system when workers are outside the warning line • Warning line system and safety monitoring system when workers are outside the warning line	Note: On roofs 50 feet or less in width, using a safety monitoring system alone (without the warning line system) is permitted only with H&S Manager approval.

For construction work on this surface:	The fall protection system required is:	Additional information
All work activities on low-slope **roofs** that are not related to roofing work.	Use fall arrest equipment if you are close enough to the edge to fall unless the perimeter is enclosed with guardrails or parapets at least 39 inches high.	
Inspections prior to the actual start of construction work or after construction work has been completed.	Fall protection is not required for inspection, investigation or assessment of workplace conditions prior to the actual start of construction work or after all construction work has been completed. Fall protection is required for inspections that take place while construction work is underway.	
Steep roofs with unprotected sides and edges 6 feet or more above lower levels.	Guardrail systems with toe boards, safety net systems or personal fall arrest systems.	
When working on, at, above or near **wall openings** (including those with chutes attached) when the outside bottom edge of the wall opening is 6 feet or more above lower levels **and** the inside bottom edge of the wall opening is less than 39 inches above the walking/working surface.	Guardrail system, safety net system or personal fall arrest system.	

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For construction work on this surface:	The fall protection system required is:	Additional information
Where the potential for **falling objects** exists, use hard hats and do **one** of the following:	1. Erect toe boards, screens or guardrail systems to prevent objects from falling from higher levels. 2. Erect a canopy structure, and keep objects far enough from the edge of the higher level to prevent their falling over the edge if they are accidentally displaced. 3. Barricade the area to which objects could fall, prohibit workers from entering the barricaded area and keep objects far enough from the edge to prevent their falling over the edge if they are accidentally displaced.	
Scaffold erection and use.	See the H&S Manual Scaffold Program.	

Firefighting Foam/PFAS

1. Per- and Polyfluoroalkyl Substances (PFAS) are a class of man-made chemicals with emerging concerns to Health, Safety and the Environment; they are very persistent and can accumulate in the environment and the human body over time.
2. PFAS are commonly found in Class B Firefighting Foams (specifically aqueous film forming foams or AFFF) where flammable liquids are stored.
3. Duke Energy prohibits the use of PFAS firefighting foams for testing and training purposes at all facilities. These products may be used for emergency fire response.
4. Ordinary fire extinguishers do not contain PFAS.
5. SDS documents often do not list PFAS ingredients. A product such as a firefighting foam can be listed as “PFOA- or PFOS-free” but can still contain other types of PFAS that are proprietary or trade secret. Contact the EHS Field Professional for assistance regarding PFAS ingredients, particularly if the SDS is unclear.

Fire Prevention and Protection/Means of Egress

Means of Egress

1. Keep exits and exit routes unobstructed.
2. Do not lock exit doors closed.
3. Do not store any flammable or combustible material in an exit route or stairwell.
4. Fire doors and dampers must never be tied or blocked in the open position or otherwise made inoperative.
5. Do not block emergency alarm pull stations.

✪ Fire Prevention

1. Be familiar with site procedure for reporting and/or responding to a fire.
2. Know how to recognize and report hazardous conditions and fire hazards associated with the materials and processes to which workers are exposed.
3. Practice good housekeeping in all buildings and vehicles to prevent the accumulation of flammable and/or combustible material.
4. Keep flammable liquids in approved containers that are properly labeled, and store them in designated cabinets or storage areas away from ignition sources.
5. Do not store combustible materials in a flammable liquids storage area.

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6. Return flammable liquids and aerosols to designated storage areas when not in use.
7. Keep flammable liquids storage cabinets and rooms closed.
8. Smoke only in designated areas.
9. Know and follow the site's requirements for the handling of oily rags.
10. Do not transfer flammable liquids into containers unless the nozzle and container are electrically interconnected (bonded).
11. Do not dispense flammable liquids by gravity from tanks, drums, barrels or similar containers except through a listed self-closing valve or self-closing faucet. Listed means tested and listed by a recognized testing laboratory (e.g., UL).
12. Do not use flammable or combustible liquids for general cleaning purposes.
13. Wash parts with flammable or combustible liquids in equipment UL/FM listed/approved for that purpose.
14. Locate and use parts-washing equipment in areas adequately ventilated to prevent accumulation of vapors.
15. Keep parts-washing equipment closed when not in use and ensure that the fusible-link closure is intact and operational.
16. Do not store compressed gas cylinders with flammable or combustible liquids.
17. Do not place objects or materials in front of electrical panels or disconnects that restrict clear access to the equipment.
18. Keep electric control panel covers in place and/or doors closed.
19. Do not store supplies or other materials inside electrical rooms or cabinets or on electrical cabinets, panels or fixtures.
20. Conduct spray finishing operations only in approved designated areas.

Fire Protection

1. Report any damaged or spent portable fire extinguishers to appropriate personnel.
2. Do not block or restrict access to fire extinguishers or other fire protection equipment.
3. Fire extinguishers and other fire protection equipment should not be used unless you are trained and designated to do so.
4. For vehicles that require mounted fire extinguishers, ensure that the mounting bracket safety/cotter pin is utilized. Replace a broken or missing pin. Non-manufactured items (wire, tie straps, etc.) are not to be used.

5. Fire or smoke detection devices may contain radioactive parts. Before disposing of any used or damaged fire or smoke detection device, contact the local H&S/EHS professional.

Hazard Awareness

1. Identify potential pinch points prior to each task so these hazards can be eliminated or avoided.
2. When working overhead, keep tools and small materials not in use in belts or in an appropriate container.
3. Do not throw, toss or drop tools, material or trash unless the action has been discussed and determined to be the safest method. Tools or material shall not be tossed from one individual to another at any time.
4. Workers should not position themselves where they could be struck by falling, flying and moving objects. Protection against these hazards shall be discussed and mitigated prior to starting work. Three-way communication must be maintained when entering and exiting work areas.
5. Prior to entering or exiting areas such as parking lots, offices, hallways, elevators, restrooms and break rooms, workers should be aware of the movement of others or equipment and take action to avoid injury.
6. Inspect work locations prior to beginning work to identify potential hazards from animal and insect bites and stings. Animal and insect repellents, devices and protective apparel shall be used as necessary.
7. Access job sites safely. Do not jump over fences, hedges, bushes, excavations or other obstacles. If required to climb a fence, maintain a controlled ascent and descent.
8. Horseplay is prohibited.

Hearing Conservation

1. Hearing protection shall be worn while working in all areas where the noise level is equal to or greater than 85 dB and in areas where workers cannot carry on a conversation in normal tones within 2 to 3 feet of each other.
2. Hearing protection shall be worn according to the instructions on signs and labels identifying potential high-noise exposure.
3. Contact an H&S/EHS professional for the appropriate type of hearing protection required.

Heat and Cold Exposure

General

1. Dress appropriately for the environmental conditions.
2. Effectively plan, engineer and schedule work/projects to eliminate, control or minimize exposure to temperature extremes.
3. Take adequate breaks and replace fluids frequently.
4. Maintain good physical condition to minimize and/or eliminate injuries/illnesses from exposure to heat or cold.
5. When working in adverse weather conditions (e.g., snow/ice storms), take appropriate precautions.
6. Wherever possible, utilize the buddy system if you must travel or work in extreme temperatures.

Cold Exposure

1. When working in cold environments, dress to stay warm and dry. Use rain gear, layered or insulated clothing as appropriate.
2. Observe co-workers for symptoms of frostbite and hypothermia. When necessary, use warming shelters or vehicle cabs for temporary relief. Seek medical attention when appropriate.
3. Walk carefully on snowy and icy surfaces.
4. When shoveling snow, be very careful to avoid overexertion.
5. Soaking wet clothing contributes to hypothermia and should be removed as soon as possible.
6. Use of vibrating tools in extremely cold weather could aggravate a circulatory condition called Raynaud's Syndrome and should be avoided by affected individuals.

Heat and Hydration

1. When working (or preparing to work) in areas of extreme heat, consume adequate and appropriate amounts of fluids to help avoid heat-related illnesses and dehydration. Use hydration coach to monitor fluid intake.
2. Plan work based on recommended action times. Provide adequate personnel for rotation.
3. Slowly build up a tolerance for heat exposure. Where appropriate, reduce action time accordingly.
4. When working in hot environments, wear appropriate clothing. Use ice vests if desired.

5. Take precautions to protect from overexposure to the sun (e.g., clothing, eye protection, sunscreen).
6. Permit workers to interrupt their work if experiencing signs of heat-related disorders or extreme discomfort.
7. Observe co-workers for symptoms of heat-related illness. When necessary, seek medical attention. Symptoms include but are not limited to dry, hot skin and high body temperature (heat stroke), moist clammy skin (heat exhaustion) and heat cramps.
8. Avoid heat-related illness by maintaining adequate levels of hydration.

This table provides general length of time between water breaks and corresponding hydration target for each hour of work. Have a hydration coach to ensure adequate fluid intake for workers.

Temperature	Work Level	Maximum Minutes Worked Between Hydration Breaks	Hydration Target
<80	Normal	–	8-12 oz./hr.
80-85	Normal	–	8-16 oz./hr.
86-90	Normal	50	12-20 oz./hr.
91-95	Normal	45	16-24 oz./hr.
>96	Normal	40	24-32 oz./hr.

* Note: For additional information, see the Thermal (Heat-Cold) Stress Program, ADMP-SAF-HSF-00059.

Hexavalent Chromium

1. Be aware of activities that may produce or disturb hexavalent chromium. Activities of concern include:
    a. Hot work on or demolition of coated surfaces or chromium-containing steels (particularly stainless steel)
    b. Welding or thermal cutting of any steel in confined spaces
    c. Spray painting with chromium-containing paints and primers
    d. Work activities disturbing fly ash, ceramic bricks or treated wood
2. Use exposure control methods including:
    a. Substitution of less hazardous materials or processes

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   b. Engineering controls (such as ventilation)
   c. Work practices (such as equipment wash downs or vacuuming, vacuuming and proper disposal of work clothing, hand/face washing, etc.)
   d. Establishing “regulated areas” where the exposure limit might be exceeded
   e. Use of personal protective equipment (such as coveralls, gloves, safety glasses, face shield, etc.) and respiratory protection

3. Where protective clothing and equipment is required for hexavalent chromium protection, use change rooms with separate, contamination-free storage facilities for street clothes.
4. Do not take contaminated clothing or equipment home.
5. Do not wear dusty or contaminated clothing into break rooms, lunch rooms or other areas where food and drink are stored or consumed.
6. Remove contamination from coveralls (or other clothing) using a method that does not put dust into the air (e.g., vacuum dust from work clothing with a HEPA vacuum).
7. Where used, empty or change HEPA vacuums/filters in accordance with department or site work practice, to minimize exposure and to ensure appropriate waste disposal.
8. Do not blow off, shake off or do anything else to contaminated materials that could send dust or particulate debris into the air.
9. Do not dry sweep, shovel or brush hexavalent chromium dust or contaminated products. Use HEPA vacuum.
10. Wash face and hands before breaks and at end of shift.
11. Do not eat, drink, smoke, chew gum or tobacco, or apply cosmetics in regulated hexavalent chromium areas.
12. Label waste bags, bags of clothing or equipment or other containers of hexavalent chromium materials according to the Hazard Communication program in the H&S Manual. See your environmental professional for disposal guidance.
13. Do not enter a hexavalent chromium regulated area unless you are properly trained.

High-Risk Tools

1. Duke Energy has set minimum safety features for many of the high-risk tools used across the enterprise. Some examples of high-risk tools' minimum safety features include but are not limited to:
   - Constant Pressure Switch
   - Pressure Relief Valve
   - Emergency Shut-Off
   - Anti-Kickback Clutch
2. Follow the business unit-specific High-Risk Tool Program concerning the use of high-risk tools and high-risk tasks.

Hot Work Program

1. A hot work permit is required for any temporary operations involving flames or producing heat and/or sparks.
2. Before performing welding, cutting or grinding, evaluate and safeguard the work area for combustible items.
3. Use the business unit or location-specific hot work permit.
4. Assign a fire watch with a suitable fire extinguisher for welding, cutting and grinding operations in work areas with combustible materials, or where the fire or sparks cannot be contained in the immediate work area.
5. When using a fire watch, continuously monitor the hot work area while work is taking place and for 30 minutes following the completion of work. Then perform final inspection.
6. Ensure the person assigned to fire watch has been properly trained.
7. Keep welding, cutting and grinding areas clean and free from accumulations of trash, rags and other combustible items.
8. All combustible materials must be removed or covered within a 35-foot radius of the hot work location.
9. For welding/cutting operations, use only friction or stationary striker to light a torch. Do not light torches with matches, cigarette lighters or hot work.

Housekeeping

1. Keep work areas, job sites and vehicles orderly, clean and free of hazards.
2. Properly dispose of scrap and waste materials at frequent and regular intervals or at the end of each shift.

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3. Keep walkways and work surfaces free of oil, grease, water, ice and other slippery materials. Clean up spills immediately.
4. Keep walks, aisles, stairways and all other passageways clear of obstructions.
5. Identify, cover or mark temporary cables, cords or hoses passing through work areas.
6. Eliminate tripping hazards or restrict access to the area.
7. For trash, recyclables and other waste materials, do the following:
    a. Keep waste in approved and properly labeled containers.
    b. Keep oily rags in a covered, listed (UL/FM) container.
    c. Safeguard sharp objects before disposal.
8. Ensure staged equipment does not create unsafe conditions by obstructing safe access to operating or emergency equipment, or by exceeding weight limits.
9. Use adequate lighting to perform work safely.
10. Keep access to emergency exits clear.
11. Keep rolling stock (pipe, conduit, all-thread, etc.) off of the floor or designate a marked storage area to eliminate trip hazards.

Hydrogen Safety (Generating Facilities)

1. Hydrogen is a colorless, odorless, flammable gas with a wide combustible range of 4% to 75% in air. Hydrogen is lighter than air and can collect in overhangs, ceilings or other confined areas.
2. Hydrogen gas burns with a nearly invisible flame and can be difficult to detect. Use extreme caution approaching areas where you suspect a hydrogen fire. If possible, isolate the source of the gas and allow it to burn out.
3. Never allow hydrogen to mix with air inside a system or component. Always use an inert, intermediate purge gas such as carbon dioxide during purging and refilling operations on generators and hydrogen systems.
4. Follow all applicable station-specific continuous use procedures and Job Hazard Analyses (JHA) when purging, refilling generators and hydrogen dryers or transferring hydrogen gas.
    a. Make a public address system announcement at the site prior to venting/purging hydrogen generators into and out of service.
5. Use non-sparking tools when working on systems containing hydrogen prior to venting and purging.

6. Only use replacement parts in hydrogen systems that are rated for hydrogen service.
7. Ensure purge gas flow rates do not exceed manufacturer's recommendations or procedure requirements when performing generator purges. Also, ensure flow regulators do not freeze up during gas transfer.
    a. Verify the quantity of gas used for the purge (either by cylinder count or flow meter reading if installed).
8. Delivery vehicles, stationary tanks and tube trailers shall be properly grounded before transferring bulk hydrogen gas. Hydrogen gas is ignitable by static electricity.
9. Only use gas detection instruments that have specifically been calibrated for use with hydrogen when performing leak checks or verifying purge conditions. Hydrogen detection instruments must read below 4,000 PPM at the vents for a system to be considered safely purged.
    a. Only trained personnel shall utilize gas detection instruments to verify gas levels during tasks associated with hydrogen gas and associated systems.
    b. Training will include the completion of a Job Performance Measure (JPM) and an understanding of the specific monitor's limitations and maintenance requirements.
10. Hydrogen-containing systems shall be conspicuously posted per signage requirements from Hazardous Classified Locations assessments or as a minimum with "Danger Hydrogen Gas" signage that warns of the hazard and prohibits smoking, flames or hot work within 35 feet.
    a. If hot work must be performed inside these boundaries, continuous hydrogen monitoring shall be performed in accordance with any additional hot work permit requirements.
11. In Regulated & Renewable Energy (RRE) Power Generation facilities, hydrogen system piping shall be painted yellow with "HYDROGEN" in black lettering, flow arrows and black bands every 20 feet.

Industrial Hygiene Monitoring

1. Where health hazards exist, contractors are required to comply with OSHA regulations.
2. Where contractor's work may potentially expose other personnel working near or adjacent to potential health hazards, the contractor is required to communicate this information.

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3. Contractors are to provide existing exposure assessments relevant to the tasks being performed to Duke Energy when requested and shall contain the following:
    a. Administrative and engineering controls used to reduce exposure
    b. Existing monitoring data to validate exposure levels
    c. Names of contractors shall be expunged, and the data shall include no personally identifiable information
4. When exposure monitoring is needed for Duke Energy tasks, contractors are required to conduct their own exposure monitoring using validated methods as defined by OSHA or NIOSH. As with existing exposure assessments, contractors are to provide the results of this monitoring to Duke Energy when requested.
5. Duke Energy may elect to conduct exposure monitoring on contractor's workers based on business need.

Inorganic Arsenic

1. Do not enter inorganic arsenic regulated areas unless you are properly trained.
2. Use respirators, coveralls, gloves and other personal protective equipment as specified by supervision or H&S/EHS professional for work in inorganic arsenic regulated areas.
3. Vacuum coveralls (and shoes) with a HEPA vacuum before breaks and lunch and at the end of the work shift if wearing coveralls out of the regulated area. Note: Leave the respirator on while removing contaminated coveralls.
4. Wash face and hands prior to break. Shower at the end of the work shift.
5. Never shovel or sweep fly ash and dust particulate (slag) unless vacuuming or other relevant methods (wet wash down) have been tried and found ineffective. Note: Obtain management approval before using one of these methods.
6. Where protective clothing and equipment is required for inorganic arsenic protection, use change rooms, with separate contamination-free storage facilities for street clothes.
7. Do not wear dusty or contaminated clothing into break rooms, lunch rooms or other areas where food and drink are stored or consumed.
8. Ensure that inorganic arsenic-contaminated protective clothing is:
    a. Removed at end of the task or end of the shift (whichever comes first)
    b. Placed in closed container that is labeled per the requirements of 29 CFR 1910.1018 – Inorganic Arsenic

9. Where used, empty or change HEPA vacuums/filters in accordance with department or site work practice to minimize exposure and to ensure appropriate waste disposal.
10. Never blow off, shake off or do anything else to contaminated materials that could send dust or particulate debris into the air.
11. Never carry coveralls, protective equipment or other contaminated materials home.
12. Certain activities, such as eating, drinking, smoking, chewing gum or tobacco, or applying cosmetics, are prohibited in regulated areas.

Inspecting and Testing Poles and Structures

1. Metal poles and structures require visual inspection only.
2. All wood poles, including adjacent poles, require an inspection before climbing, working or changing load. Visually inspect all poles for any of the following signs of instability:
    a. Decay, loose soil, green growth, damp conditions or other signs of possible weakness in the surrounding soil.
    b. Pay attention to the landscape. Runoff can saturate poles with water, accelerating decay.
    c. Wood separation, bird holes, rot, cracks and burn marks.
    d. Rust or rot in guy wires and anchor rods that are attached to the pole or structure.
    e. Any sign of instability in adjacent poles and structures.
    f. Any slack in guy wires, which could indicate that anchor rods have pulled up out of the ground.
3. Hammer Sounding – Rap the pole sharply with a hammer weighing about 1.4 kg (3 pounds), starting near the ground line and continuing upward circumferentially around the pole to a height of approximately 1.8 meters (6 feet). The hammer will produce a clear sound and rebound sharply when striking sound wood. Decay pockets will be indicated by a dull sound or a less pronounced hammer rebound.
4. Prodding – Prod the pole as near the ground line as possible using a pole prod or a screwdriver with a blade at least 127 millimeters (5 inches) long. If substantial decay is present, the pole is unsafe.
5. “Rocking” of a wood pole to determine stability is NOT an owner-approved method and shall not be used during wood pole inspections.

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6. If the pole is found to be unsafe to climb or to work from, it must be secured so that it does not fail while a worker is on it. The pole can be secured by a line truck boom, by ropes or guys, or by lashing a new pole alongside it. If a new one is lashed alongside the defective pole, work should be performed from the new one. Tag unsafe poles to prevent others from climbing.

Lead in Construction

1. Identify activities that may disturb lead-containing materials. Activities of concern include:
    a. Disturbance (e.g., sanding, scraping, grinding, cutting, abrasive blasting, welding) of painted surfaces
    b. Spray painting with lead-containing paints and primers
    c. Maintenance on, or demolition of, structures that contain lead (paints, solder and roof flashing)
    d. Removal or encapsulation of lead-containing materials
    e. Construction, alteration, repair of structures or components that contain lead
    f. Lead contamination/cleanup
2. Contact H&S/EHS professional to determine if lead exposure exists.
3. As practical, substitute less hazardous materials (such as lead-free paints) or processes (such as paint strippers) as the first choice to eliminate the airborne exposure potential.
4. When indicated by exposure assessments, use exposure control methods, including:
    a. Engineering controls (such as mechanical ventilation)
    b. Work practices such as limiting work time, vacuuming and proper disposal of work clothing, hand/face washing, etc.
    c. Limiting access into work areas for activities where the exposure limit might be exceeded
    d. Use of personal protective equipment (such as coveralls, gloves, safety glasses, face shield, etc.) and respiratory protection
5. Where protective clothing and equipment are required for lead protection, use change rooms, with separate contamination-free storage facilities for street clothes.
6. Do not wear contaminated clothing into break rooms, lunch rooms or other areas where food and drink are stored or consumed.

7. Do not carry coveralls, protective equipment or other contaminated materials home.
8. Ensure that lead-contaminated protective clothing is:
    a. Removed at end of the lead task or end of the shift (whichever comes first)
    b. Placed in closed container, which is labeled per OSHA requirements
9. Remove contamination from coveralls (or other clothing) using a method that does not put dust into the air (e.g., vacuum dust from work clothing with a HEPA vacuum).
10. Wear respiratory protective devices for any lead-disturbing work, unless the work activity has specifically been identified as not requiring respirators.
11. Follow all company and/or location-specific requirements of the respiratory protection program.
12. Empty or change HEPA vacuums/filters in accordance with department or site work practice to minimize exposure and to ensure appropriate waste disposal.
13. Do not blow off, shake off or do anything else to contaminated materials that could send dust or debris into the air.
14. Do not sweep, shovel or brush lead dust or contaminated products (e.g., paint debris, abrasive blast) unless vacuuming or other equally effective methods have been tried and found ineffective. Note: Obtain management approval before using one of these methods.
15. Wash face and hands before breaks and at end of shift.
16. Shower at the end of the shift after performing activities where lead concentrations exceed the PEL.
17. Do not eat, drink, smoke, chew gum or tobacco, or apply cosmetics in areas where lead work occurs.
18. Do not enter a lead-regulated area without proper training.
19. Collect, handle, label and properly dispose waste material (such as removed paint, abrasive blast material, HEPA filters, contaminated clothing or equipment). See H&S/EHS professional for specific requirements.

Lightning

1. Where applicable, use lightning detection to determine distance of lightning strikes. Detection may be determined by:
    a. Duke Energy Meteorology guidance.
    b. Government sources (Vaisala NLDN National Lightning Detection Network).

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   c. Lightning detection application.
   d. Flash to bang calculation. Count the number of seconds from seeing a lightning strike to the time thunder is heard. Divide by 5. This is approximate distance in miles.
2. Follow business unit guidance where applicable.
3. Business unit-specific recommendations for work suspension:
   a. Transmission and plant-based employees: Prepare for suspension at 15 miles. Suspend work at 8 miles.
   b. Solar locations: Prepare for suspension at 30 miles. Suspend work at 10 miles.
   c. Wind locations: Prepare for suspension at 60 miles. Suspend work at 30 miles.
4. Resume work activities when it is clear that the storm has passed or 30 minutes after the last strike.

Line of Fire Hazards

1. Eliminate injuries from line of fire hazards through job planning, engineering controls and body positioning.
2. Identify line of fire hazards before and while performing tasks. Discuss line of fire hazards. Take appropriate mitigation actions.
3. Consider these as line of fire hazards: struck by; struck against; caught in/under/between; sharp objects; falling/flying/rolling objects; and hand tools/machinery.
4. Eliminate line of fire hazards where possible. Methods include use of physical barriers, machine guards for moving and rotating equipment, and falling object protection including toe boards/netting; exclusion zones; and tool securement.
5. Be aware of body position of all workers where line of fire hazards are present.

Lockout/Tagout for Facilities Other than Transmission, Distribution and Generation

1. Workers shall be trained in the requirements of the lockout/tagout procedure in use prior to performing tasks that require lockout/tagout.
2. Identify all hazardous energy sources associated with the work activities.
3. Isolate hazardous energy sources, tag and lock energy-isolating devices, and relieve stored hazardous energy.

4. Before beginning work, verify by testing or operation that hazardous energy has been relieved.
5. Before restoring hazardous energy, verify work is complete and persons are clear of equipment.
6. Management and contractors will inform each other of their respective lockout/tagout procedures.
7. For transmission, distribution and generation activities, follow business unit-specific lockout/tagout procedures.

Machine Guarding

1. Before starting work using powered machinery or tools, ensure that guards:
    a. Are in place so as to prevent contact with dangerous, moving parts by any part of the body (e.g., hands, arms) or clothing
    b. Are firmly secured to the machine, if appropriate, and are made of durable material that will withstand the conditions of normal use
    c. Are appropriate for the equipment
2. Do not create new hazards by having a shear point, jagged edge or an unfinished surface that can cause a laceration.
3. Report any broken or missing guards to your supervisor. Do not use the tool until repairs have been made.
4. Before working, perform a pre-job safety assessment to identify potential hazards to the hand.
5. When working around machinery with rotating or moving parts:
    a. Ensure gloves, ties, loose clothing and long hair do not become entangled in equipment.
    b. Tie back long hair.
    c. Button or properly roll up long sleeves, and tuck in shirttails.
    d. Do not wear loose jewelry (e.g., necklaces, chains, cords) outside clothing.
    e. Wear badge straps of a breakaway-type of material.

Methylene Chloride

1. Review SDS and labels to determine if products contain methylene chloride (Dichloromethane).

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2. If you have the potential to be exposed to methylene chloride, you will need to have specific methylene chloride training beyond normal hazard communication requirements.
3. Use methylene chloride-free products where possible.
4. Seek approval from senior leadership to procure methylene chloride. Contact the field environmental/EHS professional for assistance submitting a Highly Restricted Exemption Form.

Motorized Equipment, Vehicle, and Trailer Operations

Distracted Driving

1. Texting, reading texts and emailing are prohibited when driving company vehicles and when driving personal vehicles on company business.
2. Follow state and local regulations on use of cellphones when driving.
3. Focus on the primary task of driving. Keep the number of calls to a minimum amount and duration.
4. Use radios responsibly while driving.
5. Avoid other activities that may distract you from driving. Pull over or park in safe locations to perform these activities.
6. For commercial motor vehicles: Driver use of hand-held cellphones is prohibited when driving commercial motor vehicles. Hand-held cellphones shall not be used to make or receive phone calls, send or receive text messages or emails or access other applications (internet, etc.) while driving commercial motor vehicles. Use of hands-free devices is allowed but only if the call can be initiated, answered or terminated by touching a single button on the cellphone or headset.
7. For non-commercial motor vehicles: Do not use a cellphone when driving unless necessary. If you have to use a cellphone, it must be used with a hands-free device.

Operational Practices

1. Operate all vehicles in a safe manner, using defensive driving techniques. Do not drive aggressively.
2. Drivers/operators shall have full responsibility for vehicles in their possession. This includes observing all federal, state and local laws and ordinances.
3. Drivers shall be properly licensed when operating company-owned, rented, leased vehicles or equipment, and personal vehicles when used on company business. Any license restrictions shall be adhered to at all times.

4. Before operating a vehicle, visually inspect it to determine whether the vehicle is safe to operate. Perform a “360-degree” inspection around the vehicle to be sure the area is clear. Do not operate unless equipment is in safe condition.
5. Seat belts shall be worn by the driver and passengers at all times when the vehicle or equipment is in motion. This shall apply to all company-owned, rented, leased vehicles or equipment, and to personal vehicles when used on company business or on company property.
6. Passengers in motor vehicles shall be in approved riding positions and restrained where occupant restraints are provided.
7. Lock unattended vehicles and remove keys to reduce the risk of vehicle theft.
8. Adhere to the following practices while fueling a vehicle: a. Turn off the engine. b. No smoking. c. 100% attendance. d. Do not fill the tank beyond the auto shutoff level. e. Report and clean up any fuel spill immediately.

Backing and Chocking

1. When parking, use the following order of preference:
    a. Pull through into the parking space.
    b. Back into the parking space, except for diagonal spaces.
    c. Park front-end in only when (a) and (b) create a greater hazard.
2. Refer to additional business unit-specific procedures on backing and use of wheel chocks.
3. Complete a survey around the vehicle (“360 degrees”) before entering the vehicle to identify and remove any hazards. Visualize your exit path.

Utility Vehicles

1. Operators should be familiar with the utility vehicle’s operations and should read the owner’s manual and follow the manufacturer’s guidelines for recommended maintenance, usage, operation and personal protective equipment.
2. When operating utility terrain vehicles (UTVs) or all terrain vehicles (ATVs), wear the appropriate PPE (i.e., head, eye) for the operating conditions to be encountered. Refer to Business Unit specific guidance/PPE Hazard Assessment for UTV/ATV specific requirements.
3. When operating utility vehicles, conduct pre-operational functional check of seat belts. Utility vehicles include tractor and utility cart type vehicles such as Polaris, Kubota, Deere, etc. It does not include amphibious vehicles or golf carts.

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4. Driver and passengers shall use seat belts at all times when the vehicle is in transit.
5. Assess the terrain to be traveled prior to departure. Utilize the business unit terrain assessment procedure or checklist prior to proceeding into an area that has unknown terrain. Stay alert for holes, rocks and other hidden hazards in the terrain.
6. Workers riding in utility vehicles shall use safety handholds and proper foot positioning as needed to maintain stability while en route.
7. Operators of utility vehicles shall ensure that passengers are aware of changes in road/terrain conditions so they do not get thrown out of the cart. Avoid sudden starts, stops or turns.

Emergencies

1. In case of any vehicle trouble, pull off to the right side of the road, if possible, and use emergency flashers and warning signals.
2. Refer to the “Use in Case of an Accident Form” located in the company vehicles for actions to take if you have an accident in a company vehicle.

Incident Reporting

1. If an employee is in an accident involving a company-owned vehicle, leased (rental) or personal vehicle being used on company business, supervision or designee contacts H&S/EHS professional to enter information into eTRAC Incident.
2. If an employee has an injury from any kind of vehicle accident while on company business, supervision or designee contacts H&S/EHS professional to enter information into eTRAC Incident.
3. Employees who receive any citation or warning ticket while operating a Duke Energy-owned vehicle shall report the citation immediately to their supervisor and to the email DOT/CMV@duke-energy.com.

Vehicle Ingress/Egress

1. Safely enter and exit vehicles by following these recommendations:
    a. For larger vehicles, open door while on the ground.
    b. For all vehicles, enter and exit using three points of contact and slide into/out of seat.
    c. If using the steering wheel as a point of contact, ensure it is locked.
    d. Scan ground for uneven or slick areas when exiting vehicles.
    e. For RRE fleet vehicles, confirm the vehicle parking brake is on and holding before exiting.

Operations in High Winds

1. Stop travel in vehicles and stop work, including climbing, when sustained winds reach tropical storm force velocity of 39 mph (18 mps).
2. Once sustained winds drop below 39 mph (18 mps), monitor wind gusts in excess of 39 mph (18 mps) for safe working and traveling.
3. In addition, follow vehicle and equipment manufacturers' requirements associated with prohibited operation in high wind conditions.

Recovery of Stuck/Mired Vehicles and Equipment

1. If a vehicle or equipment becomes stuck or mired, immediately stop all work in the area and ensure personnel are safe.
2. Notify site management and H&S/EHS professionals as appropriate.
3. Where recovery requires the use of rigging to extract or place in an upright position, utilize professional recovery services. If professional recovery service is not available or desirable, develop detailed plans/job hazard analysis for the recovery.
4. Consider all stuck vehicle recovery operations to be non-routine and plan accordingly.
5. Consider recovery efforts to be high-risk depending on factors such as depth of vehicle, recovery slope or angle, unstable load, or vehicle overturned. High-risk extractions require significant planning and possibly engineering review including review of extraction rigging and rigging load ratings.
6. Once rigging equipment has been used for extraction purposes, remove it from service and conduct an evaluation to ensure it is safe before reusing. Consider identifying extraction rigging equipment and using it only for that purpose.
7. Follow business unit procedures where available.

Natural Gas/Fuel Gas Safety (Generating Facilities)

1. Natural gas is a colorless, odorless, flammable gas with a combustible range of 5% to 15% in air. Natural gas is lighter than air and can collect in overhangs, ceilings or other confined areas. Mercaptan is typically added to natural gas (at the local gas yard) as an odorant for safety at most generating facilities.
2. Never rely on your sense of smell to detect natural gas leaks. Always use combustible gas detection equipment calibrated for use with natural gas when performing tests or leak checks.

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3. Natural gas-containing systems and equipment shall be conspicuously posted per signage requirements from Hazardous Classified Locations assessments or as a minimum with Danger Flammable Gas signage prohibiting hot work within 35 feet.
    a. If hot work must be performed inside these boundaries, continuous gas monitoring shall be performed in accordance with any additional hot work permit requirements.
4. Extinguish natural gas fires by shutting off the source of the gas.
5. Use non-sparking tools when working on systems containing natural gas prior to venting and purging. Only use replacement parts in fuel gas systems that are rated for natural gas service.
6. In Regulated & Renewable Energy (RRE) power generation facilities, natural gas piping shall be painted yellow with “Natural Gas or Fuel Gas” in black lettering and flow arrows with green bands every 20 feet.
7. Follow RRE Technical Program FHO-PRG-NA-0011 (Purging Natural Gas Piping Systems) requirements for purging, venting and line breaking on natural gas containing systems and piping.
8. Before performing hot work on a fuel gas-containing system, it must be purged with an inert, intermediate gas such as nitrogen.
9. Combustible gas meter readings read at vents shall be per tech program requirements to be considered purged.
    a. Only trained personnel shall utilize gas detection instruments to verify gas levels during tasks associated with methane gas and associated systems.
    b. Training will include the completion of a Job Performance Measure (JPM) and an understanding of the specific monitor’s limitations and maintenance requirements.
10. Hot taps required at RRE facilities shall only be performed by qualified natural gas company personnel.

Natural Gas Safety

General Information and Properties of Natural Gas and Propane

1. Natural gas and propane are odorless in their native state. Both are mixed with odorant that allows small quantities to be readily detected by a “rotten egg” smell.
2. Both natural gas and propane are colorless and nontoxic, but either can displace oxygen when concentrated, such as in confined spaces.

3. Natural gas will tend to rise since it is lighter than air, while propane is heavier than air and will tend to sink.
4. Natural gas is typically delivered to Duke Energy customers by underground pipelines. Above-ground facilities on the Duke Energy natural gas system include pressure regulating stations and metering facilities. Propane is delivered by truck to customer storage tanks.
5. Natural gas has a combustible range of 5% to 15% in air and requires a 1,100 to 1,200 degree Fahrenheit ignition source. The combustible range of propane is 2% to 10% in air and the ignition source must be 920 to 1,020 degrees Fahrenheit. Common sources of accidental ignition include open flame, switching lights or appliances on or off, smoking, or even static electricity.
6. The products of gas combustion normally include water vapor and carbon dioxide. However, when oxygen is limited or venting is inadequate, poisonous carbon monoxide can form. Other sources of carbon monoxide include cars warming up in garages and charcoal grills operated indoors.

Emergency Actions for Duke Energy Employees – Gas Odor Outdoors

1. If you smell natural gas outside, move to an area where the odor is no longer detectable.
2. If you are traveling, find a safe place to pull over to report the odor.
3. Report natural gas odors by calling the Duke Energy gas hotline at 800.752.7504 (NC, SC, TN) or 800.634.4300 (OH, KY). A Duke Energy Natural Gas Operations first responder will be dispatched to ensure that the area is “made safe” and that repairs are initiated.
4. Symptoms of carbon monoxide poisoning include:
   - Sudden flu-like symptoms
   - Dizziness, headache, sleepiness
   - Nausea or vomiting
   - Fluttering or throbbing heartbeat
   - Cherry red lips, unusually pale complexion
   - Unconsciousness

Intrinsically Safe Devices for Gas Operations

1. Electronic devices used in the presence of combustible gas must be rated as intrinsically safe. Electronic devices that are not intrinsically safe are not allowed where combustible gas is present.
2. Intrinsically safe electronic devices are designed so that they will NOT become a source of ignition in the presence of a combustible gas.

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3. Many electronic devices are NOT intrinsically safe. This may include flashlights, cellphones and doorbells. Electronic devices that are intrinsically safe will be marked.
4. Do not use electronic devices in gaseous environments unless you have been trained on the correct tool and know the meaning of the term intrinsically safe.

Encountering Damaged Natural Gas Facilities

1. If you encounter natural gas “blowing” from excavation damage or damage to an above-ground facility, it is important for personnel to remain at a safe distance away from the blowing gas.
2. Recognize that natural gas will rise.
3. If there is equipment near the blowing gas, do not attempt to move the equipment or cut off equipment that is still running until qualified Natural Gas Operations personnel can access the situation and the equipment can be moved safely.
4. Valves to cut off blowing gas should only be operated by trained Natural Gas Operations personnel.
5. Report the leak to the Duke Energy gas hotline by calling 800.752.7504 (NC, SC, TN) or 800.634.4300 (OH, KY).
6. If the natural gas has ignited, do not attempt to extinguish it. If natural gas is extinguished improperly, it may migrate to other areas and create a more dangerous situation.
7. If an explosion has occurred, be aware that there may still be active leaks in the area. Maintain a safe perimeter until a Duke Energy Natural Gas Operations first responder informs you that it is safe to reenter the area.
8. Do not allow anyone to enter an unsafe excavation in order to expedite a repair of a natural gas line. The safest way to repair these leaks is for trained Natural Gas Operations personnel to stop the flow of natural gas by turning valves or digging remotely from the leak and “pinching off” the line.

Damage Prevention

1. Locate underground utilities prior to soil disturbance. Where appropriate, contact 811 and follow the local regulations. Once you have approval to dig, proceed cautiously, checking positive response and looking for evidence of underground utilities that have not been marked. Hand dig as required.
2. If you see others digging or about to dig in an area where there are underground utilities, be proactive, take time to check the situation, and ensure that a utility locate has been requested and facilities marked. It may be appropriate to stop work.

3. A Natural Gas Operations representative should be on-site when anyone is digging near natural gas transmission pipelines. These critical high-pressure pipelines are usually in maintained rights of way with pipeline markers indicating their approximate location. If you see excavation in these locations, and there are no Natural Gas Operations personnel present, stop work and notify the Duke Energy gas hotline by calling 800.752.7504 (NC, SC, TN) or 800.634.4300 (OH, KY).

Office Safety

1. All desk and cabinet drawers shall be closed when not in use.
2. Only one file drawer shall be open in a cabinet at any one time.
3. Place heaviest loads in lower drawers or shelves to avoid “top loading.”
4. Objects that present a hazard shall not be stored on top of cabinets or other storage enclosures.
5. Paper cutter blades shall be down and secured when not in use.
6. Electrically powered office equipment shall be operated in accordance with manufacturer’s instructions.
7. When using power strips, follow manufacturer’s recommendations. Do not overload outlets. **Do not connect** multiple power strips in series.
8. Keep combustibles to a minimum. Do not store combustible material, such as paper, in close proximity to electrical outlets and connections.
9. Use office chairs appropriately to avoid tipping. Do not tilt chairs such that they could tip over.
10. Do not stand on chairs, desks or tables. Use appropriate step stools or ladders.
11. Use handrails when available.
12. Do not use portable space heaters without prior approval from your supervisor and site/facilities management. If portable electric space heaters are used, they must be attended to while in use, turned off after normal working hours and kept at least 3 feet away from combustible materials.
13. Avoid distractions while walking in office areas such as using cellphones, reading materials, etc.
14. Begin meetings with safety/environmental (Level 1) assessment and include emergency evacuation procedures, location of exits and AEDs, and site-specific emergency information.

Paint Chip Cleanup

1. Before cleaning up paint chips, determine if the chips contain asbestos, lead or hexavalent chromium. If chips contain asbestos, lead or hexavalent chromium, this section applies.
2. Determine how long the cleanup is expected to last.
3. If cleaning paint chips is expected to take more than 15 minutes, contact your H&S/EHS professional.
4. If cleanup can be accomplished in less than 15 minutes, use one or more of these options:
    a. Manually pick up paint chips and place in a plastic bag.
    b. Use a high-efficiency particulate air (HEPA) filtered vacuum that has been designated for use with asbestos or lead.
    c. Use a tack cloth or roller.
5. In addition, comply with these requirements:
    a. Do not sweep, blow or vacuum paint chips with a vacuum that is NOT a high-efficiency particulate air (HEPA) filtered vacuum.
    b. Do not use compressed air to clean paint chips.
    c. Do clean the area in a pattern that will minimize stepping on or crushing the paint chips.
    d. As necessary, have the vacuums emptied and cleaned by appropriately trained personnel.
    e. Store the collected paint chips and waste in waste receptacles labeled for asbestos or lead.

Pedestrian Safety

1. Cross at crosswalks and obey crossing signals.
2. Look for oncoming vehicles and cyclists in both directions.
3. Do not text or talk on a cellphone when crossing the street. Take your time.
4. Make eye contact with approaching drivers and wait for them to acknowledge you before you cross in front of a turning vehicle.
5. Stop in parking garages and wait for oncoming traffic.

Personal Protective Equipment

General

1. Workers shall maintain PPE in a sanitary and reliable condition.

2. Workers shall inspect PPE prior to use and immediately following any incident that could have caused damage.
3. Damaged PPE shall not be used.
4. Workers shall adhere to manufacturer's recommendations for PPE.
5. A hazard assessment must be performed before PPE is issued or used in order to ensure that the prescribed PPE is appropriate for the hazard(s) present.

Eye and Face Protection

1. Eye and face protection that complies with acceptable consensus standards (such as ANSI Z87.1) must be used whenever there is a hazard due to particles, fumes, liquids, gases or radiation that could injure the eyes.
2. Minimum eye protection shall consist of safety glasses with side shields. Side shields must meet applicable standards for quality, durability and protective ability.
3. When face shields are used, safety glasses with side shields or equivalent must be worn beneath the face shield.
4. Goggles or other tight-fitting eyewear (i.e., spoggles) that creates a seal around the eyes shall be worn when safety glasses with side shields will not provide adequate protection.
5. Goggles must be worn where a risk of chemical splash is present. Unvented goggles should be used where chemicals that are eye irritants and have high vapor pressures are used. Goggles may be worn over regular prescription glasses to provide protection equivalent to safety glasses with side shields.
6. Welders and helpers (e.g., fire watch) must use shaded eye protection appropriate for the type of welding or cutting being performed. A table with minimum shade requirements can be found in OSHA regulations at 29CFR1910.133 (a) (5).
7. Work with lasers above Class IIIa requires eye protection as determined by the laser safety officer assigned to evaluate the work and shall be appropriate to the type of laser and energy of the beam being used.
8. When using a portable grinder, the operator shall wear sealing eyewear such as spoggles or goggles, and an impact-rated face shield.

Head Protection

1. Hard hats and suspensions should be replaced periodically.
2. Head protection is required when workers are exposed to injury due to falling objects, electric shock, burns or other head injury hazards.

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3. Hard hats or helmets must meet or exceed acceptable standards such as ANSI Z89.1.
4. Where electrical hazards are present, hard hats must be nonconductive and meet requirements for Class E (up to 20 kV).
5. Hard hat caps (partial brim) shall be worn in the forward orientation unless other PPE is in use that requires the helmet to be reversed and the orientation is made in accordance with manufacturer requirements.

Foot Protection

1. ANSI/ASTM-rated protective footwear (safety boots or shoes) shall be worn where there is danger of foot injury from falling and rolling objects or from objects that might pierce the foot.
2. Safety boots/shoes shall have safety toe caps (metallic or composite) and steel shanks when protection against penetration is required. Clip-on protective toe caps may be worn over regular shoes.
3. Wear protective footwear as required by business unit procedures.

Hand Protection

1. Gloves appropriate to the task shall be used to protect the hands and wrist area from exposure to heat, cold, water or steam, chemicals, sharp edges, abrasion, electrical contact and other hazards as necessary. Gloves may also be used for protection against dirt and other nontoxic materials, as well as while operating machinery or handling materials.
2. When protection is needed from cuts and punctures, ensure gloves with the proper cut and puncture resistance are selected and used.
3. Inspect hand protection prior to use, ensuring that it is appropriate for the task and in suitable condition. Consider that additional inspections or tests may be set forth by standards for certain gloves prior to use.
4. Consult with H&S/EHS professional for assistance in selecting proper hand protection.

Clothing/Body Protection

1. Protective clothing shall be used when hazards that could cause injury to the body are present, including but not limited to arc flash, chemical exposure, protection against cold or heat, exposure to dirt, dust or liquids or exposure to hazardous materials in an emergency response action.
2. Don and doff protective clothing in a manner that prevents contaminating your body. Dispose of contaminated clothing properly.

Traffic Vests/Rain Gear

1. A minimum of Class 2 traffic vest/rain gear apparel shall be worn while working on the ground within the road right of way or any other work location where visibility of worker is needed for protection.
2. Class 3 traffic vest/rain gear apparel shall be worn when flagging traffic within the road right of way. Class 2 traffic vest/rain gear is acceptable for flagging traffic inside company-controlled access zones not related to public roads.
3. Flame-resistant reflective vests/rain gear apparel shall be worn when there is potential to be exposed to an arc, flash or contact.
4. Traffic vest/rain gear apparel shall be visible at a minimum distance of 1,000 feet.
5. When traffic vest or rain gear apparel is required to be worn, it must be securely closed using zippers, Velcro, etc.

Powered Industrial Trucks

1. Powered industrial trucks shall be operated in accordance with manufacturer's recommendations.
2. Only trained and authorized operators shall be permitted to operate a power lift truck.
3. Powered industrial trucks shall be inspected before being placed in service each day or before each shift. Document inspection.
4. A powered industrial truck shall not be used when deficiencies are found. The truck shall be repaired before it is placed back into service.
5. The rated capacity shall not be exceeded. When these vehicles are equipped with supports, they shall be utilized to prevent tipping.
6. When the powered industrial truck is parked or left unattended, the operator shall place the forks flat on the floor, neutralize controls, turn off the ignition key, set hand brake and follow business unit-specific policies or procedures to prevent unauthorized use.
7. While loading or unloading material, forks may be left in the raised position if there is an ergonomic concern and precautions have been discussed.
8. The operator may leave the seat of a parked and running powered industrial truck only if:
    a. the operator is in line of sight
    b. the operator is within 25 feet of the controls
    c. the powered industrial truck is secured

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   d. the forks are flat on the floor
   e. the controls are neutralized
   f. the hand brake is set

9. Modifications shall not be made without written approval from the manufacturer.
10. Attachments must be considered as part of the load and used in accordance with manufacturer's recommendation.
11. Free-rigging (suspending load from the tines with a sling) is prohibited.
12. Operator shall not allow anyone to stand or pass under the elevated portion of any powered industrial truck.
13. The operator shall not descend a ramp or grade with the load in front.
14. Only the driver shall ride on the powered industrial truck, unless an additional seat is provided.
15. Powered industrial truck operators and passengers shall wear seat belts where provided.
16. The operator of an order picker or high-lift rider truck shall wear approved fall protection and lanyard. The lanyard shall be attached to an approved overhead member of the platform at a point located above and near the center of the platform.
17. When forward view is obstructed, the driver shall travel with the load trailing or get assistance.
18. Operators shall properly stack and balance loads for transporting.
19. Operators shall travel with forks or load as close to the ground as possible and spread the forks for proper balance and stability of load.
20. Arms, legs and feet shall remain inside the powered industrial truck when moving.
21. Hands and arms shall not be extended through openings in the mast area.
22. The charger shall be turned off before connecting or disconnecting from an electric powered industrial truck.
23. Before driving or operating a powered industrial truck inside or on a truck or trailer, those trucks or trailers shall be secured and wheels chocked.
24. Travel speeds shall be dictated by surface conditions.
25. The horn shall be sounded before going through doorways or around blind corners.
26. Plan routes to avoid hazards of overhead power lines. Use spotters when necessary.

Spotters

1. A spotter shall be used when a powered industrial truck's operator field of vision is restricted or a blind spot prevents safe operation.
2. The spotter must be visible and remain visible to powered industrial truck operators when the equipment is moving. Everyone in the work area should be aware of his/her role and the desired path for equipment movement. Comply with business unit-specific spotter requirements if applicable.
3. If the operator cannot see the spotter, the operator must stop moving immediately and not resume movement until visual contact has been reestablished.
4. Spotters must be aware of their surroundings and never walk directly into the path of a moving vehicle or in areas of close proximity between the vehicle and the equipment.
5. Both the spotter and operator should agree on the communication method and use it consistently while working, incorporating an emergency communication method (e.g., whistles, hand signals).
6. Spotters shall have no other assignments that interfere with their ability to perform their spotter duties.

Pressurized Equipment

1. Before dismantling valves, flanges, access covers and similar apparatus associated with boilers, pressure vessels or pressure piping, the pressure shall be relieved and the equipment adequately drained. Follow lockout/tagout and line-breaking procedures as appropriate.
2. When an access cover to a pressurized system is bolted into place, a minimum of two loosened bolts should be left in the cover prior to breaking the seal. Additional bolts may be required based on expected pressures and shock load. When possible, the bolts shall be opposing.

Preventing Slips, Trips and Falls on Same Level

1. To avoid slips, trips and falls on same level, workers should consider these best practices:
    a. When walking, workers should identify all pathway hazards and take the necessary precautions to avoid injuries associated with overhead obstacles and slip and trip hazards. These hazards may include: shelving, awnings, limbs, thresholds, floor matting, uneven ground, elevated curbing, and icy or wet surfaces.
    b. Wear footwear that is in good condition and correct for the situation.

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Consider using high-traction footwear on icy surfaces where appropriate. Remove high-traction slip-on footwear prior to driving and before entering buildings.

c. At building entrances, where provided, use adequate means to clean water/snow/debris from shoes.
d. When on potentially slick surfaces, slow down, take shorter steps and keep center of gravity within stride.
e. Keep walking surfaces clean. Address spills immediately.
f. Do not walk with an obstructed view.
g. On stairs, hold handrails where available.
h. Avoid multitasking while walking. This includes dialing/texting on cellphones and mobile devices or reading materials.

Radio Frequency Devices and Exposure

Exposure

1. Radio frequency (RF) exposures from cellphone antennas on transmission towers, antennas on rooftops, wireless devices on distribution poles and other devices shall be evaluated and communicated to exposed workers.
2. Workers shall obey all posted signs, assume all antennas are active and maintain safe distance from all antennas.

Wireless Devices and Antennas on Poles and Streetlights

1. The following procedures shall be followed when working on or around distribution or streetlight poles with antennas or wireless equipment attached:
    a. Look for signage related to “Radio Frequency Exposure.”
    b. Where no signage or a blue “Notice” signage exists, workers may work without restrictions.
    c. Where yellow “Caution” or red “Warning” signage exists, workers shall:
        i. Turn off the power to the wireless devices at the disconnect switch.
        ii. Complete all work on the pole.
        iii. Turn on the power to the wireless device at the disconnect switch.

Transmission Towers with Antennas and Microwave Devices Attached

1. Minimum requirements for ascending towers:

   a. Do not climb towers with unsafe accumulations of snow and ice.
   b. At least two workers shall be on-site when anyone is on a tower.
   c. At least two workers trained in first aid/CPR shall be on-site.
   d. Workers shall inspect base of tower prior to climbing.
   e. Complete Communication Tower Daily Safety Checklist initially and every day work is performed (required by North Carolina OSHA only).
   f. Fall protection is required at heights above 4 feet.

2. Before beginning work on a transmission tower with cellphone antennas or microwave devices, workers or supervisors planning the work shall:
   a. Assess potential RF hazards. Identify type of antennas on the tower.
   b. Determine if RF Safe Distances described below can be maintained for the work planned. If RF Safe Distances cannot be maintained, contact supervisor to have antennas de-energized.
   c. Ensure yellow RF Caution signs are in place at the base of the tower.
   d. Be trained and authorized to perform the work.
   e. Obey all posted signs.
   f. Assume all antennas are active.
3. Workers shall maintain safe distances from antennas. Safe distances from transmitting antennas have been determined by calculation and field measurements. The following are working distances by workers spending extended periods of time in the area. Momentary passage through an area, as would be the case of a worker climbing a tower, has been determined to be safe.

Panel Antennas

Maintain a 6-foot working distance from the front of panel antennas. Being less than 6 feet from the side, bottom, top and back of the antennas is not a hazard.

Omnidirectional Vertical Antennas

Maintain a 4-foot safe working distances around omnidirectional antennas in all directions.

Microwave Dish Antennas

Microwave dish antennas are to be assessed on an individual basis. Do not look directly into a microwave dish or into an open waveguide.

Workers are not to access areas within 10 feet of the front of a transmit microwave dish in the main beam of the signal. If work is to be performed in these areas, contact the Telecommunications department.

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If antennas are on the tower, assume a yellow RF Caution environment exists even if the yellow RF Caution signs are not in place at the base of the tower. Notify supervisor if the signs are not posted at the location.

Antennas and Microwaves on Rooftops

1. Obey all signs and barricades. Contact Duke Energy Telecommunications department for assistance.

Radio Use in Generating Facilities

1. When adding new equipment or performing services that require a radio at a Duke Energy generation station, the radio frequencies to be utilized by the equipment must be approved by the Duke Energy job sponsor (engineer, project manager, equipment owner, etc.) prior to installation or use.

Radiography

1. Radiation restricted areas must be established by the radiographer before exposing the radiation source.
2. Restricted areas must be identified by magenta and yellow “DANGER RADIATION” signs and tape. Radiation signs should be posted in English and Spanish if the workforce includes non-English speaking persons.
3. Only authorized individuals are permitted inside established radiation restricted areas.
4. Authorized individuals entering radiation restricted areas must be appropriately trained and have personal radiation dosimeters (e.g., film badges, alarming rate meters).

Respirable Crystalline Silica

1. Identify activities that may disturb respirable crystalline silica materials. Activities of concern include:
    a. Disturbance (e.g., cutting, sawing, drilling, jackhammering, chipping) of concrete, cement, bricks or asphalt
    b. Demolition of structures that contain concrete, cement or bricks
    c. Activities involving coal or fly ash that cause significant dust releases
2. Contact H&S/EHS professional to determine if respirable crystalline silica exposure exists.
3. Do not perform respirable crystalline silica disturbing activities unless you are properly trained.
4. Never aggressively shovel or sweep fly ash, slag, cement, concrete or coal dust.

5. When indicated by exposure assessments, use exposure control methods, including:
    a. Engineering controls (such as wetting, tools with shrouds and vacuum attachments).
    b. Limiting access into work areas for activities where the exposure limit might be exceeded.
    c. Use of respiratory protection, where required. Contact H&S/EHS professional to determine if needed.
    d. Follow all company and/or location-specific requirements of the respiratory protection program.
6. Empty or change high-efficiency particulate air (HEPA) vacuums/filters in accordance with department or site work practice to minimize exposure and to ensure appropriate waste disposal.
7. Never blow off, shake off or do anything else to contaminated materials that could send dust or particulate debris into the air.
8. Certain activities, such as eating, drinking, smoking, chewing gum or tobacco, or applying cosmetics, are prohibited in areas where respirable crystalline silica exposure can occur.

Respiratory Protection

1. Use acceptable engineering controls to control harmful gases, smokes, dust, mist, sprays or vapors as the primary method to prevent airborne exposures.
2. If engineering controls are not feasible, appropriate respiratory protection must be used.
3. Before using a respirator or being fit tested, the user must receive training and medical clearance.
4. Any time a respirator is worn, the wearer must be free of facial hair between the respirator sealing surface and the face or any that may interfere with valve function. Note: Employees are not required to be clean-shaven while donning respirators during training activities, if the employees are not training in a hazardous atmosphere that would require the use of a respirator. Business units may determine whether employees are required to be clean-shaven while donning a respirator during training activities if the training does not take place in a hazardous atmosphere that would require the use of a respirator.
5. The user must inspect respirator before use.
6. The user must perform positive and/or negative pressure fit checks each time the respirator is put on to verify proper fit and operation.

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7. The user must exit the contaminated work area and remove respirator if it malfunctions or any breathing difficulty occurs, or they can taste or smell any contaminant.
8. Do not leave respirators unprotected in contaminated work areas during breaks or intermittent work.
9. The user must be familiar with proper care and cleaning requirements for the type of respirator being used.
10. Store the respirator in a sealed plastic bag.
11. Do not store respirators on their sealing surface. This may distort the face mask.
12. Where appropriate, the worker must return issued respirators to the designated collection point after each use for proper cleaning, inspection, maintenance and storage.
13. Employees must only use company-approved respiratory protection.
14. The employee will notify his or her supervisor, an H&S/EHS professional or management of medical changes or concerns when wearing respiratory equipment.
15. Follow appropriate instructions for voluntary use of disposable respirators where it has been determined that no respiratory hazard exists.
16. Provide suggestions, feedback and concerns about the respirator program or your respirator to your supervisor or H&S/EHS professional.

Safety Signs and Barricades

1. Make safety signs readily visible at all times when a hazard exists.
2. Use safety signs that can withstand the environment.
3. Use red “Danger” tape and signs to warn of immediate hazards. Locate tape and signs at such distance from the hazard that persons cannot accidentally come into contact with the hazard.
4. Use yellow “Caution” tape and signs to designate potential hazard areas. Locate tape and signs so as to restrict access into the area.
5. When using warning tape or signs if the hazard is not obvious, add specific information about the hazard and the name of a contact person.
6. When working within a posted area, use the personal protective equipment necessary to avoid injury.
7. Neatly install the ribbon/tape and keep it intact as long as the hazard exists.
8. Promptly remove the ribbon/tape and/or signs when the hazardous condition is corrected.

9. Do not use ribbon/tape as a barrier to prevent a fall or in place of a required guardrail. If ribbon/tape is used to warn of a floor opening or hole, ensure a person is continuously stationed at the opening to prevent accidental entry.
10. In switchyards, use red Danger tape to designate energized areas next to work areas. Attach red tape or red flags to designate safe heights on structures, columns or poles above, which workers may not climb because of energized equipment or circuits.

Scaffolds

1. Workers who perform work from and who erect or disassemble scaffolds shall be trained and supervised by a competent person.
2. Inspect all scaffolds before use or at the start of each work shift. Pay attention to footing, planking, guardrails, mesh, connections, weld, rust, ladder condition or other access means.
3. After significant events that could affect a scaffold's structural integrity, re-inspect the scaffold.
4. Read and follow instructions on scaffold tags.
5. Use personal fall arrest equipment on supported scaffold platforms higher than 6 feet if guardrails are not installed.
6. Use personal fall arrest equipment and a guardrail system on single-point and two-point adjustable suspension scaffolds.
7. Use personal fall arrest equipment on boatswain's chairs, catenary scaffolds, float scaffolds, needle beam scaffolds and ladder jack scaffolds.
8. Where tools, materials or equipment falling from a scaffold could strike workers below, do one or more of the following:
    a. Install a toe board or equivalent.
    b. Mark the area below with caution tape or install barricades to prohibit workers from entering.
    c. If materials are piled higher than the toe board, use a screen around the perimeter of the scaffold.
9. Do not use a defective scaffold. Ensure a WARNING, DO NOT USE tag is put on a faulty scaffold until it is repaired.
10. Use the access provided with the scaffold.
11. Do not accumulate too many tools, materials and debris on the scaffold or overload it beyond the rated capacity.
12. Do not stand on or lay tools, materials and equipment on any scaffold railing.

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13. To prevent tipping, do not load a scaffold unevenly.
14. During high winds and storms, do not work on an outside scaffold.
15. Do not work on an ice-coated scaffold.
16. Ensure mobile scaffolds rest on a solid, level footing.
17. If wheels or casters are provided, lock them to prevent accidental movement on scaffolds.
18. When asked to move mobile scaffolds:
    a. Get help so the pressure of pushing is not concentrated in one point on the scaffold.
    b. Ensure scaffold will move across level floors that are also free of obstructions and openings.
    c. Apply force close to (5 feet or less) the base of the scaffold.
19. Workers may “ride” a mobile scaffold if all of the following conditions are met:
    a. The floor or surface the scaffold will travel on must be within 3 degrees of level and free of obstructions and openings.
    b. The minimum dimension of the base of the scaffold must be at least half its height.
    c. Tools, materials, equipment and debris must be removed before the scaffold is moved.
    d. Workers riding the scaffold know about the movement of the scaffold.
    e. Workers are not on any part of the scaffold that extends over wheels, casters and other supports.
20. For other than mobile scaffolds, scaffolds shall not be moved horizontally while workers are on them, unless they have been designed by a registered professional engineer specifically for such movement.
21. Before performing work from a scaffold, identify and safeguard any electrical hazards.
22. When moving mobile scaffolding, avoid obstructions including piping equipment, instrumentation, electrical lines, etc.
23. Keep the scaffold’s working surface clean and organized at all times.
24. When the job is completed, remove all items from the scaffold.
25. Do not erect, use, dismantle, alter or move scaffolds so that they or any conductive material handled on them comes closer to exposed and energized power lines than:

   a. For lines under 300 volts – 3 feet
   b. For lines over 300 volts under 50 kV – 10 feet
   c. For lines over 50 kV – 10 feet plus 4 inches for every 10 kV over 50 kV

26. Do not use the frame on welded tubular frame-supported scaffolds to access upper levels unless the distance between the integral rungs is 16 ¾ inches or less and the rungs are uniformly spaced. Otherwise, use an extension or hook-on ladder.
27. Use tag lines or equivalent measures to control swinging loads when they are being hoisted onto or near scaffolds or if the loads could contact the scaffold.
28. On suspension scaffolds inspect wire ropes for defects before each work shift and after any occurrence that could affect a rope's integrity. Replace ropes if:
    a. Any physical damage impairs the function and strength of the rope.
    b. Kinks impair the tracking or wrapping of rope around the drum(s) or sheave(s).
    c. Six randomly distributed broken wires are in one rope lay, or three broken wires are in one strand in one rope lay.
    d. Abrasion, corrosion, scrubbing, flattening or peening causes the loss of more than one-third of the original diameter of the outside wires.
    e. Heat damage caused by a torch or any damage caused by contact with electrical wires is evident.
    f. The secondary brake has been activated and has engaged the suspension rope during an over-speed condition.
29. Ensure that wire rope clips on suspension scaffolds are inspected and retightened to the manufacturer's specifications at the start of each work shift.
30. Ensure the wire ropes on suspension scaffolds are shielded, treated to resist corrosive substances or made of material that is not adversely affected by the substance being used when acids and other corrosive substances are used.
31. Shield suspension ropes when performing a heat-producing process.
32. Reduce the possibility of welding current arcing through the suspension wire rope when welding is performed on suspension scaffolds by ensuring the following:
    a. An insulated thimble is used to attach each suspension wire rope to its hanging support.
    b. Excess suspension wire rope and any additional independent lines from grounding are insulated.

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c. The suspension wire rope is covered with insulating material extending at least 4 feet above the hoist.
d. The tail line below the hoist is insulated to prevent contact between it and the platform.
e. The portion of the tail line that hangs free below the scaffold is guided and/or restrained so that it does not become grounded.
f. Each hoist is covered with insulated protective covers.
g. In addition to a work lead attachment required by the welding process, a grounding conductor is connected from the scaffold to the structure.
h. Ensure that the size of this conductor is at least the size of the welding process work lead, and this conductor is not in series with the welding process of the work piece.
i. If the scaffold grounding lead is disconnected at any time, the welding machine is shut off.
j. An active welding rod or uninsulated welding lead is not allowed to contact the scaffold or its suspension system.

Telecommunications Hazards

1. Optical fibers shall never be examined with the naked eye.
2. Ultraviolet-curing lamps shall only be used in designated holders. Special ultraviolet-proof glasses shall be worn when it is necessary to examine energized lamps.
3. When splicing optical ground wire (OPGW) cable, the cable shall be grounded to the transmission structure ground.
4. Bare wire communication conductors on power poles or structures shall be treated as energized lines unless protected by insulating materials.
5. Rubber matting and appropriate rubber gloves with glove protectors shall be used when working on the telephone company side of the high-voltage isolation equipment.
6. Fiber-optic splicing glass shards shall be removed from worksites daily.

Tools

General

1. Work only with safe, approved tools and equipment properly suited for the job. Use tools in accordance with manufacturer's recommendations.
2. Modification and fabrication of tools and equipment shall be made only if approved. Consult engineering and business unit-specific procedures as appropriate.
3. Inspect tools before use.
4. Handle tools so that contact with exposed energized conductors is avoided.
5. Use proper tools for aligning bolt holes. Remove bolts or rivets in a controlled manner that will prevent injury or damage.
6. Do not tamper with safety switches. Do not modify, tape, bypass or otherwise defeat critical safety functions of devices, including dead man switches and other safety switches.

Grinders

1. Use proper personal protective equipment. Utilize manufacturer's safeguards.
2. Portable grinders shall be equipped with constant pressure control switches.
3. Keep grinding wheels properly dressed and turning true. Use the correct grinding wheel.
4. Inspect grinding wheels before use.
5. Immediately before mounting, all wheels shall be closely inspected and sounded by the user (ring test) to make sure they have not been damaged.
6. Secure bench grinders properly. Adjust work rest to within 1/8 inch to the grinding wheel. Adjust tongue guard to no more than 1/4 inch between wheel and guard.
7. Drills, grinders and sanders shall not be placed on work surfaces or handed to another worker until the equipment has stopped rotating.

Chain saws

1. Follow manufacturer's instructions for operation and adjustment.
2. Wear PPE including hard hat, goggles or face shield with safety glasses, hearing protection, hand protection and foot protection.
3. Wear leg protection (chaps) when using a chain saw while standing on the ground. Does not apply to pole saws.
4. Chain saws, when not in use, shall be properly protected with a blade guard or case.

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5. Gasoline-powered chain saws shall be equipped with a continuous pressure throttle control system that will stop the chain when pressure on the throttle is released.
6. The chain saw shall be started with the chain brake engaged.
7. Chain saw shall not be operated with one hand.
8. Do not cut any material other than wood.
9. Ensure that the chain saw chain does not contact any obstruction such as other logs or branches, rocks, fencing, nails or spikes.
10. Avoid contact of the bar tip with any object to reduce kick-back hazards.
11. The chain saw shall be started at least 10 feet from the fueling area.

Hand tools

1. Keep knives and other cutting tools properly stored when not in use. When using knives and cutting tools, use proper hand protection and avoid drawing the knife/cutting tool toward you or others.
2. Tools with sharp edges should have guards when stored or in transit to work locations.
3. When using a screwdriver or any tool with puncture capability, do not hold work in the hand in such a manner that slipping will cause injury.
4. Tools shall not be painted, taped or modified in any way that would prohibit inspection for visual cracks or defects.
5. Coverings on the handles of tools shall not be considered as electrical insulation or as a substitute for voltage-rated gloves.
6. Chisels, punches, drifts and similar tools shall be kept free of mushroomed heads to prevent flying fragments.
7. Nonconductive measuring devices shall be used when taking measurements on or near exposed energized electrical conductors or equipment.
8. Handles shall be installed on files and rasps before use.
9. Open-end wrenches shall be used with force in the proper direction. Do not overload wrenches by using extensions. Ensure good footing when using wrenches to prevent slipping.
10. When using a striking face (slugging) wrench, use caution. The wrench, when being struck by a hammer, shall not be held by the hand or foot. Use a properly tied rope or other practical means to secure the wrench.
11. Do not throw tools between other workers or locations. Use a tool bucket or other suitable container and handline to raise or lower tools between elevations.

12. Use diagonal or side cutters or other suitable tool to cut tie wraps.
13. Use of personal knives is prohibited for work-related tasks. Use appropriate cutting tools.

Transmission Substation/Switchyard Entry Requirements

This section applies to all Duke Energy Transmission employees and contractors. It defines requirements for entry into Duke Energy substations and switchyards. Additional information can be found in the Site Entry Policy for Transmission Substations and Switchyards document.

Job Briefing, Personal Protective Equipment, and Jewelry and Conductive Articles

1. Follow requirements as described in this Duke Energy EHS Handbook and/or applicable business unit Work Methods or procedures. Contractors are to comply with this Duke Energy EHS Handbook and applicable business unit EHS Supplemental Requirements and Contractor-owned programs or procedures.

Training/Escort

1. Individuals requesting unescorted physical access to substations and/or switchyards must complete the required Substation Entry Training provided in that region. However, training alone should not be relied upon as the sole determination that an individual understands the hazards that exist within a substation but is one prerequisite to granting unescorted access.
    a. Qualified persons that have Substation Entry Training as part of their job qualifications and routine training are exempt from this requirement.
    b. Training completion must be able to be verified through the applicable learning management system as being completed.
    c. Contracted companies shall ensure all employees/subcontractors who require unescorted entry to substations are trained to comply with OSHA 1910.269(a)(2)(i) and (ii) and supply documentation upon request.
2. Site-specific training may also be required for certain generation sites, customer sites or other specialized sites. Always check with local management for additional training or entry qualification prior to arriving on-site.

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3. Individuals who do not meet the training and experience requirements for unescorted access shall be permitted on-site as long as they are escorted by a qualified individual during the duration of their visit.
4. Lock substation entrances that are not observed by an attendant.

Notification of Arrival/Departure

Individuals are required to immediately notify an employee in charge (EIC) or site contact if applicable upon arrival and departure if other work is taking place at the substation. Individuals should consult the EIC or site contact regarding other work taking place in the substation and review the pre-job briefs (PJB) of the other work taking place to ensure understanding of all work taking place and potential impact to the work or task being completed by the individual or group arriving on the site.

1. Upon arrival, if no other group is already on-site, the EIC of the group shall hang a sign on the entry gate that indicates the EIC name and contact number in order to alert future individuals arriving on-site that there are already individuals working on the site. If other workgroups arrive to perform work, the workgroup leads will determine who will act as the overall Site Contact for the site. Note that only a single sign is required with a single name and number indicating the EIC or Site Contact.
2. Employee-In-Charge or designated primary contact person is required to provide notification to the appropriate control centers immediately upon arrival at a site as well as immediately upon departure from the same site through the Substation Integrated Voice Response (IVR) system. The IVR system can be reached by dialing 800.820.9365. Note: This does not replace the requirement for additional contacts that may be needed for coordination of work with any Transmission Control Center, Distribution Control Center, Plant Control Room or other operating authority.
3. Specific locations may have additional site-specific procedures for arrival and departure such as local alarms that may need to be disabled/enabled upon arrival/departure.

Vehicle Movement

1. Upon entering the gate of any substation, a travel plan discussion must be initiated. If the substation is occupied, notification to the lead person must be made upon entry. Prior to entering, ensure the substation gate is secured so it does not swing into the moving vehicle. The travel plan discussion must include the proposed drive path, the vehicle positioning plan and the obstacles to maneuver. Only vehicles needed for the job task should enter the substation.
2. A 360-degree view of all sides, front and back, over and under, of the vehicle shall be made just prior to vehicle movement.

3. A vehicle guide shall be used when available if vehicle is operated within 10 feet of any device within the substation fence. Backing should be avoided unless:
    a. It is the safest option
    b. All hazards are identified and avoided while backing
    c. The vehicle is operated at a safe speed

    When backing, all applicable company rules must be followed.
4. A maximum travel speed of 10 mph shall be observed.
5. Duke Energy Transmission employees must adhere to this Substation Driving Plan (GDLP-SAF-TRM-00006 Substation Driving Plan).

NERC Critical Infrastructure Protection (CIP) Qualifications

1. All personnel entering a Duke Energy substation or switchyard are required to follow all applicable Duke Energy Transmission requirements.

Ground Inspection Requirements Upon Entering a Substation or Switchyard

1. All personnel entering a Duke Energy substation or switchyard are required to follow the “Duke Energy – Transmission Missing Grounds Policy.”

Traveling Under Power Lines on Projects

1. This section does not apply to travel for the purpose of accessing work on electric transmission and distribution lines and equipment to be worked on. When power lines are above or adjacent to haul or designated routes and an assessment has determined that the max height of equipment (unstowed position) would not breach the minimum approach distance listed in the EHS Handbook, goal posts may be excluded from this requirement with documented Duke Energy approval.
2. Warning signs and goal posts shall be installed on haul roads and designated routes where equipment travel below power lines is required for more than 30 days.
3. Goal posts shall be installed a minimum of 20 feet on each side of the power line(s) and high enough to maintain minimum OSHA required clearances per Table 2: Minimum Clearance Distances While Traveling With No Load and Boom/Mast Lowered.
4. When goal posts are required and there is not sufficient space to install goal post(s), then signage and designated spotters must be utilized when equipment is traveling the haul roads/designated route.

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Trenching and Excavations

Avoiding Underground Utility Dig-ins

1. Locate underground utilities prior to soil disturbance. Where appropriate, contact utility locating services. Examples include digging, trenching, excavating, post hole digging, drilling, augering, pile driving, grading, etc.
2. Follow the safe digging practices and business unit specific procedures.
   - Identify the presence of underground utilities.
     a. Contact 811 to secure valid locates.
     b. Obtain a positive response.
     c. Exercise due diligence.
        - Note: Substations are exempt from 2a and 2b. When digging in a substation, other effective means will be used to determine if energized wire or other underground hazards exist before digging.
   - Dig safely.
     a. Visually identify all marked underground facilities and determine tolerance zone.
     b. Soft dig in the tolerance zone to fully expose all facilities before utilizing any mechanized equipment. Soft digging does not include use of power tools. It can include vacuum excavation. It does not include probing soils with metal rods.
     c. Utilize the right tool for the job, including vacuum technology.
   - Evacuate and establish safe perimeter.
     a. Do not create ignition sources.
     b. Immediately report damages.

Trenching and Excavation Requirements

1. Ensure a trench and excavation permit is completed prior to trench and excavation activities. Business units may have equivalent alternatives.
2. Each worker in an excavation shall be protected from cave-in by an adequate protective system to consist of either sloping, shielding or shoring, except when:
   a. Excavations are made entirely in stable rock or
   b. Excavations are less than 5 feet in depth and there is no potential for cave-in.
3. A competent person is one who is capable of identifying existing and predictable hazards in trenching and excavations and who has authorization to take prompt corrective measures to eliminate them.

4. Regardless of the depth of the excavation, OSHA requires a competent person to inspect conditions at the site daily. Inspections must be made as frequently as necessary during the progress of the work, to assure that the hazards associated with excavations are eliminated, before workers can enter the trench/excavation.
5. If utilized, ladders must be secured and extend a minimum of 36 inches above the landing. Secure the ladder firmly at the top to prevent it from slipping sideways or the foot from slipping outward. If there is no structure to tie off to, use a stake in the ground.
6. For trenches 4 feet or deeper, provide a stairway, ladder or ramp and locate it so that no more than 25 feet of lateral travel is required to access.
7. Workers who are exposed to vehicular traffic must wear high-visibility vests or other garments.
8. Workers shall not work under loads handled by lifting or digging equipment.
9. A warning system must be utilized when mobile equipment is operated adjacent to an excavation, or when such equipment is required to approach the edge of an excavation, and the operator does not have clear and direct view of the edge.
10. If an excavation is deeper than 4 feet, or has the potential to have a hazardous atmosphere, atmospheric testing must be done before workers can enter.
11. When ventilation is used to reduce the level of atmospheric contaminants to an acceptable level, testing must be conducted as often as necessary to ensure continuing safety.
12. A written rescue plan shall be prepared when supplied-air required atmospheric conditions exist in an excavation or could reasonably be expected to develop. The rescue plan must correspond with the level of risk associated with the task and specify:
    a. Who will be called if a trench emergency should occur. On-site and/or off-site emergency and rescue services must be summoned in a timely manner.
    b. Who will provide necessary emergency services to rescued employees.
    c. Who will prevent unauthorized personnel from attempting a rescue.

    The rescue plan, if required, must be attached to the Permit or documented at the bottom or back of permit (or business unit equivalent form).
13. Emergency rescue equipment must be readily available where hazardous atmospheric conditions exist or can reasonably be expected to develop.

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14. Workers must be protected when working in excavations where water has accumulated or is accumulating. If water is controlled or prevented from accumulating using water removal equipment, the water removal equipment and operations shall be monitored by a competent person to ensure proper operation.
15. Daily inspections of excavations, adjacent areas and protective systems must be made by a competent person for evidence of a situation that could result in possible cave-ins, failure of protective systems, hazardous atmospheres or other hazardous conditions. The soil for all excavations should be treated as Type C soil unless the competent person has analyzed the soil and completed the Soil Analysis Form and attaches it to the permit or business unit equivalent form.
16. If evidence of a possible cave-in, failure in the protective system, hazardous atmosphere or other significant concerns are found, all affected workers must be removed from the hazardous exposure until rendered safe.

    Employees shall not be allowed in shields when shields are being installed, removed, or moved vertically.

    Employees must be protected from the hazards of cave-ins when entering or exiting the shield.

    Trench box must extend 18 inches above the vertical part of the excavation walls to keep dirt, rocks and other debris from rolling into the trench. If the top of the trench box is at ground level, however, the trench box does not have to extend 18 inches above grade.
17. Inspections shall be conducted by the competent person prior to the start of work and as needed throughout the shift. Inspections shall also be made after every rainstorm or other hazard-increasing occurrence. These inspections are only required when worker exposure can be reasonably anticipated.
18. Employees shall be protected from excavated or other materials or equipment that could pose a hazard by falling or rolling into excavations. To keep spoil piles and materials from falling into the trench, clear edges of excavations back to at least 2 feet.
19. Provide walkways where workers or equipment are permitted to cross over excavations that are greater than 30 inches in width. Use guardrails where walkways are 6 feet or more above lower levels.
20. Barricade open, unattended excavations.

Vegetation Management

Cardinal Vegetation Rules (CVR)

Follow the Cardinal Vegetation Rules (CVR) in Transmission and Customer Delivery.

1. Adhere to drop/fell zone procedures.
2. Utilize fall protection.
3. Adhere to minimum approach distances (MAD) and grounding requirements.
4. ✪ Adhere to Chainsaw Safety procedures.
5. ✪ Adhere to Chipper Safety procedures.

General

Vegetation management contractors shall comply with applicable OSHA standards and with ANSI Z133, Safety Requirements for Arboricultural Operations. In addition, this section applies.

Every worker is responsible for his or her own safety while on the job site and shall comply with all requirements.

Drop/Fell Zones

1. The drop zone is defined as the area beneath workers aloft involved in arboricultural operations and/or where the potential exists for struck-by injuries from falling objects.
2. Workers overhead shall visually check the drop zone prior to making the first cut.
3. Ground staff shall inform arborists/lineworkers aloft before entering the drop zone under the tree or operating noisy machinery such as chain saws and brush chippers, which could hinder communication.
4. In manual tree felling operations, non-involved workers shall be positioned at a distance from the tree at least two times the height of the tree or trunk being removed until the chain saw operator and/or crew leader communicates it is safe to approach more closely.
5. When felling a tree, the Manual Tree Felling Procedure steps defined by ANSI Z133 Arboricultural Operations shall be followed.
6. The drop/fell zone must be identified and secured. The drop/fell zone shall be marked by the use of suitable signs, cones and/or barriers.
7. A pre-established form of communication shall be agreed upon and documented in the Job Briefing prior to commencing with work aloft or manual tree felling.
8. All verbal, audible or visual communications between ground personnel and the arborists/saw operator require command-and-response.

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9. Communication can be whistle, hand signals, etc. and must be effective during the operation of equipment or machinery that hinders audible communication.
10. If more than one aerial lift is working within one span, for instance to access vegetation over or under the conductors, both lifts shall operate either above or below the conductors at one time.

Personal Protective Equipment

1. Contractors are to ensure the appropriate use of eye and face protection in accordance with ANSI Z87.1 and the Personal Protective Equipment section of the EHS Handbook. Exception: Contractors operating chain saws are not required to use face shields unless the need for an additional level of protection is identified during the hazard assessment.
2. Contractors regularly supporting Transmission and Distribution Vegetation Management shall use non-gauntlet work gloves when performing ground work. This includes, but is not limited to, activities that expose individuals to cuts and punctures. Examples include: sharpening blades, working on chain saws, handling materials and tools in beds of trucks, and handling timbers, logs or cut debris. Exception: Glove use when operating a chipper is at the discretion of the contractor. Due to the potential risk of snagging when gloves are used, they shall be non-gauntlet gloves.
3. Contractors regularly supporting work on Transmission and Distribution Vegetation Management shall wear protective footwear that complies with ASTM F2413. At least one shoe of each pair shall be clearly marked. Requirements include impact resistance (IR or I/75) and compression resistance (CR or C/75). In addition, footwear shall be: safety toe of either steel or composite; substantial over the ankle, lace-up uppers made of leather or other puncture-resistant material; non-slip soles with a substantial heel. Exception: When performing tasks that are subject to walking in water and wet land conditions, contractor is exempted where rubber boots or waders are used. Also, dedicated herbicide applicators shall use over-the-ankle boots and are not subject to other requirements.

Water Safety

Working Over or Near Water

1. Workers working over or near water, where the danger of drowning exists, shall be provided with U.S. Coast Guard-approved personal flotation devices (PFD).

2. When fall protection is used and will prevent persons from falling into water, personal flotation device is not required.
3. Prior to and after each use, personal flotation devices shall be inspected for defects that would alter their strength or buoyancy. Defective units shall not be used.
4. Ring buoys with at least 90 feet of line shall be provided and readily available for emergency rescue operations. Distance between ring buoys shall not exceed 200 feet.
5. At least one lifesaving skiff shall be immediately available at locations where workers are working over or adjacent to water.

Working on Watercraft/Boats

1. When in a watercraft, wear Coast Guard-approved personal flotation devices.
2. Prior to and after each use, the personal flotation devices shall be inspected for defects that would alter their strength or buoyancy. Defective units shall not be used.
3. Ensure operators have demonstrated skills and/or training to operate watercraft safely.
4. Ensure all watercraft meet U.S. Coast Guard requirements.
5. Ensure watercraft longer than 16 feet are equipped with a Coast Guard-approved throwable personal flotation device.
6. Prior to getting on or off boats, survey the boat ramp or dock for surface conditions. Look out for slippery surfaces and take precautions as appropriate.
7. Prior to use, ensure all gear and equipment is in good working condition.

Working on Barges

The following applies to barges used for maintenance and construction around dams and generating plants. It does not apply to coal barges.

1. When working on a barge, wear U.S. Coast Guard-approved Type I, II, III or V personal flotation devices. Inspect personal flotation device before and after each use. Note: Personal flotation devices must be available, but are not required, to be worn on structural barges with substantial guardrails.
2. Ensure availability of a Coast Guard-approved 30-inch life ring with 90 feet of line and at least one permanent ladder that will reach the surface of the water from the top of the barge.
3. When combustible materials are present, ensure fire extinguishers are readily available.
4. Ensure safe access/egress to and from barge.

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5. Ensure handrails are installed, unless their presence is a more significant safety hazard.
6. Ensure all barge connections are in place before operation (e.g., barge-to-barge, spud wells-to-barge and anchorage).
7. ✪ When operating a mobile crane on a barge, develop stability and lift plans in compliance with the requirements of OSHA 1926.1437 and ASME B30.8.
8. Ensure engineering has approved all barge modifications.
9. Designate and maintain unobstructed walkways/aisles at all times.
10. Use and maintain anti-skid material on walking/working surfaces.
11. Safely secure all material/equipment to the barge deck.
12. ✪ Mooring plans for barges and supporting watercraft shall account for changing water conditions and severe weather.
13. Ensure operators of equipment (e.g., cranes, drills) are familiar with the equipment and can demonstrate the skills necessary to safely operate it from an unstable platform (e.g., barges, dredges).

Diving

1. For diving requirements, see OSHA 29CFR 1910 Subpart T, Commercial Diving Operations.
2. Remotely Operated Vehicles (ROVs)
    a. The use of ROVs is the preferred means to perform underwater inspections and some diving operations.
    b. The use of ROVs eliminates the hazards to the human diver.
    c. Involve the site or business unit in any decision to place a diver in the water.

Welding and Thermal Cutting – Health Exposures

1. Do not clean with compressed air. Use HEPA vacuums or other means that do not disperse dust into the air.
2. Do not take contaminated protective clothing or equipment home.
3. Do not enter eating/drinking area with work clothing unless dust has been removed (in a manner that does not disperse dust into the air).
4. Whether working in the weld shop or in the field, position yourself so that your breathing zone is not in the fume plume, regardless of the relative hazard of the process. If work conditions prevent proper positioning, use ventilation to direct fumes away from welders, cutters and helpers.

5. When using welding screens around your work, arrange screens so that restrictions to ventilation are minimized.
6. Position the exhaust system takeoff near enough to the electrode to ensure maximum fume capture when using local exhaust ventilation.
7. Do not use oxygen for ventilation purposes.
8. Where ventilation or natural conditions do not control welding/cutting fumes from reaching the workers' breathing zones, use appropriate respirators.
9. When possible, remove all coatings by chemical or mechanical means (with local exhaust ventilation) prior to thermal cutting. Remove a "cut line" of at least a width of 3 to 4 inches. If coating removal is not possible, use ventilation and/or respiratory protection.
10. Prior to removing or disposing of coating materials, determine if the coating contains lead or hexavalent chromium. See your location H&S/EHS professional for guidance.
11. Before applying heat, verify that metal is clean of any solvents (particularly chlorinated solvents), oils or other materials that may cause hazardous decomposition products.
12. Do not perform welding or thermal cutting activities within 200 feet of the storage of chlorinated solvents.
13. Evaluate potential fume hazards. Review product SDS and/or hazard label, ventilation options and PPE. Coordinate with H&S/EHS professional to determine if adequate exposure assessments have been completed for the specific activity.

Work Zone Safety

General

1. Follow state and local regulations for establishing safe work zones.

✪ Temporary Traffic Control Zones

1. Temporary traffic control zones established by Duke Energy employees shall comply with the Duke Energy Work Zone Field Manual or the applicable Business Unit Work Zone Program annex and business unit-specific procedures.
2. Temporary traffic control zones established by the current Duke Energy contractors for Duke Energy employees shall comply with the current Federal Manual on Uniform Traffic Control Devices (MUTCD), applicable state and local work zone requirements, and the following:

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a. Temporary traffic control zones shall include traffic cones a minimum of 36 inches in height and a minimum of 12 pounds in weight.
b. A work zone whistle or air horn for flagger(s) shall be used as an audible alert system to warn of vehicles encroaching the work zone.
c. Flaggers shall use retro-reflective stop/slow paddles or retro-reflective red flags when permitted by state or local regulations. The paddles shall be 24 inches by 24 inches on a 7-foot staff.
d. Except in emergency situations, flagger stations shall be illuminated at night.

3. Notify your supervisor of issues related to work zone safety or to request assistance of a work zone contract company to provide work zone and/or flagging services in the event employees are not trained or do not have the required equipment to set up work zones in a manner to comply with all applicable federal, state, county, city, and local maintenance of traffic (MOT) requirements for the location of the work.

Worker Security in Public Places

1. Practice situational awareness and always have a mental action plan when arriving at the job site with respect to security hazards, including presence or signs of dogs, signs of aggression or hostile intent by customers or members of the public.
2. If a hostile event occurs with a customer, leave the scene immediately. For all life-threating incidents, call 911 immediately and report all incidents to your immediate supervisor.
3. In addition, for employees, ensure these events get reported to the Duke Energy Enterprise Security Command Center at 800.951.9924. Include any suspicious activity in or near Duke facilities. Contractors should contact their Duke Energy Point of Contact.

Environmental Sections

Aerosol Can Management

1. Aerosol cans are a hazardous waste at the time they are discarded or when they can no longer be used for their intended purpose, unless empty.
    a. Aerosol cans may be managed as a universal waste in states that have added aerosol cans to their list of universal waste materials.
2. The aerosol can is considered empty when all of the following conditions are met:
    a. If all material that can be removed is expelled
    b. If it has no more than 3% by weight of the total capacity of the container
    c. When the pressure of the aerosol can approaches atmospheric
3. Empty aerosol cans can be recycled as scrap metal or disposed in the regular trash.
4. Defective aerosols can be collected and stored as hazardous waste (or as universal waste if adopted by the state) in a Department of Transportation-approved container in a Satellite Accumulation Area (SAA) or other acceptable storage area. Do one of the following:
    a. Hazardous waste: Place a hazardous waste label on the storage container and mark it “Defective Aerosols for Disposal.”
    b. Universal waste (if adopted by the state): Place a universal waste label on the storage container and mark it “Universal Waste Aerosol Cans.” Date the container when the accumulation starts.
    c. To be punctured: Collect as nonhazardous waste to be punctured by a “qualified individual” to remove the contents and recycle the metal. Contact your field environmental/EHS professional to determine what constitutes a qualified individual.
5. Return defective aerosol cans to the manufacturer, if possible.
6. Treat defective aerosol cans that cannot be returned as hazardous waste or universal waste if applicable.

    The defective aerosol can must have a secure cap or the valve stem removed, and it must be lab packed for disposal.
7. Damaged aerosol cans may be transported to a location for proper disposal.
8. If you have questions about the proper handling of aerosol cans, contact your supervisor or field environmental/EHS professional.

Air Permitting

1. Proper air permits for the site must be secured prior to starting any work activities that may potentially result in new or modified air emissions.
2. Partner with an environmental/EHS professional to complete the Duke Energy Environmental Requirements Review Checklist (see page 135) to identify potential activities that could trigger the need for an air permit. Generally, the following activities need to be considered:
    a. New/modified stationary combustion sources (e.g., emergency generators, engines, boilers, turbines, etc.)
    b. New/modified material handling equipment (e.g., storage silos, tanks, transfer points, conveyors, etc.)
    c. New/modified sources of fugitive dust (e.g., storage piles, haul roads, material drop points, etc.)
    d. New/modified process with the potential to emit any pollutant to the atmosphere
3. Contact an environmental/EHS professional with any questions regarding the proposed project or activity, including whether existing permits cover the scope of work or if a modified permit will be required.
4. Construct and operate all equipment in accordance with the specifications and locations provided in the air permit application. If a change is needed, notify your supervisor and an environmental/EHS professional immediately.
5. Ensure that an environmental/EHS professional is provided copies of all documentation required in the permit (i.e., as-built drawings, certificates, etc.). Some permits allow for the construction of equipment and others allow for the operation of emission sources. Additionally, the permit may require notifications and submittals based on operation dates. Prior to initial operation of any piece of equipment addressed in a permit, contact an environmental/EHS professional to confirm proper permissions have been obtained.
6. Because the permit may require a number of notifications or submittals based on different project milestones, ensure that an environmental/EHS professional is included on all project communications related to schedule. Also, notify them of any changes or delays.
7. Route all regulatory correspondence or interpretation questions related to the project and/or permit through an environmental/EHS professional. Only designated environmental/EHS professionals are allowed to interface directly with regulatory personnel.

Containment Systems

1. Identify all oil/water separators and other containment systems at your location and know what drains into them and where they discharge.
2. Include oil/water separator operation inspections in operator rounds. Review regularly for any unusual conditions, alarms or petroleum discharge, where applicable.
3. Operate and maintain these systems as recommended by the manufacturer and document in the approved work management system.
4. Never use oil/water separators for oil storage.

Oil and Chemical Containment Structures

1. Conduct preventive maintenance of oil and chemical containment structures, including non-earthen berm containment structures that are greater than 55 gallons in volume, as prescribed within the work management system. At a minimum, a visual evaluation of the containment structure is to be performed annually and if the evaluation reveals deficiencies, further evaluations or repairs must be initiated.
2. Focus inspections and maintenance activities on risks associated with spills or unpermitted discharges.
3. All bulk storage container drainage valves must be securely sealed closed when not in use by qualified personnel. Manual valves for high-risk containments must be locked closed when containments are not being drained by qualified personnel.
4. Drain oil or chemical bulk storage containment areas in a manner to prevent an uncontrolled discharge.
5. Inspect secondary containments periodically (as specified in the Spill Prevention, Control and Countermeasure (SPCC) plan). Inspections must be performed on all oil sources, secondary containments, loading/unloading areas and spill response equipment, as specified in the SPCC plan. If the containment is not dry, the area must be inspected by a person trained in oil handling and the facility SPCC plan (i.e., eLearning activity ENC005 or equivalent) prior to opening the valve or emptying by pump, ejector, etc.
6. Document all inspections and containment draining activities. Inspections records must include the name of the inspector and date of the inspection.
7. Retain inspection records at the facility (or at a designated location for unmanned facilities) for a minimum of three years.
8. It is critical that facility personnel actively verify the position of manual drain valves located at high-risk containments. At fully staffed sites this shall

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be done and documented daily. At unstaffed generation sites they shall be checked and verified on a biweekly basis. At other unstaffed facilities they shall be checked and verified monthly. Containments are designated as high-risk if they meet one or more of the following criteria:

a. Listed as a large spill potential (LSP) containment in RRE
b. Located adjacent to a storm drain with unimpeded flow to waters of the U.S. (WOTUS)
c. Located within 125 feet of WOTUS
d. Stores chemicals where the storage volume exceeds the regulatory Reportable Quantity (RQ) or
e. Identified by the business unit

✪ Note: The use of engineered drainage devices and/or systems that prevent the passage of petroleum products but allow rainwater to discharge from spill containments may be used in lieu of or in addition to drain valves if approved by the business unit and the certifying professional engineer if associated with an SPCC plan.

Contaminated Soil

✪ 1. Any known soil contamination, environmental restrictions, recorded land use restrictions, engineering controls or environmental investigation results should be discussed with your supervisor, the field environmental/EHS professional and the environmental remediation professional prior to performing any clearing, grubbing, stripping, excavation, potholing, trenching or other land-disturbing activities.

✪ 2. During project planning for land-disturbing activities, partner with the field environmental/EHS professional to complete an Environmental Requirements Review Checklist (see General Environmental Requirement Review section).

✪ 3. Prior to disturbance of soil in a substation, the requirements of the Substation Soil Procedure (ADMP-ENV-EVS-00165) must be met.

✪ 4. Contact your supervisor, environmental/EHS professional and environmental remediation professional immediately if you encounter previously unknown contaminated soil during land-disturbing activities. If an environmental/EHS professional is not immediately available to respond to the discovery, contact the Spill Coordinator Hotline as described on page 157.

✪ 5. Any soil planned to be removed from a site must be properly tested and disposed of at a Duke Energy-approved facility. Contact the environmental/EHS professional or Waste SME for testing and disposal requirements.

6. Do not spread, stockpile, transport or dispose contaminated soil prior to contacting your supervisor or the field environmental/EHS professional.
7. A site Environmental, Health and Safety Plan must be reviewed and approved by an environmental/EHS professional before disturbing contaminated soil.

Cultural Resources

1. Cultural and archaeological resources include older cemeteries and monuments, historic buildings and structures (50-plus years) or their remains, historic or Native American sites, ancient artifacts (e.g., pottery, arrowheads, etc.), clusters of brick, stone or shell, rock paintings or rock carvings and other items of historic or archaeological value.
2. At all times, avoid or minimize the impacts to cultural resources due to land-disturbing activities such as new construction, expanding, clearing rights of way, etc.
3. Stop work immediately if you discover a cultural resource (e.g., grave, artifacts, archaeological site, human bones, etc.) while working on a Duke Energy project or property.
4. Do not handle or touch the discovered item and notify your supervisor and an environmental/EHS professional.
5. The environmental/EHS professional will provide the appropriate procedures, assist with Environmental Requirements Review, refer the appropriate Duke Energy professionals responsible for performing cultural resource evaluations, and provide further instruction as needed. There may also be state and/or federal reporting requirements.

Decommissioning/Demolition Activities

1. Prior to mobilizing a crew to complete decommissioning or demolition activities at a site or facility, partner with an environmental/EHS professional to complete an Environmental Review Checklist (see page 136). Also, review the sections in this Handbook on Used Oil and Waste Management and Disposal since those are common issues that arise during demolition projects.
2. Evaluate or address the following prior to beginning the job:
    a. Review activities that may affect surface waters, streams, rivers, wetlands or other sensitive environments to determine if permits will be needed. Follow all applicable regulations.
    b. Submit applicable permits and notifications prior to beginning work if project will disturb one or more acres of land.

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c. Complete an endangered species evaluation. Contact the field environmental/EHS professional for guidance.

d. Complete a hazardous material building inspection if warranted.

e. Develop a stormwater management plan and implement best practices, or develop a stormwater pollution prevention plan if required by permit.

f. Determine a process for interacting with an agency inspector.

g. Determine a process for handling material containing asbestos, including making any necessary notifications to the appropriate environmental/governmental agencies (county/state) and proper disposal.

h. Contact the field environmental/EHS professional before initiating a coal pile closure project to ensure that the appropriate procedures are incorporated and the necessary permits are obtained.

✪ i. Contact the environmental remediation professional prior to initiating demolition of a substation. Substation demolition must be performed in accordance with the Substation Retirement: Environmental Assessment and Remediation procedure (ADMP-ENV-EVS-00186).

j. Contact the field environmental/EHS professional before initiating the reuse of demolition materials to ensure appropriate procedures are incorporated and any necessary material tests or reuse approvals are obtained.

k. Determine a process to minimize fugitive dust.

l. Determine paint removal and sampling processes.

m. Determine piping inspection and removal processes.

n. Determine a proper storage and management process for solid and hazardous waste. Consider station generator status when planning work to eliminate hazardous waste that could accumulate over station limits.

3. Evaluate materials such as caulk, paint, mastics and sealants to determine PCB content through proper sampling procedures. Results will determine disposal requirements. Evaluate materials for lead and asbestos on a case-by-case basis. Contact the field environmental/EHS professional for guidance with material disposal testing requirements.

✪ 4. Evaluate and test oil from transformers and other oil-filled electrical equipment for PCB content. Any oil with PCB content greater than 50 parts per million (PPM) must be disposed of as PCB-containing oil.

5. Contractors are responsible for developing their own Spill Prevention Control and Countermeasure (SPCC) Plan for the decommissioning/demolition activity if there is not an existing SPCC plan in place or the plan is insufficient per 40 CFR 112. This should include the quantity of on-site petroleum, location, type of secondary containment, and frequency of secondary containment and container inspections.
6. Determine the location of all waste accumulation areas. Ensure that all required regulatory inspections are conducted. Work with an approved Duke Energy waste vendor to ensure proper disposal of all site wastes, including materials to be recycled.
7. Contact the field or project environmental/EHS professional for additional support.

Domestic Sewer Lift Stations

1. High-risk domestic sewer lift-stations shall be monitored continuously (level indication alarm in a control room or audible alarm for sites that are staffed 24/7). Such equipment must have preventive maintenance performed at a minimum frequency of three years to limit failure.
2. Domestic sewer lift-stations are designated as high-risk if they meet one or more of the following criteria:
    a. Located adjacent to a storm drain with unimpeded flow to waters of the U.S. (WOTUS)
    b. Located within 125 feet of WOTUS
    c. Identified by the business unit
3. Conduct preventive maintenance of all domestic sewer lift-stations, as prescribed within the work management system (maximum frequency of three years).
4. Focus inspections and maintenance activities on risks associated with spills or unpermitted discharges.
5. Drain domestic sewer lift-station areas in a manner to prevent an uncontrolled discharge.

Drinking Water

1. As practicable, drinking water at Duke Energy facilities should be purchased from municipal water sources and should not be supplied by Duke Energy-owned systems.
2. Where Duke Energy is the system owner and operator, the system is regulated and must follow specific regulations before being placed into

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service and during operation. The assigned field environmental/EHS professional must be consulted prior to the construction or modification of a drinking water system or if there are any questions regarding operation.

3. Construction, expansion or modification of a regulated drinking water system requires a permit before work begins. General system maintenance does not require a permit but may require follow-up disinfection and testing.
4. Operation permits may be required before placing a new drinking water system in operation.
5. Each regulated drinking water system is required to have a minimum of one certified operator assigned. The operator must hold a valid certificate issued by the State Environmental Certification Board, which is equivalent to or exceeds the class of the facility. Re-certification requirements vary.
6. It is the responsibility of the certified operator to be knowledgeable of federal, state and local regulations that are applicable to their system and ensure that the system is operated in compliance with these regulations. This includes verifying all materials used in construction, expansion, operation, treatment and maintenance of a system are certified by NSF International.
7. Systems are required to be sampled at representative locations in the system. Each system shall have a sampling plan that is reviewed and adjusted to describe compliance sampling required and the proper time period. Samples shall be taken in accordance with procedures acceptable to agency with regulatory authority.
8. Laboratories performing compliance analytical work MUST be certified for drinking water testing by appropriate agency and approved by Duke Energy.
9. Cross connection between potable and nonpotable systems is prohibited.
10. Appropriate backflow prevention is required between any public system and Duke Energy-owned systems.
11. Required records (which vary by state) must be kept at the system location.
12. If illness is suspected to have been caused by drinking water, contact the field environmental/EHS professional.

Environmental Requirements Review Checklist

General Environmental Review Requirements

1. Duke Energy complies with all regulations, statutes, permits obligations and policy requirements applicable to its existing and planned facilities. Environmental requirements can be complicated and often require interpretation. State and local permitting and regulatory requirements may

also be applicable. As a result, project managers, engineers and work leaders must consult with an environmental professional before starting a project.

2. The Environmental Requirements Review Checklist procedure (ADMP-ENV-EVS-00021) assists individuals in identifying potential environmental requirements when planning projects for new or existing facilities, process modifications and maintenance activities, or when making changes at Duke Energy facilities.
3. The checklist covers many compliance considerations such as chemical control, hazardous and nonhazardous waste management, air compliance and permitting for new construction, new waste water discharges, water permitting, drinking water, asbestos and lead abatement, stormwater erosion and sedimentation, wildlife management and demolition activities.

Completing an Environmental Requirements Review Checklist

1. Complete the checklist early in the project development process to ensure that any environmental permitting or design considerations can be included in the project scope and schedule. Contact an environmental/EHS professional to start the environmental review process.
2. Fill out the checklist based on knowledge of the project work scope.
3. Include work scope details in the checklist comments section or as an attachment (specifications, work plans, drawings, aerials, SDS sheets, etc.) that will provide environmental personnel with enough project detail to perform an adequate review.
4. Submit the completed checklist and supporting information to the field environmental/EHS professional for the project location. If a field environmental/EHS professional is not available, send the checklist directly to Environmental Services for review.
5. The field environmental/EHS professional will review the checklist and provide the project manager with timely feedback regarding any necessary actions to build into the project scope to ensure compliance from a permitting and facility compliance perspective. This feedback can be delivered in many forms but must be timely and accurate to avoid delays in project performance.
6. It is the field environmental/EHS professional's responsibility to involve Environmental Services subject matter experts (SME) as needed to ensure complete and accurate feedback to project management personnel.
7. Notify the field environmental/EHS professional if the project scope changes following the initial review. The project manager must ensure that any additional environmental compliance actions are identified and completed.

Environmentally Acceptable Lubricants (EALs)

1. Environmentally Acceptable Lubricants (EALs), if accidentally released into the environment, result in reduced environmental harm, compared to conventional hydrocarbon-based lubricants. Depending on the EAL product, spill reporting requirements vary considerably from non-reportable to similar to conventional lubricants. Fines and penalties resulting from spills of an EAL may be less severe than for the same release of conventional lubricants, though the regulatory response cannot be predicted with certainty.
2. There are three properties that an EAL must meet to be classified as "EAL compliant." The lubricant must have all three of these properties:
    a. Be readily biodegradable
    b. Have low aquatic toxicity
    c. Be non-bioaccumulative
3. Consider evaluating the following equipment for possible conversion to EALs:
    a. Equipment with a history of reportable lubricant spills
    b. Equipment over "Waters of the United States"
    c. Equipment within 100 feet of "Waters of the United States"
4. Changing the lubricant type in a component shall be considered a configuration change. Prior to converting a lubricant to an EAL, the following factors must be addressed to avoid problems as a result of the change:
    a. Is the new lubricant "EAL compliant"?
    b. Is the EAL listed as "Approved" in the Duke Energy Approved Chemical List?
    c. Does the EAL meet the lubrication needs of the machine?
    d. Will spill reporting requirements change?
    e. Is the new lubricant compatible with the existing lubricant?
    f. Are there potential changes in sampling, relubrication and filter replacement interval(s)?
    g. Are there potential changes to lubricant storage and handling requirements?
    h. What is the lubricant cost?
    i. Are there potential changes to contamination control measures (moisture, particulate, other lubricants)?
    j. Are there potential changes to waste disposal/handling procedures?
5. Contact your field environmental/EHS professional for guidance and recommendations on evaluating equipment for conversion to an EAL.

Fugitive Dust

1. All Duke Energy facilities have a general requirement to minimize fugitive dust emissions. Fugitive dust is particulate matter that becomes airborne and is not emitted from a stack or vent.
2. Follow documented fugitive dust management plans and procedures at all times. The field environmental/EHS professional can provide copies of plans upon request.
3. Take additional control measures if the existing controls are ineffective (e.g., changing site or atmospheric conditions).
4. Use Best Management Practices (BMPs) applicable to the job, project or site to reduce the possibility for fugitive dust to leave the property. The field environmental/EHS professional can provide a list of BMPs upon request. A few examples:
    a. Maintain good housekeeping throughout the plant/project site.
    b. Ensure material storage piles are properly stabilized.
    c. Consider using stabilizers if the material or pile will be inactive for an extended period of time. Contact the field environmental/EHS professional for a list of approved materials.
    d. Water the roads frequently enough to prevent dusting conditions. Implement an on-site speed limit to minimize the amount of dust generated from vehicle movement.
    e. Ensure that trucks and other material handling equipment (e.g., conveyors) are covered to minimize exposure to wind.
    f. Minimize exposed surfaces for storage piles. Items to consider include pile height, slopes and total area exposed.
5. Contact your supervisor or the field environmental/EHS professional for guidance and recommendations on controlling fugitive dust emissions.

Hazardous Material Transportation

1. Hazardous material pre-transportation functions pertain to Department of Transportation (DOT) hazardous materials processes that occur prior to the carrier being present and relate to the classification, packaging, filling, marking, labeling, loading of hazardous materials, preparing/reviewing/certifying shipping papers, and providing emergency response information. Hazardous materials transportation is any process while the carrier is present that involves the loading, segregating, placarding, movement, storage during transportation, and unloading of hazardous materials.

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2. A hazardous material is a substance or material that the Department of Transportation (DOT) has determined to pose an unreasonable risk to health, safety and property when in transport. Examples of DOT hazardous materials are:
   - Paints or solvents
   - PCBs (≥20ppm AND ≥1 pound)
   - Asbestos (≥ 1% friable)
   - Lithium and lead-acid batteries
   - Pesticides or herbicides (mosquito repellent)
   - Compressed gas cylinders (propane, sulfur hexafluoride)
   - Fuels (gasoline or diesel)
   - Laboratory chemicals
   - Acids or bases
   - Aerosols
   - PVC cement
   - Pressurized SF6 equipment (≥29 psig)
3. DOT regulations govern hazardous material (HAZMAT) pre-transportation and transportation. Employees and contractors performing these functions are considered DOT HAZMAT employees. Always consult business unit procedures for guidance before shipping equipment or storage containers that contain hazardous materials (e.g., compressed gas).
4. All DOT HAZMAT employees are required to have training that includes general awareness, safety, security awareness and function-specific training. Initial training must be given within 90 days of employment with Duke Energy.
5. Employees may perform HAZMAT transportation functions before completing the initial training if they are under the direct supervision of a properly trained and knowledgeable employee. All HAZMAT employees must receive recurrent training at least once every three years.
6. DOT's Materials of Trade (MOT) exception is widely used within Duke Energy. The MOT regulations provide exceptions from the hazardous materials regulations for certain small quantities of hazardous materials when transported and used by Duke Energy. Duke Energy requires individuals who ship materials, per the MOT regulations, are trained within 90 days of employment by Duke Energy and take refresher training at least once every three years. Duke Energy's MOT material cannot be transported by contract carriers (e.g., ACT, MRC Global).

7. Contact your supervisor or the field environmental/EHS professional for more information regarding hazardous material transportation.

Land-Disturbing Activities

General

1. Consult with an environmental/EHS professional when planning any land disturbance. Any use of land that results in a change in the natural cover or topography and that may cause erosion or contribute to sedimentation is considered a land-disturbing activity. This includes altering the quality and quantity of stormwater runoff.
2. Notify the field environmental/EHS professional of any regulatory site visits or inspections as soon as practical. Environmental Services will coordinate any communications with regulatory agencies (federal, state or local).
3. Ensure any site inspection documents are maintained and readily available on-site.
4. Review any permits to ensure that the required documentation and inspections are being completed and that there are clear owners for these activities.
5. Ensure that the erosion control measures are installed in accordance with approved plans. If changes are made or the project scope extends past the existing limits of disturbance, contact an environmental/EHS professional prior to implementing the change.

Erosion Control in Environmentally Sensitive Areas

1. Minimize any land-disturbing activities within the project area. No land grubbing or grading shall occur when working in environmentally sensitive areas such as streams and wetlands or floodway/floodplain areas without applicable approvals.
2. Minimize any soil, spoil stockpile and/or laydown areas adjacent to environmentally sensitive areas. If stockpiles or laydown areas are needed, provide reinforced, double-row silt fence or high hazard silt fence depending on field conditions. Consult with an environmental/EHS professional for guidance and compliance assistance around such areas.
3. Construction equipment should be appropriate for the intended purpose – use of appropriate/engineer-approved matting and low ground pressure tracks/tires should be used to minimize any adverse wetland impacts or to minimize soil disturbance. Heavy equipment should only be used in environmentally sensitive areas if there is no other option, and if best practices such as appropriate matting are used.

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4. Any wetland activity should include the use of appropriate/engineer-approved matting and/or temporary bridging materials for access or crossings (riprap, soil, gravel or other such materials should not be used). No logs or cleared vegetation should be used for matting and/or temporary bridging.
5. Any stream crossings should include the use of temporary bridges (riprap, soil, gravel or other such materials should not be used). Use air bridges to cross active gas lines and other buried utilities as necessary.
6. It is highly recommended that a minimum 30-foot vegetative buffer zone on both sides of streams should be maintained such that only vegetation that poses a hazard or has the potential to grow tall enough to interfere with lines is removed. Consider hand-clearing woody vegetation immediately adjacent to environmentally sensitive areas. Check with the field environmental/ EHS professional for state riparian rules and regulations to ensure adequate measures have been installed to ensure compliance.
7. Any streambed or streambank disturbance should be minimized and any rutting or similar impacts should be returned to pre-project contours and topography to the greatest extent possible immediately after work is complete. All restoration should be tied into existing contours.
8. If conditions warrant, protective measures such as silt fences, wattles or other appropriate erosion control measures should be utilized to protect sensitive environmental areas.
9. Felled trees in environmentally sensitive areas should be removed by chain or cable, and not dragged if possible, unless the agency agrees to let fallen timber remain. If stumps are to be removed, do so only by grinding in upland areas, outside of wetlands or other environmentally sensitive areas.
10. Installation of aerial infrastructure should be set back at least 30 feet from a stream, where practical. Refer to State Riparian Buffer rules for specific stream requirement.
11. Transformers or other equipment that has potential for adverse environmental impacts due to potential leaks or spills should be located a minimum of 125 feet away from environmentally sensitive areas.
12. Immediately repair and report any erosion control measure failures, deficiencies and/or concerns to Environmental Services.

Directional Drills/Borings

1. Consult with the field environmental/EHS professional for all current guidance and procedures regarding utility line installation.
2. Pipeline/utility line construction through jurisdictional waters and wetlands should be accomplished utilizing directional drilling/boring methods to the maximum extent practicable. Based on site conditions and analysis by project experts, it could be more protective of a resource to open cut in a wetland area.
3. When directional boring or horizontal directional drilling (HDD) under surface waters or wetlands, personnel shall closely monitor the entire project area for hydraulic fracturing ("fracking") or releases along the length of the bore.
4. Any discharge or release of hydraulic materials or sediments into surface waters or wetlands will need to be reported to the applicable regulatory agency. Contact the field environmental/EHS professional to report a release as soon as possible.
5. Drilling entry and exit points shall be managed to avoid any significant buildup of sediments or other materials that could impact the environment. Silt fences, wattles, liners or other such devices shall be installed at these locations to ensure no releases to environmentally sensitive areas occur.
6. Entry/exit points should be located a minimum of 30 feet away from streams, wetlands or other sensitive environmental areas. For directional boring or HDD under surface waters or wetlands, the line should be at least 5 feet below the streambed.
7. The environmental/EHS professional will provide the appropriate directional drilling/boring guidance and procedures, assist with Environmental Requirements Review, and provide further instruction as needed.

Project Closeout/Maintenance

1. Remove erosion control measures after vegetation stabilization and project closeout conditions are met to ensure diffuse flow of stormwater through the project areas.
2. Restore wetland to pre-construction conditions as much as possible. This includes restoring ruts and/or fill. Mulched material should not be placed in streams or wetland areas.
3. Consult with the field environmental/EHS professional for all current guidance and procedures, and to ensure all permit required activities are completed and the permit is closed, as necessary.

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Maintenance Tasks Located Over or Near Water

1. Conduct inspections of equipment, components, secondary containments, spill catchments and surrounding areas for any fluid leaks before and after conducting maintenance.
2. Perform weekly inspections of oil-filled equipment, while it is running, to ensure that there are no leaks while oil and fluid are moving. Report all leaks and equipment deficiencies to your supervisor and write a work order to complete repairs, as necessary.
3. Any oil-filled equipment found to have leaks or improper maintenance should be taken out of service immediately, repaired and re-inspected before being put back into service.
4. Properly maintain all equipment per business unit standards or manufacturer instructions. Ensure preventive and corrective maintenance activities are logged or recorded in an approved work management system such as Maximo or Passport.
5. Properly store and handle oils, lubricants and other fluids. Follow all Environmental Health and Safety requirements for oil and chemical storage, such as secondary containment and maintaining and storing material 125 feet outside of the boundary of concern, whenever possible.
6. Obtain the SDS for fluids in the equipment being serviced, and become familiar with associated hazards.
7. If possible, conduct all non-oil filled electrical equipment maintenance in a single location that is far from storm drains and greater than 125 feet from waterways. This may require moving the equipment to a new location that is better suited for spill containment.
8. Conduct major maintenance such as disassembly in a location that is at least 20 feet from storm drains, at least 125 feet from waterways and suitable for proper spill containment.
9. Use drip pans, catchments, plastic coverings, booms and/or drop cloths to catch any potential drips and spills when equipment is being serviced. If equipment is not moved before conducting maintenance and the work is conducted within 125 feet of a waterway, cover any nearby openings to waterways, such as grates or intakes, with plastic, and place oil absorbent booms around grates and in possible paths of flow to bodies of water in case of spills.
10. Make sure that spill cleanup kits are well-stocked, on hand and readily available when performing maintenance.

11. If a spill occurs, be prepared to respond quickly. Immediately notify your supervisor and the field environmental/EHS professional and, if safe, take steps to prevent spill from spreading or reaching water bodies.
12. Contact your supervisor or the field environmental/EHS professional for any site-specific requirements.

Migratory Birds

1. Over 1,000 species of birds are protected by federal law (including very common species like robins, crows and vultures). This law makes it illegal to injure, kill or capture any protected bird, possess any feather or part of a protected bird, or destroy an active nest.
2. Duke Energy operates 24-hour Migratory Bird hotlines to assist employees and contractors with bird-related incidents that occur during your daily work activities. Hotline numbers are below:

Carolinas: 800.573.3853	Florida: 727.386.3084	Wind: 704.430.7946
Midwest: 317.430.4497	Solar: 704.657.5775	

3. When planning to conduct an outdoor work activity or project, ask the following questions:
    a. Are there birds or nesting birds in the project area?
    b. Are there large birds or a large bird nest in the project area?
    c. Will this action affect important bird habitat (nesting, foraging, hunting, congregating and overwintering areas)?
    d. Will this action take place adjacent to a river, lake or other natural feature where eagles or osprey are reasonably expected to be present?
    e. Could this action/work unintentionally injure or kill any birds?
    f. Do we need a bird protection plan or bird permit?
4. If you answered “Yes” to any of these questions, call the Duke Energy Migratory Bird Hotline for guidance.
5. Notify your supervisor or the field environmental/EHS professional and call the appropriate Duke Energy Migratory Bird Hotline if you find a dead or injured bird on Duke Energy property or rights of way.
6. Notify your supervisor or an environmental/EHS professional and call the appropriate Duke Energy Migratory Bird Hotline if a bird nest with eggs or chicks or an empty nest is in danger of being disturbed. Do not proceed with the work until advised to do so by the Duke Energy Migratory Bird Hotline or the environmental/EHS professional.

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7. Notify your supervisor or an environmental/EHS professional and call the appropriate Duke Energy Migratory Bird Hotline if a bird is trapped inside a Duke Energy facility.
8. If you suspect you have come into contact with a dead or injured bald or golden eagle, or a large nest near work activities (which may be a bald or golden eagle nest), stop work immediately. Notify your supervisor or an environmental/EHS professional and call the appropriate Duke Energy Migratory Bird Hotline.

Operating and Maintaining Mobile Equipment

1. Immediately inspect equipment when it arrives on-site for any fluid leaks or maintenance issues.
2. Inspect equipment daily, before and after each use. Be sure to log all inspections per business unit procedures. Pay particular attention to proper installation of hoses, seals, fittings, filters, etc.
3. Any equipment found to have leaks or improper maintenance must be taken out of service immediately, repaired and re-inspected prior to placing back into service.
4. Properly maintain all vehicles and heavy equipment. Ensure preventive and corrective maintenance activities for equipment not managed by Fleet Services are logged or recorded in an approved work management system per business unit procedures.
5. As possible, routinely inspect all fuel, oil or fluid-containing fittings, hoses and seals during machinery operation to detect leaks. Note: Over-tightened hoses or using incompatible fittings/fluids can cause leaks.
6. Ensure proper storage and handling of fuel, oils, lubricants and other fluids such as antifreeze. Keep in mind the appropriate Environmental, Health and Safety requirements for fuel and chemical storage such as secondary containment and container labeling.
7. If possible, conduct all vehicle/equipment maintenance and refueling at one location, away from storm drains, and greater than 125 feet from waterways.
8. Perform major maintenance, repair jobs and vehicle/equipment washing off-site or in an approved location such as a fleet garage.
9. Assess the risk of an oil spill occurring and use drip pans, catchments, plastic, drop cloths, or other general spill prevention methods to catch any potential drips and spills under mobile equipment to prevent an oil discharge, especially when sitting idle during periods of non-use (during breaks and overnight).

10. Perform a weekly inspection of parked equipment to check for leaks, spill containment installation deficiencies or environmental concerns. Keep a record of the most recent inspection at or near the home base of the equipment to allow for supervisor review of results and actions taken for problems found.
11. Conduct a pre-use inspection before operating equipment. This includes checking for hydraulic fluid gathered on the ground barrier, fluid around fittings and fluid on the hydraulic cylinder.
12. Watch out for any fluid leaks that develop during operation, regardless of how minor.
13. Recognize hydraulically supported equipment “drifting” during operation (i.e., if operator must continually fight to keep the load raised on an excavator) as a potential hydraulic fluid leak. Even if no external leaks are apparent, drifting is a sign of internal leaking.
14. Maintain the proper level in all hydraulic reservoirs. Hydraulic systems operated with a low fluid level will run hot, which will shorten the life and/or damage system components such as seals and hoses.
15. Carry portable spill cleanup kits in trucks or heavy equipment to lessen the amount of time needed to respond in case of a problem.
16. If a spill occurs, be prepared to respond quickly. Promptly notify the appropriate personnel and take steps to prevent the spill from spreading or reaching water bodies in the United States.
17. Contact the field environmental/EHS professional for any site-specific equipment requirements.

Open-Burning

1. Open-burning activities (the burning of any materials where contaminants resulting from combustion are emitted directly into the air, without passing through a stack or chimney from an enclosed chamber) may occur as a result of land-clearing activities, conducting fire-fighting training or right-of-way vegetation management.
2. Open-burning activities are generally discouraged if other cost-effective and regulatory-approved options are available.
3. Consider controlled options for open burning in lieu of placing material in an open pile. Consider curtain burners with clean fuel for startup. Contact an environmental/EHS professional to discuss potential permitting requirements related to the use of controlled options.

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4. Restrictions on materials combusted, distance from residences and structures, time of day and burning during periods of drought are specified in state and local regulations. Any type of open burning can also trigger the need for advance permits or approvals from county, local or municipal officials, including the local/state fire marshal, state Division of Forestry and USDA Forest Service (fire towers) that may have jurisdiction. Since these local approvals vary considerably, contact an environmental/EHS professional for support.
5. Prior to an open-burning event, ensure a communication plan has been developed and executed. This includes communications with the appropriate environmental/EHS professionals, and Corporate Communications and Community Relations organizations. Courtesy notifications may be made to regulatory agencies by the designated environmental/EHS professional, even if they are not a regulatory requirement.

Polychlorinated Biphenyl (PCB) Management

✪ Equipment and certain manufactured products that contain ≥ 50 parts per million (ppm) PCBs or materials that have been contaminated with PCBs are regulated by the Environmental Protection Agency under the Toxic Substances Control Act (TSCA). Items with PCBs content ≥ 500 ppm are more stringently regulated. Contaminated soil and other materials may be regulated as PCB Remediation Waste at PCB concentrations ≥1ppm.

The following list includes equipment, manufactured products and materials that may be regulated under TSCA:

- Equipment – transformers, bushings, capacitors, ballasts, natural gas pipe, breakers, switches, fluorescent lamp ballasts and paper-insulated lead-covered cable
- Manufactured products – paints, caulks, mastics and wire coatings
- Materials – oil, soil, water, natural gas pipeline liquids (condensate), truck beds, plastics, paper, wood, cardboard and debris

1. PCB spills ≥ 50 ppm and the collection of PCB gas condensate should be reported immediately to your supervisor and an environmental/EHS professional to ensure cleanup protocols and regulatory reporting requirements are met.
2. ✪ All PCB shipments, cleanup records and analysis should be completed by or submitted to the field environmental/EHS professional and compiled for a site's PCB Annual Document Log/Records as applicable. The following must be kept in mind when shipping PCB materials:

- Regulated PCB materials must be shipped on a Hazardous Waste Manifest to an approved PCB disposal facility.
- Separate facilities may not share EPA Identification numbers for purposes of shipping PCB wastes; facilities with no EPA Identification number must use the generic identification number “40 CFR Part 761” on manifests.
- Contractors must deliver all PCB and PCB-unknown materials from the field to a Duke Energy operations center prior to shipment for disposal.
- PCB shipments must only be overseen by staff who have completed DOT Shipper Training.
- Do not assume that oils being shipped for disposal do not contain PCBs or other hazardous materials; material must be analyzed by a certified laboratory and results reviewed prior to shipment.

Regulated PCB Materials must be shipped on a Hazardous Waste Manifest to an approved PCB disposal facility.

3. Non-leaking PCB or potential PCB materials must be marked per the company procedure and can be temporarily stored at a location, such as an operations center, for 30 days prior to being shipped to a long-term PCB storage warehouse location (Plainfield, Brecon, Toddville, Garner or Wildwood).
4. Contact your supervisor or the field environmental/EHS professional if you have questions or concerns regarding PCBs or PCB-containing equipment.
5. The Substation Soil Procedure (ADMP-ENV-EVS-00165) should be followed when conducting soil disturbance activities in substations that were built in 1978 or earlier. This procedure outlines the specific soil testing processes, regulatory guidance, roles, and responsibilities to address soils potentially impacted with PCBs in substations. Contact EHS Field Support or Remediation for specific requirements.

Pesticide and Herbicide Application

1. Use only “General Use” pesticides and herbicides that are approved chemicals. General Use pesticides are available for purchase for use by the general public. “Restricted Use” pesticides (RUPs) are generally prohibited for use by Duke Energy. RUPs have the potential to cause unreasonable adverse effects to the environment and injury to applicators.
2. Follow the personal protective equipment (PPE) instructions on the product label when handling, mixing or applying pesticides or herbicides. Unless otherwise specified on the product label, the minimum PPE includes a long-sleeved shirt, long pants, eye protection (safety glasses or face shield) and impermeable gloves.

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3. Ensure applications are performed by or under the supervision of a licensed applicator (unless the application is incidental as described in item #4).
4. Personnel can perform incidental applications of general use wasp, hornet, fire ant and other small insect pesticides (including tick, insect and dog repellents) for immediate personal protection as long as all label instructions are read and followed.
5. Maintain pesticide application records for at least three years from the date of application. Generally, records are not required for the following applications:
    a. Janitorial use of general use disinfectants
    b. Personal use of tick, insect and dog repellents
    c. Incidental use of the following household-type products: general use rodenticides and insecticides including wasp and hornet aerosols, fire ant killers and insect/mosquito repellents for small localized spot applications
6. Keep records for general use applications of microbiocides, molluscicides, algaecides and other antimicrobial products for water, wastewater or cooling tower treatment per the requirements of water and wastewater certifications.
7. Prepare and review a Pesticide Discharge Management Plan for applications performed under a Pesticide Discharge Permit.
8. Ensure all aerial herbicide application projects follow the Transmission Vegetation Management Aerial Herbicide Application Operational Procedures.
9. Follow best management practices when applying all terrestrial and aquatic vegetation herbicides or insect pesticides.
10. Immediately notify your supervisor or the field environmental/EHS professional if any pesticide/herbicide chemical spills occur.

Refrigerants

Chlorofluorocarbons (CFCs) and hydrochlorofluorocarbons (HCFCs) are widely used as refrigerants in industrial refrigeration, commercial and comfort air conditioning, motor vehicle air conditioners (MVAC) and small refrigeration appliances. They are also used as a dielectric gas in some load center transformers. Halon is used as a fire suppressant in waterless fire protection systems for computer, telecommunication and data/document storage facilities and hand-held fire extinguishers.

1. Avoid knowingly venting any refrigerant substance, including approved substitutes, into the atmosphere during testing, maintenance, servicing, repairing or disposal of an appliance or MVAC, except as allowed by the regulations. Minor releases of refrigerant that result from purging hoses or

from connecting or disconnecting hoses to charge or service appliances or MVAC are not considered violations of the prohibition on venting.

2. All Duke Energy and contracted technicians servicing or disposing of refrigerant-containing appliances or MVAC must have Environmental Protection Agency (EPA) certification and training for the appliances or MVAC they service. Recycling and recovery equipment certified by the EPA must be used.
    a. Documentation of the EPA certification(s) must be maintained with the field environmental/EHS professional or with project documents.
    b. Ensure that the technician certification level aligns with the work to be performed.
    c. Only certified technicians may purchase refrigerant or halon materials.
3. Follow prescribed service practices from the manufacturer to minimize the release and maximize the recycling of refrigerants and halon during appliance or MVAC servicing, repair, testing, maintenance and disposal.
4. Provide documentation to the field environmental/EHS professional sufficient to track and document annualized leak rate, timely repairs, retrofits and notifications for stationary refrigeration equipment with a full charge of more than 50 pounds, as required by law. Some documentation is also required for appliances containing less than 50 pounds of refrigerant and for MVAC activities.
5. Follow manufacturer or approved work practices to ensure refrigerants and halon are recovered from appliances or MVAC, as required by regulations. All refrigerant must be recovered prior to disposal.
6. Ensure that recovered refrigerant materials are properly disposed. Contact your supervisor or the field environmental/EHS professional with any questions regarding refrigerant management.

Reciprocating Internal Combustion Engines (RICE)

1. Evaluate the need for an air permit for any reciprocating internal combustion engine (RICE) equipment brought on-site. Equipment may include but is not limited to the following: pumps, generators, light stands and welders.
2. Provide secondary containment for each piece of equipment while on-site to prevent discharge of unanticipated leaks/spills to the environment.

    Containment must provide adequate capacity and be in good operating condition.
3. Maintain records (e.g., rental agreements, logbooks) for any portable RICE equipment brought on-site to demonstrate the engine is not stationary.

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4. Ensure the field environmental/EHS professional is made aware of any engine rented and/or owned by Duke Energy. Equipment meeting the definition of stationary RICE source may require an air permit modification prior to operation.
    a. Ensure that any equipment purchased meets the required regulatory emissions standards for the intended use.
    b. Maintain a copy of the manufacturer's emissions certification on-site.
5. Prior to operation of the equipment, ensure that required maintenance activities have been entered into an approved Work Management System for stationary equipment that is rented or owned by Duke Energy.
    a. Only use ultra-low-sulfur diesel (ULSD) fuel in the engine.
    b. Use a logbook to track run time and the reason for the run event. A non-resettable hour meter must be installed on the engine and used to record hours of operation.
    c. Ensure maintenance, including changing hoses, belts, oil and oil filters, is completed per the manufacturer's guidelines or at the default thresholds provided by EPA. Have the records readily available on-site.

Spill Prevention Control & Countermeasure (SPCC) Plans and Requirements

1. Develop and implement a Spill Prevention Control & Countermeasure (SPCC) plan for any facility that has a total above-ground oil storage capacity of 1,320 gallons or more (excluding containers of less than 55 gallons) or total below-ground oil storage capacity of 42,000 gallons, which due to its location could reasonably be expected to discharge oil in quantities that may be harmful into or upon the navigable waters of the United States or adjoining shorelines.
2. Contractors with oil containers equal to or greater than 55 gallons that have a combined total oil storage capacity of 1,320 gallons or more, and due to its location could reasonably be expected to discharge oil in quantities that may be harmful into or upon the navigable waters of the United States or adjoining shorelines, must prepare and comply with their own SPCC plan that meets the requirements set forth in 40 CFR 112. The SPCC plan and any associated records shall be made available to Duke Energy for review upon request.
3. Contact the field environmental/EHS professional for assistance with SPCC plans and their requirements.

4. Document all inspections and containment draining activities as described in the facility SPCC plan. Inspection records must include the name of the inspector and date of the inspection.
5. Retain inspection records required by the SPCC plan at the facility (or a designated location for unmanned facilities) for a minimum of three years.
6. SPCC plans must be amended when any material changes (i.e., adding/removing oil containers or piping systems, altering secondary containment or drainage, modifying operation or maintenance procedures, etc.) occur at the facility. Amendments must be made within six months of the change.
7. Ensure SPCC plans are reviewed by the assigned professional engineer, their agent or subject matter expert at least once every five years. Tier I SPCC plans can be reviewed by the field environmental/EHS professional. Any administrative updates such as changes in facility management or phone numbers should be made immediately.
8. Notify the field environmental/EHS professional prior to adding or removing any oil storage containers (excluding containers of less than 55 gallons).
9. Ensure procedures are posted at designated loading/unloading locations per the facility SPCC plan. Perform the loading and unloading of fuel- and oil-containing products in accordance with posted procedures at the unloading area with facility personnel in constant attendance. It is important that tank level be evaluated before, during and after product off-loading to prevent tank overfill (see also the Oil and Chemical Containment Systems section of this handbook for more information).
10. Train all personnel involved in the operation and/or maintenance of oil storage containers or oil-filled equipment as described in the facility SPCC plan. All personnel involved in the operation and/or maintenance of oil storage containers or oil-filled equipment shall also receive an annual discharge prevention briefing.
11. Train contractors to properly recognize and respond to a spill and be able to promptly notify facility personnel.
12. For sites having an SPCC plan, containment for oil storage containers (drums, totes, tanks, portable and fixed compressors, portable and fixed generators, portable light banks, etc.) greater than 55 gallons should be sized to contain the entire capacity of the largest container and sufficient freeboard for precipitation. The containers and containment should be inspected/checked periodically to ensure there are no issues and the containment is empty of rainwater as described in the facility SPCC plan.

    Note: General containment may be either active or passive in design. Active measures are those that require deployment or a specific action and

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could be deployed either before an activity involving the handling of oil starts (such as covering drains or putting drip pans down while refueling) or in reaction to a discharge (putting down sorbent material). The active measures must be implemented in time to prevent the spilled oil from reaching surface waters. Passive measures include dikes, berms, retaining walls or other barriers, sumps and collection systems, spill diversion ponds or retention ponds.

13. General containment should be provided for oil-filled operational equipment that is part of an SPCC plan and needs to only address the typical failure mode and the most likely quantity of oil that would be discharged.
14. For sites having an SPCC plan, inspections must be performed and documented as specified in the plan for all listed oil sources, secondary containments, loading/unloading area and spill response equipment.
15. Inspect containment structures subject to periodic integrity testing per the SPCC plan (unless it has been determined that there is no risk of impact to navigable waters if the container is compromised).
16. General containment (see note above) should be provided for oil storage containers less than 55 gallons in volume located at sites having an SPCC plan. The containment strategy utilized should be inspected/checked periodically to ensure there are no issues but there is no requirement to document the inspection.
17. Protect containers, above-ground piping and containment structures from potential vehicular traffic and mobile equipment impacts.
18. Maintain records of oil spill events, inspection reports and training records as required by the SPCC plan and applicable business unit standard operating procedures. Contact the field environmental/EHS professional for assistance with documentation requirements.
19. For sites and/or contractors not meeting the SPCC threshold of 1,320 gallons and therefore not having an SPCC plan, general containment shall (see note above) be provided for all oil storage containers regardless of volume. The containers and/or containment strategy utilized should be inspected/checked periodically to ensure there are no issues but there is no requirement to document the inspection.
20. Secondary containment for motive power containers (e.g., self-propelled crane, self-propelled heavy vehicles, cherry picker, forklift) is not regulated under SPCC. For this type of container/equipment you should assess the risk of an oil spill occurring and provide general spill prevention methods to prevent an oil discharge (see the Operating and Maintaining Mobile Equipment section for more information).

Spills

1. Prior to working with oils and chemicals discuss:
    a. Your work and the risk of spills associated with your work
    b. What could go wrong or lessons learned from past events
    c. What barriers you have put in place to eliminate or reduce the risk of a spill
    d. Distance or flow paths to water
    e. The location and access to an appropriate spill kit
    f. The location of the SPCC plan for the facility, if applicable
2. Generally, perform equipment checks on mobile equipment as follows:
    a. Inspect equipment at the beginning of a shift, at the end of a shift and when the work scope changes.
    b. Inspect hydraulic equipment before use and have defective equipment repaired BEFORE it fails.
    c. Make every effort to park and/or stage equipment more than 125 feet away from water bodies.
    d. Take precautions to prevent leaks from getting to the water/drains when working near water or yard drains.
    e. Use containment for off-road or mobile equipment if possible.
3. Equipment refueling should take place more than 125 feet from water bodies whenever possible. When not possible, extra precautions shall be taken to avoid spills.
4. Refueling must be attended through the entire refueling operation.
5. Proper precautions must be taken to prevent the spilling of any fuel.
    a. Place a bucket/containment under refueling nozzle.
    b. Have absorbent pads, socks, etc. available.
    c. Place containment under mobile equipment and refueling truck as applicable.
6. Dike or contain leaking oil-filled equipment. Properly bag or wrap oil-filled equipment for transport. It may be necessary to pump the oil from the leaking unit in some cases.
7. Provide containment or construct temporary berms under portable oil-filled equipment, such as compressors and generators, when used outside. General containment may be used in lieu of sized secondary containment if oil quantity is less than 55 gallons or located at a non-SPCC regulated site. In addition, assess the risk of an oil spill occurring and provide general

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spill prevention methods to catch any potential drips and spills under mobile equipment (such as bulldozers, excavators, and cranes) to prevent an oil discharge, especially when sitting idle during periods of non-use (during breaks and overnight).

8. In the event of a spill, your responsibilities are to first protect yourself and then SWIM:
    a. S – Secure the area.
    b. W – Warn others to leave the area.
    c. I – Inform: Contact Environmental Services IMMEDIATELY and as necessary inform the control room or site management. Time is of the essence when reporting to Environmental. Do not wait to gather information. Immediately notify your field environmental/ EHS professional.
    d. M – Monitor the area, keeping others away until qualified help arrives.

Oil Releases from Electrical Equipment (Transformers)

1. If an oil spill from electrical equipment is suspected or detected, first inspect the equipment for PCB content information:
    a. Use appropriate PPE (steel/composite shoes, reflective vest, eye and hand protection, coveralls, shoe covers, etc.) when approaching and managing an oil spill from electrical equipment.
    b. Inspect the unit for a Blue or Green Dot, Blue Non-PCB Label and/or review the unit's nameplate to determine the PCB content of the unit.
    c. If an oil spill has occurred from a transformer whose PCB status cannot be determined by nameplate, Blue Dot or Label, run a Clor-N-Oil PCB field test or assume the unit is PCB-contaminated.
2. For all leaking PCB transformers or transformers with unknown PCB content:
    a. Contain or seal the leak.
    - Control leakage to the greatest extent possible.
    - Install knockout seals when bushings are the source of the leak.
    - Use plastic steel putty such as Pig Putty or Aqua Seal pads.

    b. Bag the unit or pump the oil out of the transformer before bagging the unit, if necessary.
    - Prevent bag damage by removing wire and sharp objects, including porcelain arrestors.

    c. Wrap absorbent pads around the base of the equipment.
    d. Tag and fasten the transformer bag.

- Attach a yellow tag to PCB transformers or transformers with unknown PCB content and record on the back of the yellow tag the "Removal from Service Date."
- Tape the top of transformer bag.
- Tape the corners of transformer bag against side of equipment.
- Minimize pooling in corners of transformer bag.

e. Transport the bagged transformer to the Operations Center and palletize the unit.

Spill Reporting

1. In order to ensure the protection of human health and the environment, federal and state laws require that certain types of spills and unpermitted releases be reported to the proper regulatory agencies as soon as possible following identification.
2. Immediately notify your supervisor and the field environmental/EHS professional when a spill or unpermitted release is discovered. In some cases, reporting to an outside agency is required within 15 minutes. There are many types of events that may require reporting. The following are some examples:
   - An oil spill, in any quantity, to navigable water (any oil sheen counts!)
   - Any oil spilled to land within 100 feet of a water body (in North Carolina; other states may vary)
   - A chemical release to air, land or water
   - A release of untreated domestic wastewater
   - An oil spill to soil greater than 25 gallons
   - A coal ash release to the environment greater than 8,000 pounds in a 24-hour period
   - Any fire involving a PCB transformer
   - An unpermitted discharge of industrial wastewater to surface water
   - An unpermitted discharge of pollutants to the air
   - Any spill from an underground storage tank
3. Follow the facility event reporting procedure. Call the control room to report an event at a generating station. Refer to the SPCC plan for facility-specific procedures if available.
4. If you are unable to reach the field environmental/EHS professional, contact one of the Duke Energy spill reporting hotlines at:

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Generation – Regional Environmental SME or Generation Spill Coordinator Hotline: **800.510.7439**
Transmission & Distribution – Regional Spill Coordinator Hotline Carolinas and Midwest: **800.527.3853** Florida: **866.769.1266**

Have the following information available when making the notification:

- Location of the spill/release – facility name, address and location on-site.
- Date and time discovered.
- Your name and phone number.
- What has been spilled? Chemical name, process released from, etc.
- Estimate of the volume released into the environment.
- To what environmental media has the material been released – land, water, air?
- Has the spill reached a drain or waterway? Does it have the potential to reach a drain or waterway?
- What health risks exist?
- What actions, if any, have been taken?

5. Perform spill response activities if you are trained and it is safe to do so. Safely reduce the duration/extent of the release and any environmental impact.

Spill Cleanup

1. Ensure the event has been reported prior to beginning cleanup.
2. Do not attempt spill cleanup if you have not been trained in HAZMAT cleanup procedures.
3. Reference the Facility Response Plan (FRP) if the facility is an FRP regulated site or the SPCC plan if it is an SPCC site.
4. Confine the spill as close as possible to the source of the release. Consider type of spill, location, weather conditions and quantity to determine the best method for spill containment.
5. Identify source of spill and determine the associated hazards.
6. Eliminate all ignition sources if flammable or combustible materials are spilled and the danger of explosion exists.
7. Use experience and judgment in choosing the best approach and remedial actions. The most generally applicable rule: Confine the spill as quickly as possible to minimize environmental impact.

8. Use the tools, equipment and materials that are readily on hand at the time that a response is necessary until there is time to bring additional, perhaps more effective, material to the site.
9. Depending on the size of the spill, containment and/or cleanup strategies needed, you may need to call an Oil Spill Response Organization (OSRO) or facility (hazardous material) HAZMAT team for assistance.
10. Response to an incidental spill does not require the utilization of a HAZMAT team but requires persons to be trained on the chemicals they are using.
11. Contact the facility environmental/EHS professional for assistance coordinating spill waste disposal.

Storage Tanks/Underground Storage Tanks

1. System tanks are considered secondary chemical containers and must be labeled appropriately. Underground tanks should be labeled at the fill point.
2. Prior to filling, ensure that the contents of the storage tank are compatible with the system.
3. Always check that available volume in the tank is greater than the volume of product to be transferred to the tank.
4. Ensure releases due to spilling or overfilling do not occur by monitoring product transfer constantly (NO EXCEPTIONS).
5. Immediately contact your supervisor and the field environmental/EHS professional upon discovering a release, spill or overfill. If possible, stop and/or contain the release.
6. Contact the field environmental/EHS professional prior to installing an underground storage tank (UST), taking a UST system out of service or placing a UST in temporary closure status. Governing agencies must be notified prior to operation or a change in tank status.
7. Contact the field environmental/EHS professional prior to installing, removing or taking out of service any above-ground storage tanks (AST). Some states or local agencies have rules requiring permits or registrations for ASTs.
8. Always consult with the field environmental/EHS professional before installing or repairing UST components. Repairs may trigger new compliance requirements that could be onerous or unworkable at your location.
9. Prevent releases due to corrosion by operating and maintaining corrosion protection systems for the metal components of the tank and piping that routinely contain regulated substances and are in contact with the ground.

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10. Review printout from the tank monitoring system daily or as required and report any system alarms to your supervisor and the field environmental/EHS professional.
11. Records of a passed leak test result for each tank must be collected and filled monthly or as required.
12. Monitor USTs for releases using an approved release detection method once each calendar month, not exceeding 30 days. For exceptions, consult the field environmental/EHS professional.
13. Train personnel on Class A, Class B and Class C USTs according to state-specific training requirements.
14. Maintain a current copy of the UST operating permit and post it in a conspicuous location at the facility.
15. Keep a copy of fuel and product delivery receipts on file for at least one year.
16. Maintain a sufficient amount of product in used oil tanks (one-third capacity) to ensure a passed leak test is achievable for each calendar month. Do not allow Fleet Services or other vendors to pump tanks out completely as it will take time to allow the tank to rise to the level where the tank monitoring system can give an accurate passed result.
17. Ensure preventive maintenance and inspection requirements for the storage tank are correctly identified and tracked to completion in a work management system or compliance tracking system.

Stormwater

Industrial Stormwater

1. Industrial stormwater permits are often required to effectively manage runoff generated by rain and snowmelt events at a facility. A valid stormwater permit must be effective at all regulated locations that discharge stormwater associated with industrial activity into the waters of the state and/or United States.
2. Maintain a Stormwater Pollution Prevention Plan (SWPPP, SPPP), or equivalent, as required by the issued stormwater permit, including any required monitoring.
3. Contact the facility environmental/EHS professional for industrial stormwater permitting and compliance assistance.

Construction Stormwater

1. Obtain approval for temporary construction/land-disturbance projects under general state National Pollutant Discharge Elimination System permit programs.
2. Create a Stormwater Pollution Prevention Plan (SWPPP, SPPP) or Erosion Control Plan (ECP) as required by the general permit. The threshold amount of area disturbance, which triggers these permits, varies by the appropriate jurisdictional authority.
3. Contact the facility environmental/EHS professional for construction stormwater permitting and compliance assistance.

Stormwater Permit Requirements

1. Use best management practices at the project site to minimize the mobilization of sediments in accordance with the appropriate plan (SWPPP, SPPP, ECP, or equivalent) as required by the permit. If there is no permit required, good practices such as silt fence, wattles or other approved erosion control devices should be used in consultation with the assigned EHS field professional.
2. Some jurisdictions require long-term Operational and Maintenance Plans (O&M Plans) to be developed and followed after the construction phase of the project. These O&M Plans may require routine documented inspections of the permanent best management practices, as well as annual reports and other documentation showing compliance of maintenance requirements and verification of site conditions of the permanent practice.
3. Document any inspections and monitoring that are mandated by the stormwater permit.
4. Maintain all documentation on-site per the record retention guidelines specified by the stormwater permit and corporate requirements.

Threatened and Endangered Animals, Plants and Their Habitats

1. There are over 300 species of threatened or endangered animal and plant species in Duke Energy regions that are protected by federal law (i.e., Endangered Species Act).
2. Harming, harassing, capturing or collecting a threatened or endangered animal is illegal.
3. Collecting or harming threatened or endangered plants on federal land is illegal.

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4. Altering habitat or degrading habitat to an extent that it impacts a threatened or endangered species in any way, creates stress on a threatened or endangered species, or directly or indirectly causes death or injures a threatened or endangered species is illegal.
5. When planning an outdoor work activity or project that causes ground or habitat disturbances, ask the following questions.
    a. Are there endangered or threatened animals or plants in the project area?
    b. Could this action affect protected or important habitat?
    c. Could this action unintentionally injure or kill any animals or plants?
    d. Are there state and local laws to protect animals, plants and habitat?
    e. Do we need a protection plan or permit?
6. If you answer “Yes” or “I don’t know” to any of the questions in item 5, contact your supervisor and an environmental professional for guidance prior to conducting the work.
7. The environmental/EHS professional will provide the appropriate procedures, assist with Environmental Requirements Review, refer the appropriate Duke Energy professionals responsible for performing endangered species evaluations, and provide further instruction as needed.
8. In the event of an endangered or threatened species incident involving the injury, death or habitat impacts to listed species, the employee or contractor should call the Environmental/EHS Field Professional, who will contact the local SME or the Natural Resources team immediately to report the incident. There is a federal reporting requirement with these incidents.

Used Oil

1. Do NOT mix hazardous waste or other materials such as solvents, antifreeze or unknown liquids with used oil.
2. Only store used oil in Department of Transportation-approved storage containers.
3. Storage containers must be closed except for when adding used oil.
4. Containers holding used oil shall be clearly labeled or marked with the words “USED OIL.”
5. Store containers within secondary containment in a predetermined used oil storage location. Be mindful if the site has a Spill Prevention and Countermeasure Control (SPCC) plan, and adhere to all requirements.
6. Know the location of the closest spill kit in the event of a spill.

7. Immediately report any used oil spills to your supervisor and the field environmental/EHS professional, and take immediate action to minimize the extent of the spill.
8. Transformer oil must have a polychlorinated biphenyls (PCBs) determination made by an environmental professional.
9. Hot-drained used oil filters are considered nonhazardous waste if they are recycled as scrap metal. Except in Florida, they can be placed in the regular trash if they have been properly characterized as nonhazardous based on generator's knowledge or analytical data and that characterization is documented in the facility files. In Florida, used oil filter containers must be labeled with the phrase "Used Oil Filters."
10. Used oil from all Duke Energy facilities can only be sent to a company-approved off-site used oil recycling or disposal vendor.
11. Do not transport more than 55 gallons of used oil at one time unless the vehicle is registered as a "Used Oil Transporter."
12. Contact the field environmental/EHS professional for any state-specific requirements.

Vehicle Washing/Washout

Vehicle Washing Permits

1. Duke Energy facilities discharging vehicle wash water that drains off-site either by direct discharge or through a storm drain must be authorized by an industrial wastewater discharge permit. These permits provide for the minimization of any pollutants to surface waters. A vehicle wash water or similar discharge permit must be obtained for any Duke Energy facility that washes vehicles on-site. An individual permit may be required for each company washing vehicles at a Duke Energy facility.
2. Complete an Environmental Requirements Review Checklist prior to modifying an existing wash system or installing a new system.
3. Contact the field environmental/EHS professional for permitting and compliance assistance related to vehicle washing.

General Guidelines for On-Site Vehicle Washing

1. Discharge vehicle wash water generated during vehicle washing to a permitted waste treatment system.
2. Contact your supervisor or the field environmental/EHS professional before using detergents if you are uncertain of a facility's permitting requirements.

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3. Collected sediment from the vehicle washing system must be evaluated by the field environmental/EHS professional to determine methods for proper off-site disposal.
4. Duke Energy vehicles may be washed off-site at a commercial vehicle washing facility as authorized by management.
5. Duke Energy facilities may contract vehicle washing to vendors who provide on-site, turnkey vehicle washing (i.e., taking vehicle wash water off-site to be disposed of appropriately).

Waste Management and Disposal

General Requirements

1. Personnel involved in the management or handling of Resource Conservation and Recovery Act (RCRA) hazardous waste shall be appropriately trained and the records for such personnel shall be maintained on-site.
2. All waste streams must be characterized and documented as hazardous or non-hazardous waste; proper storage and management processes for each type of waste must be determined and followed.
3. All hazardous waste manifests are to be signed prior to shipment by either Duke Energy trained employees or vendors with a contractual agreement to sign hazardous waste manifests on behalf of Duke Energy.
4. Consider facility generator status and associated requirements before generating or shipping hazardous waste.
    a. Very Small Quantity Generators (VSQG) generate ≤ 220 lbs. non-acute hazardous waste or ≤ 2 lbs. acute hazardous waste per month.
    b. Small Quantity Generators (SQG) generate >220 lbs. but <2,200 lbs. non-acute hazardous waste or ≤2.2 lbs. acute hazardous waste per month.
    c. Large Quantity Generators (LQG) generate ≥ 2,200 lbs. non-acute hazardous waste or >2.2 lbs. acute hazardous waste per month.
5. Contact EHS Field Professional or Waste Subject Matter Expert with questions or for guidance on waste management and disposal.
6. Hazardous waste must be collected in properly labeled containers.
7. All waste disposal receiving facilities where the following materials are disposed must be Duke Energy-approved:
    a. Polychlorinated biphenyl (PCB) waste
    b. Hazardous waste and materials contaminated with hazardous waste

c. Parts washer fluids
d. Recycled drums
e. Universal waste
f. Electronic waste
g. Used oil
h. Solvent rags
i. Coal combustion residuals
j. Manufactured gas plant waste
k. Petroleum/oil contaminated materials
l. Nonhazardous industrial process waste
m. Scrap metal
n. Antifreeze
o. Selective catalytic reduction catalysts
p. Scrubber waste
q. Treated wood
r. Batteries (lead acid, lithium, nickel cadmium, mercury, silver, etc.)
s. Petroleum contact water (PCW)

8. Minimize the generation of waste by limiting the amount of materials used on a job. Do not co-mingle wastes; this may generate additional hazardous waste.
9. Any discarded or unneeded solid, liquid or gas that presents a danger to people or the environment should be considered a hazardous waste.
10. Report all spills to your supervisor or the field environmental/EHS professional and take immediate action to minimize the extent of the spill.
11. All contractor-supplied unused products should be removed from site by the contractor as product and not shipped as waste.

✪ Hazardous Waste Satellite Accumulation Areas

1. Contact EHS Field Professional or Waste Subject Matter Expert for Satellite Accumulation Area (SAA) requirements. Any type of hazardous waste generator may create a SAA; if no SAA is designated, full compliance with RCRA regulations is expected.
2. Hazardous waste can be collected indefinitely in a satellite area as long as the waste will continue to be generated and does not exceed 55 gallons.
3. Containers must be closed unless adding or removing waste to the container.

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4. Containers must be in good condition and have a container label with an indication of hazard (proper Department of Transportation shipping label may be used).
5. The words “Hazardous Waste” or words identifying the waste must be clearly marked on the container.
6. Once the 55-gallon limit is reached, the generator must mark the date it was full and move the container to the Hazardous Waste Central Accumulation Area within 72 hours.

Hazardous Waste Central Accumulation Areas

1. All containers in the Hazardous Waste Central Accumulation Area (CAA) must be labeled with the hazardous waste label and include the date the waste was moved to this location.
2. Keep aisle spaces of CAA clear and free from obstacles. Maintain at least 2 feet of aisle space.
3. A site may have more than one CAA, but all CAAs must be in the Small Quantity Generator’s Emergency Plan or the Large Quantity Generator’s Contingency Plan.

Nonhazardous Waste Management

1. Characterize each waste stream to determine if usage or interaction with other materials transitioned the material to a hazardous waste category.
    a. Seek guidance from the field environmental/EHS professional on proper disposal of the following materials:

Aerosol cans	Antifreeze	Caulking	Drip water
Electronics	Empty containers	Grease	Hydrostatic test water
Mercury	Non-PCB ballasts	Oil cleanup debris	Oily rags
Photo processing waste	Rinsates	Solvent rags	Tires
Treated wood	Universal wastes	Used oil	Wastewaters

2. Best Practices for waste receptacles (dumpsters):
    a. Dumpsters should be covered at the end of every workday and before storm events.
    b. Empty waste dumpsters as needed to prevent overflow. Clean up immediately if containers overflow.

c. Anchor all lightweight items in waste containers during periods of high winds.

Recyclable Materials Management

1. Items that are typically recycled by Duke Energy include:
    a. Asphalt and concrete (larger volumes of this material may be aggregated and used for beneficial purposes such as beneficial fill, etc. if approved prior to such use)
    b. Cardboard and paper
    c. Plastic or metal drums (recycle or recondition all drums)
    d. Electronics such as cathode ray tubes, testing equipment, printers, copiers, computers, etc.
    e. Lamps
    f. Used oil
    g. Tires
    h. Empty aerosol cans
2. Contact the field environmental/EHS professional for information on recycling programs for your facility or jurisdiction.

Universal Waste Management

1. Universal wastes (UW) are a subset of hazardous wastes that can be managed under reduced requirements. Universal wastes include the following:
    a. Lamps – mercury vapor lamps, fluorescent lightbulbs, neon lights, high-intensity discharge (HID) lamps, high-pressure sodium (HPS) lamps and incandescent lamps
    b. Mercury-containing equipment – thermostats and manometers
    c. Batteries – nickel-cadmium (NiCd), lithium (LiOH, Li-ion), silver oxide and lead-gel batteries
    d. Aerosol cans – damaged, partially full or full containers that are being disposed of
2. Follow the basic requirements for managing UW:
    a. Affix a Universal Waste label on a collection container.
    - Label lamps: Universal Waste – Lamps (in Florida, label containers Universal Waste – Mercury Lamps)
    - Label batteries: Universal Waste – Batteries

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- Label mercury devices: Universal Waste – Mercury-Containing Equipment (in Florida, label containers Universal Waste – Mercury Devices)
- Label aerosol cans: Universal Waste – Aerosol Cans

b. Keep the container closed, except when adding or removing waste.

c. Record the accumulation start date (the date when the first item is placed into the container) on the label.

d. Recycle universal wastes within one year of the accumulation start date.

3. UW batteries may be recycled through Call2Recycle – Rechargeable Battery Recycling Corporation
4. Broken lamps

   a. Broken universal waste lamps and residues are to be managed as a hazardous waste. Contact the waste subject matter expert for proper collection and disposal. (Small amounts (< 5%) of unintentional breakage may be acceptable in packaged lamps.)

Contact your supervisor or the field environmental/EHS professional for additional guidance in managing wastes.

Wastewater

1. National Pollutant Discharge Elimination System (NPDES) permits authorize the discharge of wastewater from point sources to waters of the U.S. Each individual wastewater discharge to a surface water body (lake, stream, canal, ocean or wetland area) must be covered under a NPDES permit to be in compliance with the Clean Water Act.
2. NPDES permits are issued after the permittee agrees to comply with a certain set of conditions including:
   - Limits on pollutants that may be present in the discharge
   - Requirements to monitor these pollutants
   - Requirements to report all monitoring results
3. Any samples collected and analyzed at a permitted outfall must be reported to the appropriate regulatory agency.
4. Laboratories performing compliance analytical work MUST be certified by the appropriate agency and approved by Duke Energy.
5. **NPDES permit compliance is the responsibility of ALL Duke Energy employees and contractors.** Failure to comply with NPDES permit requirements can result in notice of violation (NOV) and associated penalties such as fines, criminal liability (including imprisonment) and disciplinary action.

a. What may be considered a permit violation?
- Exceedance of permit limit
- Failure to meet monitoring and/or reporting requirements (missed samples, late report submittals, missed compliance schedules)
- Discharge of pollutants or wastewater that was not authorized by the NPDES permit
- Discharges of oil, hazardous substances or wastes
- Exceedances of water quality standards
- Bypassing required treatment
- Failure to properly operate and maintain wastewater treatment equipment/systems

6. In order to meet strict reporting requirements, **immediately** report any exceedances or unauthorized discharges to your supervisor and the field environmental/EHS professional.

Water Use and Minimization

1. Many Duke Energy facilities have environmental permits that regulate and limit the consumptive use of water. Be aware and comply with the water withdrawal and/or consumptive use permits for the facility where you are working. Contact the site environmental personnel for compliance assistance with facility permits.
2. Detect and repair all water leaks in a timely manner.
3. Use dry methods for cleaning when possible (e.g., sweep paved areas instead of using water).
4. Control water hose flow with an automatic shut-off nozzle. Do not let water run when not in use.
5. Use vehicle-washing systems that reuse water when available.
6. Contact the facility environmental/EHS professional to discuss water conservation ideas or with water-use-related questions.

Well Drilling and Closure Requirements

1. Construct and maintain all groundwater monitoring wells, drinking water wells or wells that withdraw groundwater for process water in accordance with applicable state and local statutory requirements, regulatory requirements or permit requirements. These requirements include:
    a. Obtaining authorization to withdraw groundwater and to install or abandon wells

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   b. Developing and maintaining proper documentation and records
   c. Conducting monitoring activities in accordance with applicable requirements and analytical procedures

2. Contact the facility environmental/EHS professional when well installation is planned or existing wells are being considered for modification or abandonment.
   a. Installation and use of wells in the various states may require any, or all, of the following activities, record-generating and record-keeping requirements:
   - Groundwater withdrawal authorization via registration
   - Groundwater withdrawal authorization via permit application
   - Groundwater use permit
   - Well construction, modification, repair or abandonment authorization
   - Well completion record
   - Well construction record
   - Specific requirements for monitoring wells
   - Withdrawal reporting
3. Follow well closure requirements when abandoning/closing a well to limit the risk of surface contaminants entering the groundwater. Contact the facility environmental/EHS professional for assistance with well closure requirements.
4. Coal Combustion Products (CCP) contractors installing or abandoning wells should follow the CCP Contractor Supplementals.
5. All work involved in the construction, abandonment and maintenance of wells should be conducted by certified well drillers licensed in the state in which the work is being conducted.
6. All work associated with the construction and maintenance of wells should be conducted under the direct supervision of a groundwater professional onsite to ensure the work is completed safely and in compliance with all applicable requirements, procedures and documentation.
7. All new monitoring well installations should be completed to include well locks (Duke #2) and dedicated pumping equipment. Please contact the Groundwater SME for guidance in the selection of these items.
8. Electrical safety concerns around drill rigs:
   a. Maintain proper distance between drill rig mast and power lines

- Conduct a power line hazard assessment – minimum distance between drill rig mast and power lines is based on voltage (see Table 1, Cranes and Rigging, EHS Handbook)
- Use a dedicated spotter during drilling operations within 20 feet of power lines
- Conditions are most dangerous when raising/lowering mast
- Drill rig should never be moved while the mast is in the up position

  b. Drill rigs travelling under powerlines

- Only travel with mast in the down position
- Maintain proper clearance as specified in Table 2 (Cranes and Rigging, EHS Handbook)
- While maintaining the proper clearance, the effects of speed and terrain must be considered

  c. The drill rig should be properly grounded during operation; contractors follow employer grounding requirements in accordance with grounding requirements in the Health & Safety Sections of the EHS Handbook.

Wildlife Management

1. The presence of wildlife can create safety hazards and distractions while working. Wildlife can also impede the progress of a project or become a general nuisance by:
   a. Leaving droppings in work areas
   b. Standing in or running across roadways and walkways
   c. Nesting too close to buildings
   d. Building dams
   e. Becoming entrapped or entangled in equipment or buildings
2. Employees and contractors can safely coexist with small and large wildlife by following site-specific rules and regulations and using good judgment.
3. Follow the designated facility speed limit.
4. Drive on designated site roads and walk on designated pedestrian walkways.
5. Do not attempt to approach or make contact with any wildlife. Do not try to feed wildlife or leave food scraps out that might attract wildlife to the area.
6. Be aware that state and local laws regulate trapping, relocating, harming and possessing wildlife.

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7. Notify your supervisor or an environmental/EHS professional when the presence of wildlife in a particular location becomes a danger or nuisance to employees and contractors.
8. Pay attention to signs that designate managed wildlife habitat and avoid these areas.
9. When planning an outdoor activity or project, ask the following questions.
    a. Are there managed habitat areas near the project?
    b. Will this project disturb managed habitats or come near managed habitat boundaries?
    c. Could this action unintentionally disturb wildlife nests, burrows or dens?
    d. Do we need a wildlife plan or wildlife permit?
10. If you answer "Yes" or "I don't know" to any of the questions in item 9, contact your supervisor and an environmental/EHS professional for guidance prior to conducting the work.

Contractor Supplemental Requirements

Contractors are required to comply with applicable sections of the EHS Handbook and with business unit-specific supplemental requirements which are mandatory annexes to the handbook. The supplemental requirements documents are available at Duke Energy's Suppliers Resources website at duke-energy.com/partner-with-us/suppliers. Select on Environmental, Health and Safety.

List of Business Unit Supplemental Requirements

- Administrative Services (Real Estate and Telecommunications)
- Coal Combustion Products (CCP)
- Customer Delivery
- Natural Gas Business Unit (NGBU)
- Project Management & Construction (PMC)
- Regulated & Renewable Energy (RRE)
- Transmission
- Vegetation Management

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